AL-BATTĀNĪ

The Sabian Zīj

al-Zīj al-Ṣābi'

the canons, translated from the Arabic

in Nallino's edition, Part III (Rome, 1899)

This is an English translation of the canons of al-Battānī's *Zīj*, made from the Arabic text as Nallino printed it. Every sentence of the Arabic is translated, in the order of the text; the numbers are given in figures. Words added for the sense stand in square brackets. The notes identify quotations, record where the translation follows a correction of Nallino's, and explain terms. The pages of the Arabic text are given in the margin. Nallino's Latin translation (Part I, Milan, 1903) was used as a check on the meaning throughout; its wording is not followed.

The translation is in progress. This version holds the list of chapters and chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31. It is AI-integrated work (Claude), 2026; no person has proofread it. It is dedicated to the public domain (CC0 1.0).

Ar. p. 1

In the name of God, the Merciful, the Compassionate. God bless the Prophet Muḥammad, His noble Messenger, and his family and his companions, and grant them peace.

This is the whole of what Muḥammad ibn Jābir ibn Sinān al-Ḥarrānī, known as al-Battānī — may God pardon him — set down on the computation of the stars and on the places of their courses as they have been tested. The book contains fifty-seven chapters in all, and this is an account of them.[1]

1. The opening of the book.
2. On dividing the circle of the sphere, and on multiplying degrees by one another, taking their roots, and dividing them by one another.
3. On finding the lengths of the chords of the parts of the circle, on entering in tables the halves of the chords of double the arcs, and on the procedures with them that follow from this.[2]
4. On the amount of the inclination of the ecliptic to the equator; on the division of this inclination, its directions, and its stages as it rises and descends. It is the declination of the sun from the right sphere.[3]
5. On finding the amounts of the equator that rise together with given arcs of the ecliptic [in places] under the equator, which is called the line of equality. By these same amounts, too, the signs pass over the meridian circle in every place on the earth; this is why they are called the ascensions of the signs in the right sphere.[4]
6. On the properties of each of the lines parallel to the equator that lie away from it towards the north, and an account of the places of the inhabited earth whose longitude and latitude are known from the book *The Form of the Earth*. Ar. p. 2
7. On finding the amplitudes of the winter and summer risings and settings on the horizon circles of the localities. These are the arcs of the horizon between the equator and the places of the ecliptic, and they are called the azimuths of the risings and settings on the horizon circle.
8. On finding the altitude of the north pole from the excess of the longest day, when this is given.
9. On finding the excess of the longest day from the given altitude of the pole.
10. On finding the altitude and the shadow, each from the other, when the shadow is horizontal, and on finding them when the shadow is vertical.
11. On finding the azimuth of the altitude and of the shadow on the horizon circle, in every locality, at every time of the day, for all degrees of the ecliptic. This is the part of the horizon circle, counted from the limit of rising and setting, that is cut off by the arc passing through the zenith and the sun.
12. On finding the meridian line in every locality, which is the direction of the south, and the direction of the equinoctial rising that becomes known with it; and on finding it in various ways.

[1] The heading gives the author's name as «Muḥammad ibn Jābir ibn Sinān»; the title page that Nallino set above it has «ibn Sinān ibn Jābir». «Tested» (*al-mumtaḥan*) is the word al-Battānī uses for what has been verified by observation.

[2] The halves of the chords of double the arcs are the sines; chapter 3 introduces them under this name.

[3] The print has «ونجزية»; Nallino's corrigenda read «وتجزية», «the division», which is translated.

[4] Nallino suspects that the words «in the localities that are» have fallen out before «under the equator»; the bracket supplies their sense. «The line of equality» (*khaṭṭ al-istiwāʾ*) is the terrestrial equator.

13. On finding the amount of the equator that rises together with given degrees of the ecliptic from the horizon in every place on the earth, which is called the ascensions of the signs in each locality; and what follows from this: finding the ascensions of any portion you wish in these ascensions and in the ascensions of the right sphere, finding the degrees of the ecliptic from these ascensions, the amount of the arc of the day and of the night, their equal hours, the times of the seasonal hours of the day and of the night, and converting the one kind of hours into the other.[5]

14. On finding the latitudes of the localities, which is the altitude of the north pole above their horizon, by observation.

15. On finding the altitude of the sun at noon on any day.

16. On finding how many hours of the day have passed and what is rising, by measuring the sun and knowing the altitude and the vertical shadow.

17. On finding the altitude from the hours of the day that have passed.

18. On finding the distances of the fixed or the wandering stars from the equator when they lie away from the belt of the signs in latitude, and the degrees of the ecliptic that culminate with them, from their places in longitude and latitude with respect to the ecliptic.[6] Ar. p. 3

19. On finding the half of the arc of the day of a star, which is half of its stay above the earth, and also below it, and the times of its hours above the earth and below it.[7]

20. On finding the degree of the ecliptic with which a star rises and the degree with which it sets.

21. On finding how many hours of the night have passed, by measuring some of the stars.

22. On finding the altitude of some of the stars from the hours of the night that have passed.

23. On finding the azimuth of a star from its altitude above the horizon.

24. On finding the distance of a star from the equator, and the parts of the ecliptic that culminate with it, from the azimuth of the place on the horizon circle where it rises or sets. By this the declination of a degree of the ecliptic from the equator is also known.[8]

25. On finding the degree of the ecliptic in which a star is, and the latitude of the star, from its distance from the equator and the degree that culminates with it, when these are known.

26. On finding the distances between the stars according to the arrangement of their places on the sphere in longitude and latitude.

27. On finding the length of the times of the solar year as found by observation, and from it the mean motion of the sun in days, months and years.

28. On finding the irregularity of the motion of the sun, and what appears with it of the degrees of the signs from the places of its farthest distance.[9]

29. On finding the amounts of the difference between days with their nights, when the daytime of one day together with its night is compared with the daytime of another day together

[5] Nallino notes that Plato of Tivoli's copy seems to have had «degree» (*daraja*) for «portion» (*wajh*), which may be the better reading.

[6] The number of this chapter is printed «يج», 13, a misprint for «يح», 18, which the corrigenda do not list.

[7] «The half» is supplied by Nallino; the codex and Plato have «the arc of the day».

[8] Al-Battānī distinguishes the distance of a star from the equator, when the star lies off the ecliptic, from the declination (*mayl*) of the degrees of the ecliptic and of the sun; so the last sentence of the title does not repeat the first.

[9] The Arabic is as ambiguous as Plato's Latin. Nallino takes the sense to be the longitude of the sun reckoned from its apogee, as it comes out from the irregular motion of the sun.

with its night, and how they are converted and transferred from one to the other.

30. On the description of the spheres of the moon and its motions; the irregularity that appears in them at the times of the conjunctions and oppositions with the sun, and the second irregularity that is compounded with it on account of the moon's distances from the sun; the causes of the two kinds of eclipse; the distance of the two luminaries from the earth; and the waxing and waning of the moon's light with its distance from the sun. Ar. p. 4

31. On the description of the spheres of the wandering stars and their states.

32. On finding the eras of the Arabs, the Rūm, the Persians and the Copts, and finding each of them from another.

33. On finding the place of the ecliptic in which the sun is seen, by the era of the Rūm or of the Arabs, whichever of the two you wish.

34. On finding the hours of computation in every locality. These are the mean equal hours after noon at the city of al-Raqqa, by which the motions are found at any time, so that the mean position of a star is known at that time of the day or the night; and on converting these hours into the hours of the localities.

35. On setting up the ascendant and the twelve houses from the hours of the day and the night, and finding the hours from the ascendant.

36. On finding the true place of the moon on the ecliptic on any day and at any time.

37. On finding the place of the northern and the southern node, which are the head and the tail, through which the moon passes in latitude.

38. On finding the latitude of the moon, which is its distance from the belt of the signs towards the south and the north.

39. On finding the parallax that the moon undergoes in longitude and latitude, its amounts in the quarters of the horizon, and the cause from which it arises; in various ways.

40. On finding the distance of the moon from the earth from its parallax in the circle of altitude that lies between the zenith and the horizon and passes through the place of the moon on the ecliptic.

41. On the sighting of the crescent at the beginning and the end of the months; the azimuth of the place in the sky where it is seen; its altitude above the horizon at that time; and its form, according to the light that is in it and to whether its two horns are level or tilted.

42. On finding the mean and the true conjunctions and oppositions of the sun and the moon, by the era of the Rūm or of the Copts, whichever you wish.

43. On finding the lunar eclipses, their amounts and their times in the localities; the side of the moon's disc on which the eclipse begins and the side on which the clearing takes place; and its figure, and working it out by computation and by the tables.

44. On finding the eclipse of the sun, its different amounts in every locality and its times there; the side of the sun's disc on which the eclipse begins and the side on which it clears; and its figure, and working it out by computation and by the tables. Ar. p. 5

45. On finding the places of the five wandering stars on the ecliptic at any time.

46. On finding the stations of the five wandering stars and their retrogradation.

47. On finding the latitudes of the five wandering stars and their directions.

48. On finding the appearance and the disappearance of the five wandering stars.

49. On finding the nine configurations that the fixed stars and some of the wandering stars have

with respect to the sun.

50. An account of the distances of the stars from the earth, their diameters, the sizes of their bodies and the extent of their spheres.

51. On finding the motion of the other stars by observation, and on entering in the table the places of those that are needed, in longitude and latitude.[10]

52. On what the masters of talismans have said, that the sphere has a motion of displacement going forwards and backwards, and the error that appears in this.[11]

53. On finding the times of the revolutions of the years, which take place when the sun returns to the place in which it was at the beginning.

54. On determining the amounts of the applications according to the latitudes of the stars, when they cast their rays on the ecliptic.[12]

55. On finding the ascensions of the signs between the quadrants of the sphere.

56. On making the vertical and the horizontal sundial for finding the seasonal hours of the day in every locality; on setting it up correctly and on the direction of the south; and how the direction of the qibla, which is the direction of Mecca the protected, is found on the sundial.[13]

57. The conclusion of the book, and a description of the making of the instrument in the shape of the sphere, which is called «the egg», and of the making of the two instruments for observation.[14]

This is the account of the contents of the book in detail, which comprises fifty-seven sections. Praise be to God for His help, and God bless Muḥammad.

Chapter 1. The opening of the book

Ar. p. 6

He said: The first thing with which every matter is begun and every discourse is opened is the praise of God, exalted be His mention, thanks to Him for His benefits, and the blessing on the Seal of His messengers and prophets — upon them be peace, and the mercy of God and His blessings.[15]

Praise be to God, who created the creatures by His power, ordered all things by His will and perfected them by His wisdom; who «encompasses all things in knowledge» and «has counted all things in number»; from whom «not the weight of an atom in the heavens or on the earth escapes, nor anything smaller than that or greater, but it is in a clear book».[16]

I bear witness that there is no god but God alone, who has no partner, and I bear witness that Muḥammad is His servant and His messenger, whom «He sent with the guidance and the religion of truth, to make it prevail over all religion, though the idolaters dislike it». Through him He guided the believers and «cut off the last of the unbelievers», and He made him a proof for all the worlds. God bless him and his good family, his chosen companions and those who follow his practice, until the Day of Judgement.[17]

[10] Where this list has «the other stars», the heading of the chapter itself and Plato's Latin have «the fixed stars».

[11] This is the theory later known as trepidation, an oscillation of the equinoxes. Nallino translates «the masters of talismans» (*aṣḥāb al-ṭilasmāt*) as «the astrologers».

[12] The applications (*ittiṣālāt*) are the astrological aspects between stars.

[13] The sundial is *al-rukhāma*, literally «the marble slab».

[14] Nallino suspects «making» (*ṣanʿa*) where the codex has *ḍaʿa*, «placing»; Plato has «making». The translation follows him.

[15] The Seal of the messengers and prophets is Muḥammad (cf. Qurʾān 33:40).

[16] Qurʾān 65:12, 72:28 and 34:3. The «clear book» is the book that is with God.

[17] Qurʾān 9:33 and 61:9 (Nallino's note has XLI, 9, a misprint for LXI, 9); for «cut off the last of the unbelievers», cf. 7:72 and 8:7.

To proceed. After the knowledge of the laws and the practice of religion, of which no one may be ignorant, the science of the art of the stars is among the sciences that are highest in rank and most exalted in standing, finest in adornment, that cling most to the heart and shine most in the soul, and that do most to sharpen thought and reflection, to quicken understanding and to train the mind. For there is a great share of good in it and great benefit: the knowledge of the length of the years and months, of the appointed times, of the seasons, of the lengthening and shortening of the day and the night, of the places of the two luminaries and their eclipses, of the courses of the stars in their direct and retrograde motion, of the changes in their configurations, of the order of their spheres and of all their other relations; and beyond this, what is attained through it by one who examines it closely and reflects on it constantly: the proof of the unity of God, and the knowledge of the essence of the Creator's greatness, of the breadth of His wisdom, the majesty of His power and the subtlety of His handiwork.[18]

Ar. p. 7

He has said — mighty is the Speaker: «In the creation of the heavens and the earth, and in the alternation of night and day, there are signs for those who have understanding.» He, blessed and exalted, has said: «Blessed is He who has placed signs in the sky.» He, mighty and majestic, has said: «It is He who has made the night and the day to follow one another.» He, glory be to Him, has said: «It is He who has made the sun a radiance and the moon a light, and has measured out its stations, so that you may know the number of the years and the reckoning.» And He, exalted be His mention, has said: «The sun and the moon run by a reckoning.» There is much more of this kind in the book of God, mighty and majestic, which it would take long to describe and much space to mention and cite.[19]

When I had studied this science at length and reflected on it constantly, I came to know how the books composed on the motions of the stars disagree, and into what errors some of their authors fell in the procedures they set down in them as foundations and in what they built on these. I also came to know what has accumulated in the motions of the stars over long periods of time, when their observations are compared with the ancient observations; that the inclination of the ecliptic to the equator has been found to be smaller; and what has changed with this change in the various kinds of computation, in the lengths of the years, in the times of the seasons, and in the conjunctions of the two luminaries that are inferred from the times and hours of the eclipses. So I set about correcting all this and making it exact according to the method of Ptolemy in the book known as the Almagest, after close study, long reflection and consideration, following in his footsteps and keeping to what he laid down. For he had investigated these matters from every side and had shown their causes and the reasons that occur in them by a geometrical and numerical proof whose soundness cannot be rejected and whose truth cannot be doubted. After that he ordered that they be tested and examined after him, and said that his observations might be improved upon in the course of time, as he himself had improved upon Hipparchus and others of his peers; for the art is a lofty one, and, being heavenly and vast, it can be grasped only approximately.[20]

I have composed a book on this, in which I have made clear what was obscure, opened what was closed and explained what was difficult in the principles of this science and out of the way in its branches. With it I have made the path of guidance easy for those who follow it and work by it in the art of the stars. In it I have corrected the motions of the stars and their places on the belt of

[18] The configurations of the stars are the nine configurations with the sun that chapter 49 explains (Nallino).

[19] Qur'ān 3:190, 25:61, 25:62, 10:5 and 55:5. The signs (*burūj*) of 25:61 are the word that al-Battānī uses for the signs of the zodiac throughout the book. In the third quotation the text prints «the night and the moon a light, in succession», a slip that mixes this verse with the next quotation; the Qur'ān (25:62) has «the night and the day», which Nallino translates and the translation follows.

[20] «Has been found to be smaller» renders *mā wujida … min al-taqārub*, literally «the drawing together that was found»: al-Battānī's observations gave a smaller obliquity of the ecliptic than the Greek value (chapter 4). Nallino stresses that the words do not show knowledge of a continuous decrease of the obliquity. For Ptolemy's remark see the Almagest, III 1 (in Heiberg's numbering; Nallino cites Halma's III, 2).

the ecliptic, as I found them by observation and by the computation of the two kinds of eclipse, together with all the other procedures that are needed, and I have added other things that are needed. I have arranged that the motions of the stars are found from the tables for noon of the day for which the computation is made, at the city of al-Raqqa, where the observation and the testing that establish all this took place — if God, exalted is He, wills. Success comes from God.

Chapter 2. On the division of the circle of the sphere, and on multiplication, roots and division

Ar. p. 8

He said: The ancients divided the circle of the sphere into 360 parts, and they gave more than one reason for this. One is that the number of these parts is close to the number of days of the year, which is completed when the sun passes a fixed point of the sphere until it returns to it. Another is that it is a number that has a half, a third, a quarter and other fractions that do not come out whole for many numbers. They placed the sun at four points of the sphere, which bring about two equinoxes and two solstices and divide the year into four distinct parts, spring, summer, autumn and winter; and they named each of these points after the season that begins when the sun passes it.

Since everything that has extension has a middle and two ends, each of these seasons divides into three parts, and so the circle of the sphere had to have twelve parts. They found the spring point the best of these points and the most fitting to begin with. For from it, after the equinox, the day begins to lengthen, and the sun is rising into the northern half of its sphere, so that the heat grows; and the nature of this season is moist and inclines to heat, which suits the beginning of growth and the coming into being of things. So they made the reckoning of the sphere begin from it.[21]

Then they found that the constellations next to these twelve parts, which are called signs, were twelve constellations, and they named each sign after the constellation next to it, although over a long time these constellations move away from the places of the signs named after them. So the first part became Aries, followed by Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius and Pisces.[22]

Each of these signs therefore has 30 parts as its share of the 360 parts of the circle of the sphere. These parts are also called degrees. Each degree is divided into 60 parts, called minutes; each minute is likewise divided into 60 parts, called seconds; each second is divided into 60 thirds; and so on, by the same rule of division, to tenths and the separate orders that come after them.

Ar. p. 9

As for the meaning of multiplication: it is repeating one of two numbers as many times as there are units in the other — I mean the multiplication of units by units. Multiplying fractions by units is repeating the fractions as many times as there are units, or taking of the units the part that the fractions are of one. Multiplying fractions by fractions is taking of one of the two fractions, whichever you wish, the part that the other fraction is of one.

Thus, when degrees are multiplied by degrees, the product is degrees; when they are multiplied by minutes, it is minutes; when they are multiplied by seconds, the product is seconds; and so with what is multiplied by thirds, fourths and the following orders: the product is of the order of the smaller one, by which it was multiplied. As for the orders below degrees, minutes and the others: when each of them is multiplied by itself, the product falls below it by as much as it falls below degrees. For example, minutes multiplied by minutes give seconds, and multiplied by seconds

[21] Nallino notes the same account of the spring point in Abū Maʿshar (d. 886), *Introductorium* II 5, and, almost word for word, in Abraham bar Ḥiyya's *Sphaera mundi* (1165), ch. 2.

[22] The Arabic names are al-Ḥamal (the Ram), al-Thawr (the Bull), al-Jawzāʾ (Gemini), al-Saraṭān (the Crab), al-Asad (the Lion), al-Sunbula (the Ear of Grain), al-Mīzān (the Balance), al-ʿAqrab (the Scorpion), al-Qaws (the Bow), al-Jady (the Kid), al-Dalw (the Bucket) and al-Ḥūt (the Fish). The translation uses the Latin names by which the same signs are known in English.

they give thirds, and what is multiplied by thirds and fourths falls by the same rule. Seconds multiplied by seconds give fourths, and multiplied by thirds they give fifths; and whatever lies further from these orders goes the same way, by the same rule.

When any number of one of these orders, obtained by multiplication or by addition, is divided by sixty, the number at which every order ends and to which all the fractions are related, the result belongs to the next higher order. When, in a number made up of two or more of these orders, more has to be subtracted from one of them than it contains, one unit is broken off from the next higher order for it, counted as 60 parts and added to it; then as much as needed is subtracted, and what remains is counted together with what remains of the higher order.

As for degrees: whatever accumulates of them when the parts of the motions are added is reckoned in revolutions. If the sum is more than one revolution, or than several revolutions, a revolution being 360 degrees, the revolutions are dropped and what remains is counted. When more has to be subtracted from the degrees than they contain, a revolution is added to them; then as much as needed is subtracted from the sum, and what remains is counted.

Ar. p. 10

If you wish to multiply one of the orders of degrees or of fractions by another of them, and to know from this table of which order the product will be, take on one of the two lines AB the cell in which the order you wish to multiply by any other order is written, and go straight on from that cell until you are level with the other order that you want: the cell level with it, among the orders of fractions, gives the order of the product. For example, suppose you wish to multiply fourths by thirds. Take on the line AB that runs across the width of the page either of the two orders — let it be the thirds first — and go on from it until you are level with the fourths on the line AB that runs along the length of the page; in the cell level with it you find sevenths, and this is the order of the product. Likewise, if you took the fourths on a line AB and went from them to the level of the thirds on the other line AB, you would find sevenths there. Do the same with any orders you wish, if God wills.

The orders of products[23]

degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths
minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths
seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths
thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths
fourths	fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths
fifths	sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths
sixths	sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths
sevenths	eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths
eighths	ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths
ninths	tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths	nineteenths
tenths	elevenths	twelfths	thirteenths	fourteenths	fifteenths	sixteenths	seventeenths	eighteenths	nineteenths	twentieths

[23] The print reads from right to left; here the columns are set from left to right. The first row is the line AB across the width of the page, the first column the line AB along its length. The cell of the fourth row and second column is printed «ورابع», a misprint for «روابع», «fourths».

Ar. p. 11

As for the meaning of the root: the root of any absolute number, whatever number it is, is that which, multiplied by itself, gives the given number. But the roots of these orders do not keep this rule, because of the difference that we have described in what results from multiplying these orders by one another. The rule holds only for the order of degrees: the root of degrees is also degrees, since degrees multiplied by degrees give degrees. As for the fractions below degrees, of the remaining orders: those of even rank, such as seconds, fourths, sixths and the like, have their root in the higher order whose rank is half of theirs — seconds, whose root is minutes, and fourths, whose root is seconds. Those of odd rank, such as minutes, thirds and the like, have no definite root unless they are converted into the order below them, so that they come to an even rank; then this rule applies to them: minutes are converted into seconds, and thirds into fourths.[24,25]

Division is finding how many times the smaller number goes into the larger, when the larger is measured by the smaller, and finding what part of the larger the smaller is, when the smaller is the number divided. Here we proceed in the reverse of what we did in multiplications and roots, by the same rule: if we divide degrees by degrees, the quotient is degrees. As for the other orders below degrees: when a lower order is divided by a higher one, whatever its rank and whether or not it is next to it, the quotient is of the order which, multiplied by the order divided by, gives back the order of the number divided. Thus seconds divided by minutes give minutes, and sixths divided by fourths give seconds. But when a higher order is divided by a lower one, the way to do it is to convert the higher order into the lower and then divide by it, and the quotient is degrees. So with minutes divided by sixths: when they are converted into sixths and then divided by those sixths, the result of the division is degrees, as we have described.

Ar. p. 12

If you wish to know, with the table given here, what results from dividing the lower orders of fractions by the orders above them, look up in the line AB or in the line AD, whichever you wish, the order that you wish to divide by an order of higher rank, whether or not it is next to it, and go on level with it until you are level with the higher order on the other line. The order of fractions at which you arrive is the order that you obtain by dividing the number divided, and it is the order which, multiplied by the higher order that you divided by, gives back the order of the number divided. Likewise, if you wish to divide a higher order by a lower one, convert the higher into the lower, look up in one of the two lines the order into which it has been converted, and go on level with it until you are level with the order that you wish to divide by: you obtain degrees. And whenever you divide an order by the same order, the result is degrees, if God, exalted is He, wills. Success comes from God.[26]

[24] An absolute number (*ʿadad muṭlaq*) is a number taken by itself, not counted in one of the sexagesimal orders.

[25] The print has «فتلزمه» (p. 11, line 7), a misprint for «فتلزَمه», «applies to them», which the corrigenda do not list.

[26] «With the table given here»: the Arabic says «with this table mentioned before»; the table is printed at the head of the next page.

The orders of quotients[27]

degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths	tenths
minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths	ninths
seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths	eighths
thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths	sevenths
fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths	sixths
fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths	fifths
sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds	fourths
sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds	thirds
eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes	seconds
ninths	eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees	minutes
tenths	ninths	eighths	sevenths	sixths	fifths	fourths	thirds	seconds	minutes	degrees

[27] The columns are set from left to right, as in the first table. The first row is the line AB, the first column the line AD. The cells of the second row and fourth column and of the fourth row and second column are printed «ثولنٍ» and «توانٍ», misprints for «ثوانٍ», «seconds».

Chapter 3. On finding the lengths of the chords of the parts of the circle, on entering in tables the halves of the chords of double the arcs, and on all the procedures with them that follow from this[28]

Ar. p. 13

He said: The ancients disagreed about the size of the diameter of the circle in relation to its circumference, although they came close to it. Some said that the circumference of the circle is 3 times its diameter and 1/7 of it. Others said that it is 3 times [the diameter] and a little more than 10 parts of 71. What the excellent Ptolemy and the astronomers worked with lies between these two amounts: 3 times [the diameter] together with 1/10 of it and 1/4 of 1/6 of it. We do not need to know the truth of this in setting out the chords, since arcs and chords have no known measure in relation to one another, and this is known only from their chords. Nor does it do us any harm to take the diameter as large as we wish. This is why Ptolemy made it 120 parts, because the computation comes out easily this way; and we also work with it in this book.[29]

Ar. p. 14

It has been shown by proof that the chord of a sixth of any circle is equal to its radius. A sixth of the circle of the sphere is evidently 60 parts, according to the basis of the computation, in the measure in which the circle is 360 parts; and the chord of the sixth is also 60 parts, in the measure in which the diameter is 120. If the chord of a sixth of the circle is multiplied by itself and subtracted from the whole diameter multiplied by itself, the root of the remainder is the chord of a third of the circle. In the same way, for any arc of a circle whose chord is known: if the chord of that arc is multiplied by itself, the product is subtracted from the whole diameter multiplied by itself, and the root of the remainder is taken, the result is the chord of the arc that remains to complete the semicircle. The chord of a quarter of the circle is the root of twice the product of its radius by itself. The chord of a tenth of any circle is found thus: to the radius multiplied by itself add a quarter of the diameter multiplied by itself, take the root of the sum, and subtract from it a quarter of the diameter of the circle; what remains is the chord of a tenth of that circle. The chord of a fifth of the circle is found thus: multiply the chord of its tenth by itself, add to it the radius multiplied by itself, and take the root of the sum; this is [the chord of] the fifth of that circle.

For any two arcs of a circle whose chords are known, the chord of the arc by which they differ is also known. Multiply the chord of each of the two arcs by the chord of what remains of the other to complete the semicircle, take the difference between [the two products] and divide it by the diameter; the result is the chord of the arc by which the two arcs differ.

For any arc of a circle whose chord is known, the chord of its half is also known. Subtract the chord of what remains of that arc to complete the semicircle from the whole diameter of the circle, take half of the remainder, multiply it by the whole diameter, and take the root of the product; the result is the chord of half that arc.

For any two arcs of a circle whose chords are known, when one is placed on the other and we join them so that they become one arc, the chord of the combined arc is also known. Multiply the chord of each of the two arcs by [that of] the other, and the chord of what remains of each of them to complete the semicircle by [that of] the other as well; take the difference between [the two products] and divide it by the whole diameter. The result is the chord of what remains of the combined arc to complete the semicircle. If it is multiplied by itself and subtracted from the whole diameter multiplied by itself, the root of the remainder is the chord of the arc combined of

[28] The translation follows Nallino's corrigenda in three places: «لتمام» for «لتمام» (p. 14, line 4), «وتر» for «وثر» (p. 14, line 18) and «حاجتنا» for «جاجتنا» (p. 15, line 19). It also reads «نحتاج» for «نحناج» (p. 15, line 11), «والوتر» for «واوتر» (p. 16, line 14), «يتلوه» for «بتلوه» and «فزدته» for «فردته» (p. 16, line 17) and «الوتر» for «اوتر» (p. 17, line 11), misprints that the corrigenda do not list.

[29] 3 1/7 and 3 10/71 are the two bounds that Archimedes proved for this ratio in the *Measurement of the Circle*. Ptolemy's value, 3 + 1/10 + 1/24 = 3 17/120, is 3;8,30 in sexagesimal fractions (*Almagest* VI 7). The ratio that is known only from the chords is that of the chord of one arc to the chord of another (Nallino).

the two arcs. By this rule and in these ways all the remaining chords in the semicircle that can be known are found.

As for the chords that are not known by proof, such as the chord of one degree and its doublings, two, four, eight and the like: they are not found by computation in the way of proof, as those arcs and those chords are. But it is known by proof that the ratio of the chord of a smaller arc to its arc is greater than the ratio of the chord of a greater arc to its arc. Now the chord of one and a half degrees and the chord of a half and a quarter [of a degree] are known by proof, and 2/3 of the chord of one and a half degrees is equal to the chord of a half and a quarter of a degree increased by a third of itself; there is no perceptible difference between them, and no harm to the computation comes from it. If the chord of three quarters [of a degree] is taken and a third of it is added to it, the sum is the chord of one degree. Once the chord of one degree is known in this way, all the chords of the degrees of the semicircle become known as well.[30]

Ar. p. 15

What is needed about the amounts of arcs that intersect one another in known arcs is found only through the chords of double the known arcs. When a circle is divided by two lines that intersect at its centre at right angles, it is divided into four equal quarters, at four angles; each angle takes in 90 parts of the circumference and is enclosed by two lines that go out from the centre to the circumference, each of them the radius. The two right angles under two quarters together are enclosed by one straight line, the whole diameter, which is plainly twice each of the two lines that enclose the one right angle under one quarter. So the ratio of each of the two lines that enclose a right angle to the diameter that encloses the two right angles is as the ratio of a quarter of the circle to its half. Hence the diameter divides the chords of the remaining arcs in the semicircle into two halves, and it divides the arcs with them into halves on its two sides. So the ratio of the chord of each of these arcs to the whole diameter is as the ratio of half that chord, which lies under half that arc, to the radius; and this is half the chord of twice the arc that lies on either side of the diameter, half of it in each of the two quarters. This is what we mean, and it is what we use in the procedures of computation, so that we do not need to double the arc in what we seek to find. Ptolemy did that only to establish the proof.[31]

We, however, have taken half the chord of twice each arc of the quarter of the circle and entered it under the entry for that arc in the quarter, and we have made the arcs in the tables increase by half a degree up to the full 90 degrees that make up the whole quarter. So half the chord of one degree stands under half a degree; half the chord of 60 under 30 degrees; half the chord of 120 under 60; and half the chord of 180 degrees — the semicircle, whose chord is the whole diameter — under 90, the degrees of the whole quarter: this is the radius, whose amount is 60 parts. All these halved chords entered in this book are related to it. So that we need not repeat ourselves in what follows, we state that whenever we speak of chords in this book, we mean these halved chords, except where we name one specially and call it a full chord; and for the most part we seldom need these.[32]

Ar. p. 16

If you want to find from the tables the sine of any degree you like, look in the table of sines, in the column of the numbers that increase by half a degree; where you find the number that you have, take the degrees, minutes and seconds entered opposite it in the table of sines: that is the sine of the degrees you wanted. If there are minutes with the degrees, more or less than 30 minutes, take what stands opposite the whole degrees, or the degrees and halves, whichever is nearest to the degrees and minutes you have and smaller than them, and keep what you find opposite

[30] This is Ptolemy's procedure for the chord of 1° (*Almagest* I 10). By the inequality just stated, the chord of 1° lies between 2/3 of the chord of 1½° and 4/3 of the chord of ¾°, and these two bounds agree to the precision of the tables.

[31] The first sentence is obscure; Nallino, following Schiaparelli, takes it to mean that every computation that combines several arcs is made through the chords of the doubled arcs.

[32] From here on the translation calls these halved chords «sines», as Nallino does in his Latin; al-Battānī calls them «halved chords» (*awtār munaṣṣafa*) or simply «chords». Other Arabic authors call the sine *jayb*. Since the radius is 60, al-Battānī's sine of an arc is 60 times its sine in modern terms. The full chords are needed in chapters 26, 50 and 54 (Nallino).

it in the table of sines. Then subtract the number you found in the column from the one you have, multiply the minutes that remain by the difference between the sine you kept and the sine opposite the number half a degree greater, and divide the product by the 30 minutes by which the numbers in the two columns of arcs increase. Add the minutes and seconds that result from the division to the sine you kept if it is the smaller, and subtract them from it if it is the greater; what you obtain after the addition or the subtraction is the sine of the degrees and minutes that you have.[33]

Or, if you wish, find what part of 30 minutes the minutes left over are: if they are a half, or a third, or less or more than that, take the same part of the difference of the sines and proceed with it in the same way, adding or subtracting; the result is the sine of the arc you wanted.

If you wish to find arcs from these sines, look for the sine in the table of sines; where you find it, or the nearest value smaller than it, take what stands opposite it in the first of the two columns of numbers: that is the arc you want, and keep it. Then subtract the sine you found in the tables from the sine you have, multiply the remainder by 30 minutes, and divide the product by the difference between the sine you found and the one that follows it. Add the minutes and seconds that result to the arc you kept; the arc you then have is the arc of the sine you want. Or, if you wish, see what part the minutes and seconds left over are of the difference between the sine you found and the one that follows it, take the same part of 30 minutes and add it to the arc you kept, as before; it comes to the same. This procedure is called taking the arc of the sines.[34]

If you wish to find the versed sines from the arcs: if the number whose versed sine you want is less than 90 degrees, subtract it from 90, find the sine of the remainder by the rule given above, and subtract it from 60 degrees, which is the radius; what remains is the versed sine of that arc. If the number is more than 90, find by how much it exceeds 90, find the sine of the excess, and add it to 60 degrees; the sum is the versed sine of the arc you wanted.[35]

Ar. p. 17

If you wish to find the arcs from these versed sines: if the versed sine you have is less than 60 degrees, subtract it from 60, find the arc of the remainder by that rule, and subtract the arc from 90; what remains is the arc of the versed sine. If the versed sine is more than 60, take 60 from it, find the arc of the remainder, and add it to 90 degrees; the sum is the arc of the versed sine.

You need no more for finding arcs and sines than what I have set out for you. For these sines it is enough to know the sines from 1 degree to 90 degrees, because the sine of whatever goes beyond 90, up to 180, is like the sine of the degrees up to 90 taken in reverse order. Likewise for the full chords you need no more than the chords of the semicircle, from 1 degree to 180, because the chords of the other half are like those of the 180 degrees in reverse order.

As for finding the full chords from the arcs, and the arcs from these chords: if you wish to find the full chord of any number of degrees, take half those degrees, find its sine from the table, and double it; the result is the full chord of the degrees you wanted. If you wish to find the arcs of the full chords with the tables as well, take half the chord, find its arc from the table in the way described above, and double the arc; the result is the arc of the full chord that you wanted.

Whenever one of these sines is multiplied by itself and subtracted from the radius multiplied by itself, the root of the remainder is the sine of what remains of that arc to complete a quarter of the circle. And when the sine of any degree is subtracted from the radius, the remainder is multiplied by 30 parts and the root of the product is taken, the result is the sine of half of what remains of that arc to complete a quarter of the circle, if God wills.[36]

[33] The sines are given in degrees, minutes and seconds of a radius of 60 degrees.

[34] «Taking the arc» is *taqwīs*, Nallino's *arcuatio*: finding the arc that belongs to a given chord or sine.

[35] The versed sine is *al-watar al-rāji*ʿ, literally «the returning chord». In modern terms the versed sine of an arc a is $60(1 - \cos a)$; the two rules of the text are the cases $a < 90°$ and $a > 90°$.

[36] In modern terms, writing Sin a = 60 sin a for al-Battānī's sine of an arc a: $\text{Sin}(90° - a) = \sqrt{(60^2 - (\text{Sin } a)^2)}$, and $\text{Sin } ½(90° - a) = \sqrt{(30\ (60 - \text{Sin } a))}$, because $60^2 \sin^2 ½(90° - a) = 1800\ (1 - \cos(90° - a)) = 30\ (60 - \text{Sin } a)$.

Chapter 4. On finding the amount of the inclination of the ecliptic to the equator, the division of this inclination, its directions, and its ranks in its ascent and descent; this is the declination of the sun from the right sphere

Ar. p. 17
Ar. p. 18

He said: The ecliptic is defined by the path on which the sun is seen to move; the equator is the circle along which the greatest sphere turns about its two poles. The inclination of the ecliptic to the equator is found only by observing the sun and following its passage through the two solstitial points on the meridian circle. This is the circle of the middle of the sky; it passes through the two poles of the equator and the zenith, and it cuts the horizon circle.

Hipparchus stated, and Ptolemy reports in his book, that the arc of the meridian circle between the winter and the summer solstice is 47° 42′, and that the inclination is half of that, 23° 51′. We ourselves observed it many times in our own day with the long alidade and with the brick, whose making and setting up are described in the Almagest, after dividing [their scales] precisely and setting up the instrument as exactly as could be done. We found the least distance of the sun from the zenith on the meridian circle, in the city of al-Raqqa, to be 12° 26′, and its greatest distance 59° 36′. From this it became clear to us that the arc between the two solstices is in truth 47° 10′, and that the inclination of the ecliptic to the equator is half of these degrees, 23° 35′; this is the distance between the poles of the two circles. We work with this value in this book, since we have seen it with our own eyes, while the first is known by report.[37,38]

From this we also learned that the distance of the city of al-Raqqa, where the observations were made, from the equator on the meridian circle is 36°. This is the altitude of the north pole of the equator above the horizon there, and it is also the distance of the equator from the zenith towards the south.[39]

If you wish to compute this inclination, so as to find the part of it that belongs to each degree you want, up to the full 90 degrees that complete the inclination, which is 23° 35′: take the sine of one degree, or two degrees, or more, up to the full 90 degrees, which run from the beginning of Aries to the end of Gemini. When you know the sine of the degrees you wanted, multiply it by the sine of the whole inclination, divide the product by the radius, which is 60, and find the arc of the quotient. The arc you obtain is the declination of the degree you wanted from the equator, on the meridian circle.[40]

Ar. p. 19

If you wish to enter this in a table degree by degree, find the declination of each degree you want up to 90 degrees, and enter under each degree what belongs to it. When you have done this, you know the declination of all the parts of the ecliptic. For the declination of what goes beyond 90, up to 180, is like the declination of the 90 [degrees] entered, taken in reverse order; the declination of what goes beyond 180, up to 270, is like the declination of the 90 in the same order; and the declination of what goes beyond 270, up to 360, is like the declination of the 90 in reverse order. We have entered this in the table at intervals of one degree, with four columns of numbers. In the first column we have entered the numbers of the ecliptic from 1 degree to 90; in the second, what the numbers of the first column fall short of 180; in the third, the numbers of the first column added to 180; and in the fourth, what the numbers of the first column fall short

[37] Ptolemy (*Almagest* I 12) found the arc between the solstices to be more than 47 2/3° and less than 47 3/4°, nearly the ratio of 11 to 83 parts of the meridian that he ascribes to Eratosthenes and says Hipparchus also used; his table of declinations (*Almagest* I 15) uses an inclination of 23° 51′ 20″.

[38] The long alidade (*al-ʿiḍāda al-ṭawīla*) is Ptolemy's parallactic instrument, the triquetrum (*Almagest* V 12); the brick (*al-libna*) is his plinth, a quadrant set up in the plane of the meridian (*Almagest* I 12). Al-Battānī describes both in chapter 57 (Nallino).

[39] The two observations give 36° 1′, as Delambre noted, since the equator lies midway between the two solstitial points: (12° 26′ + 59° 36′)/2 = 36° 1′. Nallino suggests that al-Battānī dropped the minute to keep his tables simple, 36° being also the latitude of Rhodes, for which Ptolemy computed several of his tables.

[40] In modern terms, $\sin \delta = \sin \lambda \cdot \sin \varepsilon$, where λ is the distance of the degree from the beginning of Aries, ε the inclination and δ the declination.

of 360. Thus, if the number falls in the first or the second column, we know that the declination is towards the north of the equator, and if it falls in the other two columns, we know that it is towards the south.

If you wish to find the declination of the sun, or of any other degree of the signs, take [the distance] from the beginning of Aries to the degree of the sun, or to whatever other degree whose declination you wish to find; the result is the argument of the declination. Look for the same number in the four columns of numbers entered in the tables of declination, and take the degrees, minutes and seconds entered opposite it: that is the declination of the degrees you wanted. If there are minutes with the degrees, take for them their share of the difference, in the way I showed you for the differences of the sines. See what part the minutes are of the 60 minutes by which the numbers increase, and take the same part of the difference between the declination of the whole degrees and the declination of the degree one greater. Then look: if the declination of the degree you have is the smaller, add what you obtained to it; if it is the greater, subtract it from it. The declination you obtain after the addition or the subtraction is the declination of the degrees and minutes you wanted from the equator.[41]

If the argument is from 0 to 90, the declination is increasing and the sun is ascending in the north; if it is from 90 to 180, the declination is decreasing and the sun is descending from the north; if from 180 to 270, the declination is increasing and the sun is descending towards the south; and if from 270 to 360, the declination is decreasing and the sun is ascending in the south. In short, when the argument of the declination is from 0 to 180, the declination is northern, and when it is from 180 to 360, it is southern. By this computation you find the declination of the sun, its direction, and its ascent and descent.[42]

They have divided the declination into six ranks of ascent and descent: they made every 15 degrees of the sun's course in each of these quarters one of the ranks of ascent and descent, up to the full 90 degrees that complete the six ranks. When the sun is in the first 15 degrees of one of the quarters, they say that it is in the first rank; when it is in the second 15 degrees, they say that it is in the second rank; and so on until the sixth rank is reached.[43]

Chapter 5. On finding the ascensions of the signs in the right sphere[44]

Ar. p. 20

He said: Suppose you wish to know how many of the 360 times of the equator rise together with given parts of the ecliptic. These are the ascensions of the signs at the place of the line of equality. This is the place that has no latitude, above which the equator turns; there night and day are always equal, on all the days of the year. In every locality the signs cross the middle of the sky with the same amount [of the equator] with which they rise on this line, and there they also cross the middle of the sky with this amount. This is why these are called the ascensions of the signs in the right sphere. Every three signs rise in the right sphere together with 90 times of the equator.[45]

If you wish to compute the ascension in the right sphere of any degree of the signs you like, take the whole inclination, which is 23° 35′, and find its sine: this is the sine of the whole inclination. Then subtract the whole inclination from 90 and find the sine of the remainder: this is the sine of the complement of the whole inclination. Then take [the distance] from the beginning of Aries

[41] «The argument of the declination» is *ḥiṣṣat al-mayl*, literally «the share of the declination»: the number with which the table is entered (Nallino's *argumentum declinationis*).

[42] The text writes these numbers in abjad numerals, with the letters of the alphabet, and 0 with the zero sign of the codex.

[43] Theon of Alexandria describes the same ranks (Greek *bathmoi*) of 15° as a usage of the astrologers; al-Battānī gives a table of them at the end of his table of the declination of the sun (Nallino).

[44] The translation reads «تلك», «that», for the misprint «تلك» (p. 20, line 14), which the corrigenda do not list.

[45] A time (*zamān*) is a degree of the equator, the measure of the daily turning of the sphere. The three signs are those of a quarter of the ecliptic that begins at an equinoctial or a solstitial point.

to the degree you want, find the declination of that degree, and find its sine: this is the sine of the declination of the degree. Then subtract the declination of the degree from 90 and find the sine of the remainder: this is the sine of the complement of the declination of the degree. Then multiply the sine of the declination of the degree by the sine of the complement of the whole inclination, divide the product by the sine of the whole inclination, multiply the quotient by the radius, which is 60, divide the product by the sine of the complement of the declination of the degree, and find the arc of the quotient. The arc you obtain is the amount of the equator that rises from the beginning of Aries to the degree you wanted.[46]

If you computed for 30 degrees, the result is the ascension of the whole sign of Aries; if you computed for 60, it is the ascension of Aries and Taurus together: subtract from it the ascension of Aries, and the ascension of Taurus remains. Then subtract the ascension of Aries and Taurus together from 90; what remains is the ascension of Gemini. When you know the ascension of Aries, the ascensions of Virgo, Pisces and Libra are equal to it; the ascensions of Leo, Aquarius and Scorpio are equal to that of Taurus; and the ascensions of Sagittarius, Capricorn and Cancer are equal to that of Gemini.

Ar. p. 21

By this rule you derive the ascensions degree by degree and enter them in the tables. Begin from the beginning of Capricorn, so that from it you know the ascensions of the signs in the middle of the sky for every locality, and so that the number by which one finds what culminates and what parts rise in every locality is one and the same number. We shall explain how the ascensions are put into tables when we come to the ascensions of the signs in the climates, so as not to repeat ourselves.[47]

Chapter 6. On the properties of the parallel lines that run parallel to the equator, on the places of the inhabited earth that are known in longitude and latitude, and on what follows from this[48]

Ar. p. 21

He said: We should begin with the equator, and then speak of the other circles that decline from it towards the north, and of the places of the earth that lie beneath these circles.

We say: The line of the earth that lies beneath the equator is the line called the line of equality. It has no latitude; the equator turns upon it and above it, from east to west, and there day and night are always equal, on all the days of the year, as we said at the beginning. When the sun passes over this line, and only then, day and night are balanced and perceptibly equal over the whole earth. The sun is then at the point common to the ecliptic and the equator, for this is where the two circles intersect: the point of the beginning [of Aries and the beginning] of Libra. Only then does the sun stand at noon above the heads of those who live on this line, so that upright objects there cast no shadow. When the sun runs in the northern half of the ecliptic, the shadows of upright objects there fall towards the south at noon; when it runs in the southern half of the ecliptic, the shadows then fall towards the north at noon. This line alone bounds the whole inhabited quarter [of the earth] on the south. Also, all the stars rise and set there, because the two poles of the sphere lie there on the horizon circle itself, and the turning of the sphere

[46] In modern terms, $\sin \alpha = \tan \delta \cdot \cot \varepsilon$, where α is the ascension of the degree in the right sphere, δ its declination and ε the inclination.

[47] Counted from the beginning of Capricorn, the ascension in the right sphere of the culminating degree is equal to the ascension at the locality of the rising degree, counted from the beginning of Aries; one number therefore gives both (Schiaparelli, in Nallino's commentary). Al-Battānī's table of ascensions in the right sphere is counted in this way. The ascensions in the climates are the subject of chapter 13.

[48] The translation reads «الشمس», «the sun», for «اشمس» (p. 23, line 18) and «تنتهي» for «ننتهي» (p. 24, line 22), misprints that the corrigenda do not list. It follows the emendanda printed in Part II in reading «الأندلس» for «الأندلس» (p. 26, line 18) and «اوقيانس» for «ارقيانس» (p. 28, line 12).

there is like that of a water-wheel, the wheel of a well.[49,50]

Ar. p. 22

It is not known for certain that this line of the earth is inhabited, for we have seen no one who claims to have reached it in our time, and Ptolemy does not mention it in his book. But people of understanding know that the climate of this line is temperate, because the sun never goes very far from it and does not stay long over its zenith, since at that time it passes quickly through the declination. So summer and winter there are both mild. Something like what we have described is seen in places near it, such as Ṣanʿāʾ, Aden and the other places of Yemen that are close to it.[51]

As for the other lines that decline from this line towards the north, which are the lines parallel to it: on any one of them, the stars that lie within the circle drawn about the north pole of the equator with a radius equal to the altitude of the pole above the horizon on that line never set on that line. Of the stars outside this circle, those that are near it may be seen twice in one night, once at the beginning of the night and again at its end, and they set in between. The stars at the zenith of one of these lines are those that cut off, on the circle through the two poles of the equator, an arc between the star and the equator equal to [the arc] between the line and the equator. The stars that are always hidden are those within the circle drawn about the south pole of the equator with a radius equal to the depression of that pole below the horizon. All the other stars, which are not of this description, rise and set.

Therefore, if one observes one of the stars within the circle drawn about the north pole of the equator with a radius equal to the altitude of the pole above the horizon, and takes its altitude at its highest, when it crosses the line of the middle of the sky above the pole, between the zenith and the pole; if one then waits until it is on the line of the middle of the sky below the pole, between the pole and the horizon, where it is lowest, and takes its altitude then; and if one finds the difference between the two altitudes and adds half of it to the smaller altitude: the result is the altitude of the north pole above the horizon there. Likewise, if the two altitudes are added together and half the sum is taken, this is the altitude of the pole, and it is the latitude of that place to the north.

Ar. p. 23

On every one of these lines whose distance from the equator is less than the inclination, the sun passes over the zenith twice a year. This is known from the table of declination, and so is the part of the ecliptic where it happens, I mean the degree in which the sun is on that day. For when the sun is at the beginning of Aries or of Libra, it passes at noon over the zenith on the line of equality, as has also been said before; and there this happens only once a year. When it is north of these two points, it reaches the zenith of those who live beneath its path in declination, that is, where the altitude of the pole equals the declination of the degree in which the sun is on that day, the declination being northern. It is known that the distance of the equator from the zenith is then equal to the declination, and that on that day upright objects there cast no shadow at noon. When the sun moves away from them, the shadows of upright objects at noon fall towards the south, until it returns to them and stands over their heads a second time. Then again upright objects have no shadow, until it moves away from them, and the shadows of upright objects fall towards the north.[52]

On the other lines, whose distance from the equator is greater than the inclination, the sun never reaches the zenith of their inhabitants, and the shadows of upright objects there never fall towards the south at noon. On them the difference between night and day in length keeps

[49] The words in brackets are supplied by Nallino; the codex lacks them.

[50] The print has «السَّنْوِيّ», «annual», which gives no sense here; Nallino's emendanda, printed in Part II, read perhaps «السّانويّ», from *sāniya*, the wheel that lifts water from a well. The sphere of the equator turns like a wheel standing upright, with its axis lying in the horizon.

[51] *Almagest* II 6 (Nallino).

[52] Once a year at each of the two points (Nallino); Plato of Tivoli has «twice», perhaps a conjecture of his own, since the codex has «once».

growing, up to the line whose distance from the equator is 66° 25′, which is what the whole inclination falls short of 90. On this line alone, when the sun reaches the summer solstitial point, which is called the beginning of Cancer, the day is longer by 12 hours, so that day and night together become one day, all of it daylight; and the night becomes the same when the sun reaches the winter solstitial point, which is called the beginning of Capricorn. This line alone is the first of the lines on which the shadows of upright objects fall towards all the directions of the horizon, because beyond this line, towards the north, the middle of the day is not bounded. On this line alone the ecliptic coincides with the horizon itself when the spring equinoctial point rises; for the beginning of Cancer then rises at the north point, and at the same time the beginning of Aries is on the eastern horizon, rising at the equinoctial rising point. Therefore, when the sun passes the summer solstitial point, it does not set for a day and a night: it travels around the horizon at varying distances from it until it returns to the north point, so that this day has no night at all.

Ar. p. 24

He said: On the other lines that decline from this line towards the north, the shadows of upright objects go round them, on every one of these lines, towards all the directions of the horizon, and the length of the day on each of them is known from the table of declination. For when the declination of the degrees at which the sun, on these lines, is distant from the solstitial point is subtracted from 90, the remainder is the distance of the parallel line from the equator, I mean the altitude of the pole on that line. The degrees that the sun traverses on the two sides of the solstitial point are either always visible or always hidden. So on some of these lines the day lasts a month, or two months, or three, or less or more, and the night lasts as long at the opposite season. This goes on up to the line whose distance from the equator is all the degrees of the quarter, where the altitude of the pole is 90 degrees. There the day, with the shadow of upright objects going round them towards all the directions of the horizon, lasts nearly six months. For there the northern half of the ecliptic, from the beginning of Aries to the end of Virgo, is never hidden below the earth, and its other, southern half never appears above the earth; so the whole year is a single day, half of it daylight and half of it night. There alone the north pole is at the zenith, and the equator is at once always visible and always hidden, for it lies in the very place of the horizon there. Clearly the turning of the sphere there is like that of a millstone. Between the line of equality and this line, the turnings vary between that of the water-wheel and that of the millstone: the turning inclines with the inclination of the place from the line of equality, according to its nearness to each of the two lines and its distance from it. God knows best.[53]

We give an example of some lengths of the day on some of these lines, to serve as a model. We take it on the line whose distance from the equator is 69° 44′, which is the altitude of the pole on this line. Subtracting this from 90 leaves 20° 16′. We find that the sun reaches this amount of northern declination when it is 30 degrees from the summer solstitial point on either side; that is, from the time it is at the beginning of Gemini it remains visible above the earth, going round the horizon, until it reaches the beginning of Leo. So on this line it does not set at all while it is in these 60 degrees on the two sides of the solstitial point, and the day, with the shadow of upright objects going round them towards all the directions of the horizon, lasts until the sun has crossed these degrees, which takes about two months. When its whole distance from the winter solstitial point lies between the two degrees opposite these two degrees, it does not appear above the earth, that is, from the time it reaches the beginning of Sagittarius until it reaches the beginning of Aquarius; so the night too lasts about two months.

Ar. p. 25

As for the line whose distance from the equator is 78° 28′: there the sun does not set below the earth when it reaches a northern declination equal to what these degrees fall short of 90, and does not appear above the earth when it reaches a southern declination of the same amount. This amount is 11° 32′. We find that the sun reaches this declination when it comes to the beginning of Taurus and to the beginning of Virgo, where its distance from the summer solstitial point is

[53] Al-Farghānī makes the same comparison with the millstone at the end of his chapter 7, and Dante took it from him into the *Convivio* (Schiaparelli, in Nallino's note).

60 degrees on either side. So the day, with the shadows of upright objects going round them, lasts nearly four months. Likewise the sun stays below the earth while its course runs from the beginning of Scorpio to the beginning of Pisces, so that the night too lasts as long.[54]

As for the known places of the earth and the inhabited localities in longitude and latitude: we have shown, by the reasoning that Ptolemy set out and in which others of the ancients agreed with him, that the earth is round, that its centre is in the middle of the sphere, that the air surrounds it on all sides, and that compared with the ecliptic it is as small as a point.[55]

As for its inhabited part, they took its limits from the inhabited islands called the Eternal Islands, in the western Ocean, which are six inhabited islands, to the farthest inhabited land of China, and they found this to be 12 hours. So they knew that when the sun sets at the farthest inhabited land of China, it begins to rise over the first of the inhabited islands said to be in the western Ocean, and when it sets at these islands, it begins to rise over the farthest inhabited land of China. That is half the circle of the earth, and it is the length of the inhabited land as far as it has been explored. In miles it is 13,500 miles, in the miles that they used in measuring the earth. Then they considered the latitude, and found that the inhabited land runs from the place of the line of equality towards the north as far as the island of Thule, which is in Britain, where the longest day is 20 hours.[56,57]

Ar. p. 26

They stated that on its course from east to west the line of equality on the earth is cut, between India and Ethiopia, at an island there on the south side of the equator; this island lies across it there and separates north from south. The line that cuts this line from north to south lies halfway between the islands said to be in the Ocean and the farthest inhabited land of China; this is the dome of the earth, known by what we have described, and its place is the place of the intersection. The latitude from the line of equality to the island of Thule is about 60 degrees, a sixth of the circle of the earth. If this sixth, the amount of the latitude, is multiplied by the half, the amount of the longitude, the inhabited land that appears in the north is half a sixth of the earth, that is, one part in twelve.[58,59,60]

They measured the Indian Sea. They said that its length, counted from west to east, from the farthest end of Ethiopia to the farthest end of India, is 8000 miles, and its breadth 2700 miles, of which it extends 1900 miles south of the island where night and day are equal. It has a gulf in the land of Ethiopia that reaches towards al-Barbar and is called the Berber Gulf; its length is 500 miles and the breadth of its channel 100 miles. Another gulf goes out towards the land of Ayla: this is the Sea of al-Qulzum. Its length is 1400 miles; the breadth of its channel, which is called the Green Sea, is 200 miles, and its breadth at its mouth 700 miles. Another gulf goes out towards the land of Persia; it is called the Persian Gulf, and it is the Sea of Basra. Its length is 1400 miles, its breadth at its mouth 500 miles, and the breadth of its channel 150 miles. Between these two gulfs, I mean the gulf of Ayla and the gulf of Persia, lie the lands of the Ḥijāz and Yemen; the distance between these two gulfs is 1500 miles. Another gulf also goes out from it, near its end,

[54] The Arabic writes 78° 28′ and 11° 32′ in abjad numerals; Nallino's Latin prints 78° 23′ in the first place, a misprint. Ptolemy (*Almagest* II 6) puts these two parallels at 69 1/2° and 78 1/3°, because his inclination is larger (Nallino).

[55] The rest of the chapter was edited and translated into French by Reinaud (1848); much of it recurs, nearly word for word, in al-Masʿūdī, Ibn Rusta and Qudāma (Nallino).

[56] The codex has «the empty islands» (*al-khāliyāt*); Nallino reads «the Eternal Islands» (*al-khālidāt*), as Plato of Tivoli and the Arabic geographers have it. They are the Fortunate Islands of the Greek geographers, the Canaries.

[57] 13,500 miles for 180° make 75 miles to the degree, a value that reached the Arabs through Syriac writers from Ptolemy (Nallino).

[58] Ethiopia (*al-Ḥabash*) is here, as in the Greek geographers, the whole south of Africa (Nallino).

[59] The dome of the earth (*qubbat al-arḍ*) goes back to the Indian prime meridian through Ujjain and Laṅkā, misplaced by the Arabs on an island on the equator; it is the «Arin» of the medieval Latin astronomers (Nallino).

[60] The computation is wrong twice over (Schiaparelli, in Nallino's commentary). The area of a zone of the sphere is not the product of its extent in latitude and its extent in longitude. And even on that rule, 60° is a third of the 180° from pole to pole, which would give 1/6, not 1/12. In fact the half of the zone between the equator and latitude 60° is (sin 60°)/4 ≈ 0.2165 of the sphere.

towards the farthest land of India; it is called the Green Gulf, and its length is 1500 miles.[61]

In this whole sea, I mean the Sea of India and China, there are 1370 islands, inhabited and uninhabited. Among them is an island at its far end, near the land of China, called Taprobane, which is Sarandīb; it is 3000 miles round. It faces India on the east. It has great mountains and many rivers, from which come rubies and sapphires. Around it are 59 inhabited islands, with many towns and villages.[62,63]

As for the western Ocean, which is called the Encircling [Sea], only its western and northern part is known, from the farthest land of Ethiopia to Britain; it is a sea on which ships do not sail. The six islands in it, facing the land of Ethiopia, are the inhabited islands, which are also called the Islands of the Blessed. Another island, facing al-Andalus and called Gadira, lies at the strait. This strait goes out from the Ocean; it is 7 miles wide where it goes out, it lies between al-Andalus and Tangier, it is called Sabṭā, and it goes out into the Sea of the Rūm. In the Ocean, on the north side, are also the islands of Britain, which are twelve islands. Beyond them it goes far away from the inhabited land, and no one knows what it is like or what it holds.[64]

Ar. p. 27

The Sea of the Rūm and of Egypt begins at the strait that goes out from the western Ocean near the island called Gadira, facing al-Andalus, and runs east to Tyre and Sidon. Its length is 5000 miles; its breadth is 600 miles in one place, 700 miles in another and 800 miles in another. It has one gulf that goes out towards the north near Rome, 500 miles long, called Adris, and another gulf that goes out towards the land of Narbonne, 200 miles long. In this whole sea there are 162 inhabited islands, five of them large: the island of Corsica, 200 miles round; Sardinia, 300 miles round; Cyprus, 350 miles round; Sicily, 500 miles round; and Crete, 300 miles round.[65]

The Sea of Pontus extends from Lādhiqa to Constantinople the Great; its length is 1060 miles and its breadth 300 miles. The river called Tanais flows into it; it comes from the north, from the lake called Maeotis, which is a large sea although it is called a lake: its length from east to west is 300 miles and its breadth 100 miles. At Constantinople a channel bursts out of the Pontus, flowing like a river, and pours into the Sea of Egypt; at Constantinople it is about 3 miles wide, and Constantinople lies on it.[66]

The Sea of Jurjān, which is the Sea of al-Bāb, is 800 miles long from west to east and 600 miles wide. In it, facing Jurjān, are two islands that were inhabited in the past. These are the inhabited places in the known seas of the earth; God knows best.[67]

The earth has been divided into three parts. The first of them is [bounded] by the Green Sea on the north side, by the channel that goes out from the Pontus into the Greatest Sea, and by what lies between the lake Maeotis and the Pontus. So the boundaries of this region are, on the west and the north, the western sea, which is the Ocean; on the south, the Sea of Egypt and of the

[61] The measures of the seas go back, through a Syriac source also used by Jacob of Edessa, to Ptolemy's map: at 75 miles to the degree they agree with it (Nallino). Al-Barbar is the country of Berbera, on the coast of Somalia. The Sea of al-Qulzum is the Red Sea: al-Qulzum is the Greek Clysma, near Suez, and Ayla is the present al-ʿAqaba. Nallino takes the Green Gulf to be Ptolemy's Prasōdēs Sea, the «leek-green sea» (*Geography* VII 2, 1).

[62] Taprobane and Sarandīb are names of Sri Lanka. «Rubies and sapphires» is literally «red *yāqūt* and the colour of the sky»: *yāqūt* is corundum, red in the ruby and blue in the sapphire.

[63] Jacob of Edessa, Ibn Rusta and al-Kharaqī give 19 islands, the number that Ptolemy lists around Taprobane (*Geography* VII 4, 11–13); Nallino concludes that al-Battānī's source had 19, written in letters (يط), which he read as 59 (نط).

[64] Gadira is Cadiz, and Sabṭā is Ceuta; the strait is the Strait of Gibraltar. The Sea of the Rūm (the Byzantines) is the Mediterranean.

[65] Adris is the Adriatic. Corsica is called by its Greek name, Kyrnos (*Qurnus*).

[66] The Pontus is the Black Sea, the Tanais the Don and the Maeotis the Sea of Azov; the channel is the Bosporus. Lādhiqa, also written Lāziqa, is a city or region of the Caucasus (Nallino). For the length of the Pontus Nallino prefers 1300 miles, the figure of Jacob of Edessa, Ibn Rusta and others.

[67] The Caspian Sea; al-Bāb is Derbent. For its length Nallino prefers 1800 miles, the figure of Ibn Rusta and Jacob of Edessa.

Rūm; and on the east, the Tanais and the lake Maeotis. This land is like an island; they called it Europe.[68]

The second part lies to the south, from the Sea of Egypt to the Sea of Ethiopia. The boundaries of this region are, on the west, the Green Sea; on the north, the Sea of Egypt and of the Rūm; on the east, al-ʿArīsh; and on the south, the Sea of Ethiopia. This part is called Libya.

Ar. p. 28

The third part is all the rest of the inhabited earth, to its farthest end. Its boundaries are, on the west, the Tanais, the river, the channel, al-ʿArīsh and Ayla; on the south, the Sea of Yemen and of India; and on the east, the farthest inhabited land of China towards the east, and China itself. This part is called Greater Asia. These three parts take in the climates, the districts and all the inhabited regions.[69]

What is not known to be either inhabited or desolate is 11 parts out of 12. The part that holds the known inhabited land, from the place of the line of equality, also holds the seas and the deserts. If someone asks whether there are plants, animals and habitation in these eleven parts, the answer can only come from inference by analogy and from opinion. The inhabited earth on our side does not go beyond the limits and boundaries we have mentioned, and no one has brought us word of what lies beyond them. But opinion and conjecture, reasoning by analogy, arrive at what no one of understanding will deny. The sun, the moon and the stars move with us, and from their motion, their nearness and their distance come summer and winter, plants, animals and habitation, and all that everyone knows. If the sun and the stars rise over every place of the rest of the circle of the earth as they do with us, then there can be plants, animals, seas and mountains there as with us; and so it should be. One degree comes to about 65 of the miles mentioned above, which is about two days' journey; God knows best.[70]

As for the longitudes and latitudes of the cities as they are set down in the *Book of the Image of the Earth*: their positions in longitude, which is the distance between west and east, were counted from the inhabited islands in the western Ocean towards the east. This was done according to what they found of the times of eclipses, those of the moon in particular, which occur earlier in some localities than in others. From this they learned that noon in each locality precedes noon in another locality to the west of it by as many times of the equator as lie between [the times of] the eclipse in the two localities. Some of it they took, approximately, from the reports of travellers.[71]

The latitudes of the cities they took from measuring the sun at noon in the localities. They found its distance from the zenith, as we have explained earlier in this book, and so learned the distance of each locality from the line of equality, which is the distance between south and north. Under each city they set down, approximately, its distance from the Eternal Islands in longitude and from the line of equality in latitude. We have entered these according to the description that we found in the well-known *Book of the Image of the Earth*, and we have also listed separately the middles of the known regions and districts, as Ptolemy did; they are 94 regions. There are errors in the longitudes and latitudes in that book; we shall come back to what is needed of this in the later parts of our book.[72]

[68] The Green Sea is here the Atlantic, as often in Arabic authors, and the Greatest Sea the Mediterranean, the «Great Sea» of Orosius (Nallino, after Reinaud).

[69] The river is the channel at Constantinople that flows like a river; the channel is the Sea of the Rūm, which reaches Tyre and Sidon (Nallino).

[70] The 13,500 miles of the half circumference give 75 miles to the degree. Nallino takes 65 for a slip for 75: the Arabic words for sixty and seventy are easily confused when their dots are left out. Plato of Tivoli has 85.

[71] The *Book of the Image of the Earth* (*Kitāb ṣūrat al-arḍ*) is not al-Khwārizmī's book of that title: Nallino shows that its author used both Ptolemy's *Geography* and al-Khwārizmī's book, and suggests that it was the Arabic recension of the *Geography* made by Thābit ibn Qurra. Plato of Tivoli's Latin adds «which is called Geography».

[72] Nallino could not find such a list in Ptolemy: the *Geography* gives no coordinates for the middles of its provinces.

Chapter 7. On finding the winter and summer risings and settings on the horizon circles of the localities, from the excess of the longest day or from the altitude of the pole, whichever of the two is known

Ar. p. 29

He said: Suppose you wish to find the amounts of the arcs of the horizon circle that lie between the equator and the ecliptic at the horizon in any locality; this is the azimuth of the rising and of the setting of any part of the ecliptic you want. Take the given excess of the longest day and find its amount in degrees, by multiplying each hour of it by 15 degrees. Take half the result and add it to 90; the sum is half the arc of the longest day. Then take the whole inclination, which is the declination of the point of the beginning of Cancer, and subtract it from 90. Find the sine of the remainder: this is the sine of the complement of the declination of Cancer. Multiply it by the sine of half the arc of the longest day, divide the product by the radius, and find the arc of the quotient. Subtract this arc from 90; the remainder is the amount [of the arc] on the horizon circle between the rising and the setting of the beginning of Cancer and the equator, to the north of the equator.[73]

Likewise, if you work with half the arc of the shortest day, which is the day of the beginning of Capricorn, the amount comes out the same, and it is known that the beginning of Capricorn rises and sets to the south of the equator. Clearly the azimuth of the beginning of Cancer in the north is equal to the azimuth of the beginning of Capricorn in the south. These are the summer and winter risings and settings: those on the north side are called the summer ones, and those in the south the winter ones.

Ar. p. 30

If you wish to find the azimuth of the rising and setting of points of the ecliptic other than these two, take the declination of any degree you like, subtract it from 90, and find the sine of the remainder. Then find half the arc of the day of that degree, or take it to be whatever you like; find its sine and multiply it by the sine of the complement of the declination of the degree. Divide the product by the radius, find the arc of the quotient, and subtract it from 90; the remainder is the azimuth of the rising and setting of that degree on the horizon circle. If the declination is northern, it lies to the north of the equator; if it is southern, to the south.[74]

If the latitude of the locality is given and you wish to find from it the azimuth of the rising and setting of any degree you like, subtract the latitude of the locality from 90; the remainder is the altitude of the beginning of Aries there. Find its sine. Then take the declination of the degree you want and find its sine; multiply it by the radius and divide the product by the sine of the altitude of the beginning of Aries. Find the arc of the quotient: this arc is the distance of the rising and setting of that degree from the rising and setting of the beginning of Aries, on the horizon circle, towards the side of the declination. This is the azimuth, if God wills.[75]

Chapter 8. On finding the altitude of the pole from the excess of the longest day

Ar. p. 30

He said: Suppose you wish to find the altitude of the north pole of the equator above the horizon, which is the latitude of the locality, from the excess of the longest day over the equinoctial day, or from the shortfall of the shortest day. Take half the excess of the longest day, the day of the first degree of Cancer, in degrees, and add it to 90; the sum is half the arc of the longest day. Or, if you wish, subtract it from 90; the remainder is half the arc of the shortest day. Whichever of the two you work with, it comes to the same. Then subtract the whole inclination from 90 and find the sine of the remainder: this is the sine of the complement of the whole inclination. Multiply

[73] The azimuth of a rising (*samt al-maṭlaʿ*) is measured on the horizon from the point where the equator rises; in modern terms it is the amplitude of the rising. The excess of the longest day is its excess over 12 hours. In modern terms the rule is $\cos A = \cos \delta \cdot \sin H$, where A is the amplitude, δ the declination and H half the arc of the day.

[74] «Half» is supplied by Nallino; the codex and Plato of Tivoli have «the arc of the day».

[75] In modern terms, $\sin A = \sin \delta / \cos \varphi$, where φ is the latitude; $90° - \varphi$ is the altitude of the beginning of Aries when it culminates.

the sine of half the arc of the day by the sine of the complement of the whole inclination, divide the product by the radius, and find the arc of the quotient. The arc you obtain is the distance of the rising of the beginning of Cancer from the north point. Subtract it from 90; the remainder is the distance of the rising of the beginning of Cancer from the equinoctial rising point, as we have explained in the preceding chapter. Then multiply the sine of half the excess of the longest day by the sine of the distance of the rising of the beginning of Cancer from the north point, and divide the product by the sine of the distance of the rising of the beginning of Cancer from the rising of the beginning of Aries. Multiply the quotient by the radius, divide the result by the sine of half the arc of the longest day, and find the arc of the quotient. The arc you obtain is the altitude of the pole where the excess of the longest day is the given amount with which you worked.[76,77]

Chapter 9. On finding the excess of the longest day, and the lesser excesses of the day, from the altitude of the pole

Ar. p. 31

He said: Suppose you wish to find the amount of the excess of the longest day, and of the shortfall of the shortest day, with respect to the equinoctial day, from the altitude of the pole when it is given. Take the latitude of the locality and find its sine. Then subtract the latitude of the locality from 90 and find the sine of the remainder: this is the sine of the complement of the latitude of the locality. Then find the sine of the whole inclination, and the sine of what the whole inclination falls short of 90. Multiply the sine of the latitude of the locality by the sine of the whole inclination, divide the product by the sine of the complement of the whole inclination, multiply the quotient by the radius, divide the result by the sine of the complement of the latitude of the locality, and find the arc of the quotient. The arc you obtain is half the excess of the longest day, and half the shortfall of the shortest day is the same. Double it: the result is the whole excess of the longest day and the whole shortfall of the shortest day. Every 15 degrees of it make one equal hour. Add the hours you obtain to the 12 hours that are the length of the equinoctial day; the sum is the hours of the longest day. Subtract the same excess from 12; the remainder is the hours of the shortest day.[78]

If you wish to know the excess of the day at degrees of the ecliptic other than these two, take the declination of any degree of the signs you like and work with it in place of the whole inclination. What you obtain at the end of the work, done in the same way, is the difference of the day at that degree: if the declination of the degree is northern, it is an excess of the day; if it is southern, it is a shortfall at that degree.

Chapter 10. On finding the altitude and the shadow, each from the other, by computation or from the table

Ar. p. 31
Ar. p. 32

He said: If you wish to find the shadow from the altitude, find the sine of the altitude and the sine of what the altitude falls short of 90. Then divide the gnomon into as many parts as you like, multiply the sine of the complement of the altitude by the number of parts of the gnomon, and divide the product by the sine of the altitude. The quotient is the amount by which the shadow extends and stretches over the level ground, in the measure in which the gnomon has that number of parts. What the calculators have worked with, and what we have worked with

[76] The equinoctial day (*al-nahār al-muʿtadil*) is the day of 12 hours. In modern terms the rule is $\sin \varphi = \sin e \cdot \cos A / (\sin A \cdot \sin H)$, where φ is the latitude, e half the excess of the longest day, $H = 90° + e$ half the arc of the longest day, and A the amplitude of the rising of the beginning of Cancer. The rule is exact, since $\sin e = \tan \varphi \cdot \tan \varepsilon$, $\sin A = \sin \varepsilon / \cos \varphi$ and $\cos A = \cos \varepsilon \cdot \sin H$.

[77] Here the codex writes 90 in Indian figures, «٩٠» (Nallino: «so the codex itself»); elsewhere the text writes numbers in words or in abjad letters.

[78] In modern terms, $\sin e = \tan \varphi \cdot \tan \varepsilon$, where e is half the excess of the longest day, φ the latitude and ε the inclination.

in this book for the amounts of the shadow, is that the gnomon has 12 parts. It may be divided into fewer or more, as the person computing wishes, because the parts of the shadow are related only to the parts of the gnomon: one says that the shadow is so many parts long, in the measure in which the gnomon was made so many.[79]

If you wish to find the altitude from the horizontal shadow that we have described, multiply the whole [shadow] by itself, and add to the product the parts of the gnomon multiplied by themselves, which, on the basis we have laid down for the computation, is 144, the gnomon having 12 parts. Take the root of the sum: it is the hypotenuse of the shadow triangle. Keep it. Then multiply the parts of the gnomon by the radius; on the basis we have laid down, this always comes to 720. Divide these 720 by the hypotenuse of the shadow triangle that you kept, and find the arc of the quotient; the arc you obtain is the amount of the altitude. If you wish to compute it another way, multiply the shadow by the radius, divide the product by the hypotenuse of the shadow triangle, and find the arc of the quotient. The arc you obtain is the distance of the degree of the sun, or of another degree, from the zenith on the circle of altitude; subtract it from 90, and the remainder is the altitude.[80]

As for the erect shadow, I mean the vertical shadow, it is the reverse of the horizontal one: it is longest at noon and shortest at sunrise. If you wish to find this shadow from the altitude, multiply the sine of the altitude by the number of parts of the gnomon, and divide the product by the sine of what the altitude falls short of 90; the quotient is the amount of the shadow in parts of the gnomon.

If you wish to find the altitude from this vertical shadow, multiply the shadow by itself, add to it the product of the parts of the gnomon by themselves, and take the root of the sum: this is the hypotenuse of the shadow triangle. Then, if you wish, multiply the parts of the gnomon by the radius, divide the product by the hypotenuse of the shadow triangle, and find the arc of the quotient. The result is the amount between the degree of the sun and the zenith on the circle of altitude; subtract it from 90, and the remainder is the altitude. Or, if you wish, multiply the shadow by the radius, divide it by the hypotenuse of the shadow triangle, and find the arc of the quotient: the arc you obtain is the altitude.

Ar. p. 33

If you wish to find either of the two shadows from the altitude with the table, and you want the horizontal shadow, look in the table of the altitude and the shadow, in the column of the altitude, for the altitude you have, and take what stands opposite it in the column of the shadow: that is the amount of the shadow of that altitude. If you wish to find the altitude from this shadow, look for the shadow you want in the column of the digits of the shadow; where you find it, take the degrees of altitude entered opposite it in the column of the altitude: that is the amount of the altitude for that shadow.[81]

If there are minutes with the degrees of the altitude, or minutes with the digits of the shadow, find the value by interpolation, as I described to you in the chapter on the declination. That is: if there are minutes with the altitude, see what part of 60 they are, and take the same part of the difference between [the shadows of] the whole degrees and of the degree one greater. Always subtract it from the digits of the shadow of the whole degrees, since every greater altitude has a smaller shadow than a lesser one; the remainder is the shadow of that altitude. If there are minutes with the shadow whose altitude you wish to find, look at the shadow that you find in the table and subtract it from the shadow you have. Find what part the remainder is of the difference between that shadow and the one that belongs to the degree of altitude one less, and

[79] In modern terms the horizontal shadow is g · cot h, where g is the gnomon and h the altitude, and the vertical shadow, described below, is g · tan h.

[80] «The whole» (*al-kull*) is the whole length of the shadow; Nallino translates «the square of the shadow». The hypotenuse of the shadow triangle is literally «the diameter of the triangle of the shadow» (*quṭr muthallathat al-ẓill*): the line from the tip of the gnomon to the tip of the shadow.

[81] The digit (*iṣbaʿ*, literally «finger») is a twelfth of the gnomon.

take the same part of the 60 minutes by which the column of the altitude increases. Subtract the minutes you obtain from the altitude that you found opposite the shadow you took in the table, the one nearest to your shadow among those smaller than it; the remainder is the altitude.

If you wish to find the erect shadow, which is the vertical shadow, from the altitude with the table, subtract the altitude from 90 and take the shadow opposite the remainder, interpolating in the same way; the result is the vertical shadow. If you wish to find the altitude from this shadow, look for the shadow you want in the table, take what stands opposite it in the column of the altitude, interpolating, and subtract the result from 90; the remainder is the altitude. The shadow has been entered in this table on the basis that the gnomon is 12 parts, so whatever you work out with it of the shadow is on the basis that the gnomon is 12 digits.[82]

Chapter 11. On finding the azimuth of the altitude and of the shadow on the horizon circle, in any locality you want, at all times

Ar. p. 33
Ar. p. 34

He said: If you wish to find the azimuth of the altitude and of the shadow at any degree of all the signs, in any locality you want, take the declination of that degree and find its sine, and the direction of the declination. Then subtract this declination from 90 and find the sine of the remainder: this is the sine of the complement of the declination of the degree. Then find the sine of the latitude of the locality and the sine of what the latitude of the locality falls short of 90. Then take the altitude at whatever time of the day you like, and find its sine and the sine of what the altitude falls short of 90. Then multiply the sine of the declination of the degree by the radius and divide the product by the sine of the complement of the latitude of the locality: the result is the sine of the amplitude of the rising. Keep it, and note its direction, which is the direction of the declination. Then multiply the sine of the altitude by the sine of the latitude of the locality and divide the product by the sine of the complement of the latitude of the locality: the result is the sine of the difference of the horizon, which is always southern. If the sine of the amplitude of the rising and the sine of the difference of the horizon are in the same direction, add them; if they are in different directions, subtract the smaller from the greater and note the direction of the remainder. Multiply what you obtain from the addition or the subtraction by the radius, divide it by the sine of the complement of the altitude, and find the arc of the quotient: the result is the azimuth of the altitude.[83]

The shadow is then at the degree for which you computed. If at that time the degree is between the ascendant and the middle of the sky, this azimuth is counted on the horizon circle from the rising point of the beginning of Aries and of Libra towards the direction that you obtained.[84]

The azimuth can also be found by another method, with the angles that are used in finding the parallax, which we shall explain later in this book.

[82] The vertical shadow of an altitude is the horizontal shadow of its complement: $g \cdot \tan h = g \cdot \cot(90^\circ - h)$.

[83] In modern terms, $\sin Z = (\sin \delta / \cos \varphi - \sin h \cdot \tan \varphi) / \cos h$, where Z is the azimuth counted from the east or west point, positive towards the north, δ the declination, φ the latitude and h the altitude. $\sin \delta / \cos \varphi$ is the sine of the amplitude of the rising, and $\sin h \cdot \tan \varphi$, always southern, is what al-Battānī calls the sine of the difference of the horizon.

[84] The codex gives only the case of the eastern half of the sky. Plato of Tivoli's Latin shows that words are missing after «direction»; Nallino restores them in Arabic in his apparatus, and his translation includes them: «[the direction] in which you found the two sines, or the greater of them if they differ; and if the degree is between the middle of the sky and the setting point, this azimuth is counted from the setting point of the beginning of Aries and of Libra on the horizon circle, towards that direction».

Chapter 12. On finding the meridian line, which is the direction of the south

Ar. p. 34
Ar. p. 35

He said: Suppose you wish to find the direction of the south, which is the meridian line, in any locality you want and at any time of the year you like, and the place of the sun is not known to you. Go to a place with an open horizon and a smooth, level surface that does not slope, and draw on it a circle of any size you like. Fix at the centre of the circle a thin rod with a pointed tip, straight and without any bend; its best length is a quarter of the diameter of the circle. Measure [the distance] from four sides of the circle to the tip of the rod, so that it stands truly upright on the centre. Then watch the shadow at the beginning of the day, when it is long. It keeps getting shorter; when it reaches the circumference of the circle and is about to enter it, mark its place on the circumference with a point, as a sign of the tip of the shadow. Then wait until noon has passed and the shadow begins to grow; when it reaches the circumference of the circle and is about to leave it, mark the place of its tip on the circumference with a second point, as [a sign of the tip of the shadow].[85]

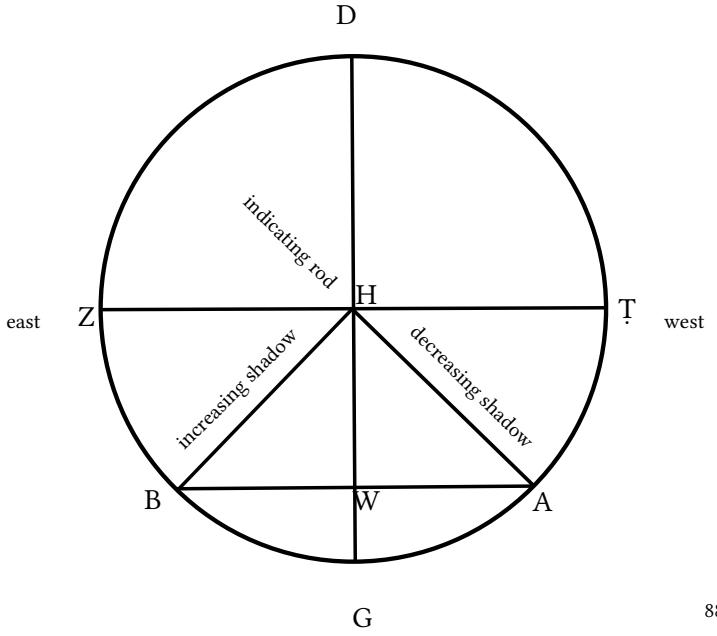
Then divide the arc between the two points into halves, mark the place of the half of the arc with a point, and from this point draw a straight line through the centre of the circle to the other side of the circumference, extending it as far as you like. To do this, lay the edge of the ruler on the point at the half of the arc and on the centre, and draw a line through both the point and the centre to the circumference on the other side. This line is always the meridian line: whenever the shadow of the rod at the centre falls on this line, it is noon, whether the shadow is long or short. This line is the direction between south and north. Then quarter the circle with another line through the centre of the circle at right angles; the circle is divided by these two lines into four equal quarters, and this line is the direction between east and west. Then write at the ends of the lines the directions of the horizon, that is, east, west, north and south.[86]

The nearer the sun is to one of the two solstitial points, the more accurate the observation, because the sun moves slowly in declination between the two observations, I mean the two observations of the shadow. It is known that the moment of noon cannot be fixed exactly, because the sun passes quickly in declination through the meridian circle; but the moment of noon, or the moment nearest to it, is found. This is a diagram of the circle described, if God, exalted is He, wills.[87]

[85] The words in brackets at the end are Nallino's conjecture.

[86] «Half», in «the half of the arc», is lacking in the codex and supplied by Nallino.

[87] «In declination», in the second sentence, is Nallino's conjecture (his emendanda, printed in Part II).



[88]

Ar. p. 36

He said: Draw a circle about the centre H, and fix the indicating rod at the place of point H. Mark point A at the place of the tip of the shadow before noon, and the sign B at the place of its tip after noon. Divide the arc AB into halves at the sign G, and draw the line from G to the sign D. The line GHD is the meridian line; point D is the direction of the south, and point G the direction of the north. Then divide the arc GAD into halves at point Ṭ, and draw through the points Ṭ and H a line that goes out to point Z. Point Z is the direction of the east, and point Ṭ the direction of the west. Whenever the shadow of the indicating rod falls on the line HG, it is always noon.[89]

If the place of the sun is known to you, find the azimuth of any altitude you like at some time of the day, and watch for the given altitude. When the altitude becomes equal to it, mark on the circumference of the circle, at the shadow of the indicator, a sign that stands in the middle of the breadth of the shadow of the indicating rod. Then divide the quarter of the circle in which this point falls into 90 parts, and count from that point, towards the side opposite to the direction of the azimuth of that altitude, as many parts as the azimuth of the altitude. The division at which you arrive is the east point or the west point, according to the time of the altitude with which you worked, before noon or after it. From this sign draw a line through the centre of the circle, and quarter the circle with another line through the centre at right angles. Then you know the meridian line by this line, and the line of east and west by the line before it.

Likewise, if the horizon is in view from this circle at sunrise or sunset, the east point or the west point is known from the azimuth of the rising or setting of the degree of the sun on the horizon circle, which we have drawn as ABGD.

Ar. p. 37

If you wish to find the line between east and west in another way, and through it the line between north and south, this is done by finding the altitude whose azimuth has no inclination from the equinoctial rising or setting point. This is possible only when the sun is in the northern signs, from the beginning of Aries to the end of Virgo. This altitude is found as follows. Find the place of the sun in the ecliptic on the day you want, and its altitude at noon on the day for which you work. Then find the sine of this altitude and the sine of what it falls short of 90. Then find the azimuth of the rising or setting of the sun, at its place in the ecliptic at the noon for which you work; by the condition we have laid down it is always northern. Then take the sine of this azimuth and multiply it by the sine of the altitude, divide the product by the sine of the

[88] As in the print, the south (D) is at the top, so that the east (Z) is on the left and the west (Ṭ) on the right. The letters of the figures are transliterated: ا A, ب B, ج G, د D, ه H, و W, ز Z, ح Ḥ, ط Ṭ, ك K, ل L, م M.

[89] The indicating rod (*al-ʿamūd al-mūrī*, «the rod that shows») is the rod set up at the centre.

azimuth and the sine of the complement of the altitude added together, and find the arc of the quotient. The arc you obtain is the altitude whose azimuth has no inclination. When you know this altitude, watch the sun; when its altitude becomes equal to this altitude, mark a point on the circumference of the circle in the middle of the shadow of the indicator. It is the east point or the west point, according to the time at which you measure, that is, the equinoctial rising or setting point. Quarter the circle at this point with two lines that intersect at the centre at right angles, and with them mark the directions of the horizon.[90]

We give an example of this, and place it in the fourth climate, where the altitude of the pole is 36° 22′. We take the place of the sun at the beginning of Cancer. The altitude of the sun at noon is then 77° 13′, and its altitude at midnight, from the northern horizon, 30° 3′; it is known that this is equal to the altitude above the earth, on the line of the middle of the sky, of the degree opposite to it, which is 30° 3′. We also find this in another way, by doubling the altitude of the beginning of Aries in the said climate and subtracting from the result the altitude of the beginning of Cancer in the middle of the sky. Clearly the altitude of the beginning of Aries in the middle of the sky in this climate is 53° 38′, and its double 107° 16′; when 77° 13′ is subtracted from this, there remains its altitude in the middle of the sky below the earth, 30° 3′. We shall show that the azimuth of the beginning of Cancer at its rising in this climate lies 30° 0′ to the north of the rising of Aries.[91]

Since this is as we have described, we draw a circle of the middle of the sky, ABL, about its centre H, with the diameter LB. Let the diameter LB be half the horizon, and let point A be the place of the zenith. We join point A to point H, so that the arc BA is the quarter of the circle between the zenith and the horizon. Point H is the place of the rising of the beginning of Aries, and point G the place of the rising of the beginning of Cancer: for HB is the southern half of the horizon, the line HL the northern half of the horizon, and the line HA the line of the quarter circle that passes through the zenith and the rising point of the beginning of Aries. We mark at the point of the beginning of Cancer on the circle of the middle of the sky the point Z. The arc BZ is the altitude of the sun at noon, and the arc ZA its distance from the zenith, which is the complement of the altitude to a quarter of the circle. We mark at the altitude of the sun at midnight the point Ṭ, so that the arc ṬL is the arc of the altitude at midnight below the earth, and we draw the line ṬZ, which passes through the point G at which the beginning of Cancer rises. The point common to the line ṬZ and the line HA is the place at which the sun, when it rises to it, is on the azimuth of H, from which the beginning of Aries rises; it then has no inclination from the azimuth of the equinoctial risings, since the line from the zenith passes through the place of the sun and through point H of the horizon. We mark the place of the sun on the line HA with the sign M.[92]

Ar. p. 38

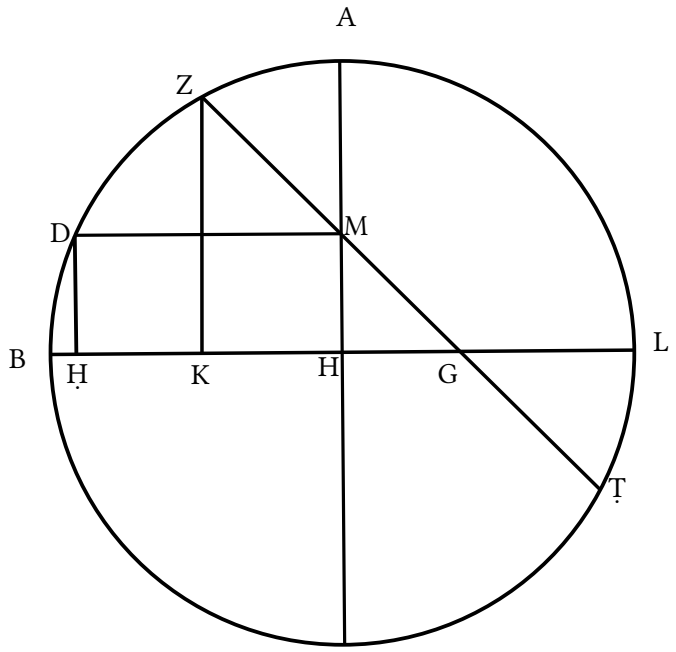
It is clear in this figure that the line HG is the line of the azimuth of the rising of the beginning of Cancer, and it is the sine of the azimuth. Further, we draw from point Z a perpendicular to the line HB, parallel to the line HA: this is the perpendicular ZK, which is the sine of the altitude at noon, so that the line KH that remains is the sine of the arc ZA, which is the complement of the altitude. We want to find the line HM, which is the sine of the altitude whose azimuth has no inclination; for the line HM is equal to the perpendicular DḤ, which is clearly the sine of the arc BD, the amount of the altitude that we seek, since the circle ABL passes through the zenith and the point of the beginning of Cancer. Now the right-angled triangle ZKG has known sides, and it is similar to the small triangle MHG, since the angle MHG is equal to the angle ZKH, the angle GMH is equal to the angle KZG, and the angle KGZ is common to the two triangles. Therefore

[90] The altitude whose azimuth has no inclination is the altitude of the sun when it stands due east or due west; al-Battānī uses it again for the sundial in chapter 56 (Nallino). In modern terms the rule is sin h = sin A · sin H / (sin A + cos H), where A is the amplitude of the rising and H the altitude at noon; it is exact, and equal to sin δ / sin φ.

[91] For 30° 0′ the codex has 30° 3′; Nallino reads 30° 0′ with Plato of Tivoli, the value used at the end of the chapter, where the sine of this azimuth is taken as 30 parts.

[92] The figure is the meridian circle, with the horizon, the vertical through the east point and the day circle of the sun seen edge on as the straight lines LB, HA and ṬZ. Here and below Nallino corrects several letters of the figure where the codex has others; the translation follows his text.

the ratio of the line ZK to the line KG is as that of the line MH to the line HG. Further, the ratio of the line HG to the line GK is as that of HM to KZ, and this is also the ratio of the line GM to the line GZ. If we take of the line KZ the part given by the ratio [of the line GM to the line GZ], there remains for us the ratio of the line HM to the line KZ; and it is the same if we take of the line KZ the part that the line GH is of the line GK: we obtain the ratio of HM to KZ.[93]



The computation of this: multiply the line HG, which has been shown to be 30 parts, by the line KZ, which is 58 parts 31 minutes and is the sine of the arc BZ; the result is 1755 parts and half a part. The line KH, which is the sine of the complement of the altitude, is 13 parts 17 minutes, so the lines HG and HK together are 43 parts 17 minutes, which is the whole line KG. When we divide that by the line KG, the result is 40 parts 33 minutes, which is the amount of the line HM that we seek, and the line DḤ is equal to it. Therefore the arc DB is 42° 32′, and this is the altitude whose azimuth has no inclination. This is what we wished to show, if God wills.[94,95]

Chapter 13. On finding the ascensions of the signs in every locality in two ways, by computation and from the table, and on the procedures with them that follow from this, if God wills

Ar. p. 39

He said: The ascensions of the signs at the place of the equator we have described above: there the signs cross the middle of the sky and the horizon with the same amount of times of the equator, and with this amount they also cross the middle of the sky in every locality. At the other places, those that decline from it towards the north, in all the localities, their ascensions differ from horizon to horizon. For when a locality has a latitude, that is, when it declines from the equator, the ascensions of the signs vary there, being greater or smaller than their ascensions in the middle of the sky, which are their ascensions in the right sphere. Every sign that rises in a locality with more than it rises with in the right sphere has a counterpart that rises in that

[93] The print names the small triangle «م ه ح», MHḤ; the angles that follow show that it is MHG, as in Nallino's translation. The words in brackets are Nallino's; the codex has «the line KG to the ratio of the line GM».

[94] «And five», in 1755, is supplied by Nallino; the codex has 1750. The quotient 1755;30 / 43;17 is 40;33,29 (Nallino).

[95] At latitude 36° 22′ the sine of the amplitude of the beginning of Cancer, HG, is 29;49 parts, not 30, and the rule then gives 42° 26′, the exact altitude of the sun on the prime vertical (sin h = sin δ / sin φ). The 42° 32′ of the text comes from taking HG as 30.

locality with less than it rises with in the right sphere, by the amount of that excess; and in every locality each sign sets with as much as its counterpart rises with there.[96]

Ar. p. 40

If you wish to know the amount of the equator that rises together with given parts of the ecliptic in any locality you want, take [the arc] from the beginning of Aries to the degree of the ecliptic you want, in ascensions of the right sphere. Find its sine, multiply it by the sine of half the excess of the longest day in that locality, divide the product by the radius, and find the arc of the quotient. The arc you obtain is the share of the difference of the day, in the quarter of the circle, that belongs to [the arc] from the first degree of Aries to that degree. Keep it. If you wish to find it another way, multiply the sine of the latitude of the locality by the sine of the declination of the degree, divide the product by the sine of what the latitude of the locality falls short of 90, multiply the quotient by the radius, divide it by the sine of what the declination of the degree falls short of 90, and find the arc of the quotient. The arc you obtain is the difference of the day in the quarter of the circle of the sphere. When you have found this share in either of the two ways, look: if the declination of the degree is northern, subtract the share you obtained from the times of the ascensions in the right sphere between the beginning of Aries and that degree; if the declination of the degree is southern, add the share to those ascensions. The ascensions you obtain after the addition or the subtraction are the ascensions from the beginning of Aries to that degree in that locality.[97]

Know that the ascension of Pisces is equal to that of Aries, the ascension of Virgo to that of Libra, the ascension of Aquarius to that of Taurus, the ascension of Capricorn to that of Gemini, the ascension of Sagittarius to that of Cancer, and the ascension of Leo to that of Scorpio. So for finding the ascensions it is enough to know the shares between the beginning of Aries and the beginning of Cancer, that is, from 1 degree to 90 degrees.

If you wish to tabulate the ascensions degree by degree, or at larger intervals, find the share of the difference of the day for one degree, for two degrees, three and four, up to the full 90 that complete the difference of the whole quarter of the circle. When you have done this, take the ascension in the right sphere of the first degree of Aries and put it in two places. Then subtract the share of the degree from one of the two places and add it to the other. The one subtracted from is the ascension of the first degree of Aries, and the one added to is the ascension of the first degree of Libra. Add the latter to 180: the sum is the ascension from the beginning of Aries to the first degree of Libra. Subtract it from 180: the remainder is the ascension from the beginning of Aries to 29 degrees of Virgo. Also subtract the ascension of the degree of Aries from 360: the remainder is the ascension from the beginning of Aries to 29 degrees of Pisces. Do the same with the share of two degrees, three and four, up to the full 90, until you have divided [the ascensions] over the whole sphere at whatever interval of the parts you want, if God wills.[98]

We have entered the ascensions of the signs for the city of al-Raqqa in full, degree by degree, and for the other climates at intervals of 10 degrees, because the ascensions differ little within an interval of this size. In the ascensions entered in the tables we have made the excess of the day increase by a quarter of an equal hour [from one table to the next], so that they are more accurate for the work with the ascensions that is needed, and more exact than ascensions computed at intervals of half an hour.

Ar. p. 41

If you wish to find the ascension of any degree you like from the table, look for the degrees whose ascension you wish to know, of whatever sign you like, in the common column of numbers in the table of the ascensions of the signs for the given climate, or in the ascensions of the right sphere, whichever you want, and take the times of ascension that stand opposite them in the

[96] The counterpart (*naẓīr*) of a sign or a degree is the sign or degree opposite to it on the ecliptic.

[97] In modern terms the share is the ascensional difference Δ: $\sin \Delta = \sin \alpha \cdot \sin e = \tan \varphi \cdot \tan \delta$, where α is the ascension in the right sphere, e half the excess of the longest day, φ the latitude and δ the declination. The ascension at the locality is $\alpha - \Delta$ for a northern declination and $\alpha + \Delta$ for a southern one.

[98] «Subtract it from 180»: the codex has «from 360»; Nallino corrects it.

column of the sign to which that number belongs. If you are working with the ascensions of the climates, this is the ascension from the beginning of Aries to that degree; if you are working with the ascensions of the right sphere, it is the ascension from the beginning of Capricorn to that degree. If there are minutes with the degrees, find what part of 60 they are, when the numbers increase by one degree, and take the same part of the difference between those ascensions and the ascensions that follow them, one degree further. Add the result to the ascensions you obtained opposite the whole degrees; the sum is the ascension of the degree and minute that you wanted. If the numbers increase by 10 degrees, see what part of the 10 degrees are the degrees and minutes that you have left over beyond what you find in the table, and take the same part of the differences of the ascensions in the table, I mean of the ascensions you found and those 10 degrees further on in the numbers. Add the result to the ascensions under the tens; the sum is the ascension of that degree.

If you wish to find the degrees of the signs from the ascensions, which is called taking the arc of the ascensions and converting them into degrees of equality, that is, degrees of the signs: look for the number of times of ascension that you have in the table of the ascensions of the right sphere or of the climate, whichever you want. Where you find it, or the nearest value smaller than it, take the degree of the signs entered opposite it in the common column of numbers: that is the degree you want of the sign in which you found the number of times. Then subtract [the times found in the table from] the times you have, and look at the remainder: if the numbers increase by one degree, multiply it by 60 minutes; if they increase by 10 degrees, multiply it by 600 minutes. Divide the product by the difference of the ascensions between that entry and the entry that follows it, and add the degrees and minutes that result from the division to the degrees you obtained first. The result is the amount of that sign that rises, or that culminates, according to which [table] you worked with. Or, if you wish, see what part of the difference of the ascensions the remainder is, and take the same part of the interval of the numbers; add the result to the degrees you had obtained.[99]

Ar. p. 42

If you wish to find the arc of the day and of the night from the table, that is, the amount of the equator that rises from sunrise to sunset, or from sunset to sunrise the next day: find the degree in which the sun is on the day you want, and take the times of ascension opposite it in the given climate whose latitude is equal to that of the city, or nearer to it than the other climates. Subtract them from the ascensions opposite the degree opposite to the degree of the sun in that climate; the remainder is the amount of the arc of the day. If the ascensions of the degree of the sun are greater than those of the degree opposite to it, which is the counterpart of the degree of the sun, add a revolution to the ascensions of the counterpart of the degree of the sun and subtract from the sum the ascensions of the degree of the sun: the remainder is the arc of the day. When you know the arc of the day, subtract it from a revolution: the remainder is the arc of the night.

If you want the arc of the day in another way, take the times of ascension opposite the degree of the sun in the climate, and also the times of ascension opposite the degree of the sun in the right sphere, and subtract 90 from the latter, so that they are counted from the beginning of Aries. When you have done this, take the difference between them and the ascensions that you obtained from the climates. Then look: if the times of the ascensions of the climates are the greater, subtract this difference from 90; if they are the smaller, add this difference to 90. The 90 after the addition or subtraction is the amount of half the arc of the day; double it, and you have the whole arc of the day. Know that this difference between the ascensions is the share of the degree of the sun in the difference, I mean the difference of the day. When you know it, look at the degree of the sun: if it is in the northern signs, add the difference to 90; if it is in the southern ones, subtract it from 90. The result is half the arc of the day, which is the part of the equator from the rising of the sun to its culmination at noon, and its double is the whole arc of the day;

[99] Degrees of equality (*daraj al-sawā'*) are degrees of the ecliptic, as against the times of the equator that rise or culminate with them; the Alfonsine Tables call them «equal degrees» (Nallino).

the two procedures come to the same.

If you wish to know the number of the equal hours of the day and of the night, divide the arc of the day or the arc of the night by 15; the result is the hours of whichever of the two you computed. When you know the hours of one of them, subtract them from 24; the remainder is the hours of the other.

If you wish to find the times of the seasonal hours of the day and of the night, of which there are always 12 in the day and as many in the night, and which are called the crooked hours, divide the arc of whichever you like, the day or the night, by 12; the result is the times of its hours. Subtract the times of the hours of whichever you computed from 30; there remain the times of the hours of the other. For these 30 are the times of two equal hours, and what the times of an hour of the night or of the day lack is added to the other.[100]

Ar. p. 43

If you wish to find the times of the hours in another way, take a sixth of the difference of the day mentioned earlier in this chapter. If the sun, or the degree you want, is in the northern half of the sphere, add this sixth to 15; if it is in the southern half, subtract it from 15. The result after the addition or subtraction is the times of the hours of the day. If you wish to find the times of the hours of the day from the table, enter the degree of the sun, or any other degree of the signs, in the common column of numbers of the table of the ascensions of the climate that is given for that city, and take the times of the hours entered opposite it in the column of the sign to which that number belongs; the result is the times of the hours of the day. If you wish to know the times of the hours of the night, enter in those ascensions the counterpart of the degree of the sun, which is opposite to it, or the counterpart of the degree you want, and take the times of the hours opposite it in the same way; the result is the times of the hours of the night. Either can also be found from the other, by subtracting one of them from 30 degrees: there remain the times of the other.[101]

If you wish to find the arc of the day from the times of the hours, or the arc of the night from them, whichever you like, multiply the times of whichever you want by 6; the result is the arc of half the day or of half the night, whichever you computed. Then double it; the result is its whole arc. If you multiply the parts of the times by 12, the result is the amount of the arc of the day or of the night for the degree for which you computed.

If you wish to convert equinoctial hours into seasonal ones, multiply the equal hours by 15 and divide by the times of the hours of the day or of the night, whichever you like; the result is seasonal hours of the night or of the day, according to what those equal hours were. If you wish to convert seasonal hours into equal ones, multiply the hours of the day by the times of the hours of the day, and the hours of the night by the times of the hours of the night, and divide the product by 15; the result is equal hours, with their fractions if a fraction remains, if God wills.

Chapter 14. On finding the latitudes of localities by observation

Ar. p. 43
Ar. p. 44

He said: If you wish to find the latitude of any locality you like, which is the altitude of the north pole there and also its distance from the equator, find the altitude of the sun at noon, in whichever way you like, when the sun passes the meridian line: with the quadrant or by finding the shadow. When you know the altitude, in either of the two ways, find the declination of the degree of the sun at that time. If it is northern, subtract it from the altitude; if it is southern, add it to it. The altitude after the addition or subtraction is the altitude of the beginning of Aries or of the beginning of Libra in that locality; subtract it from 90, and the remainder is the latitude of that locality. If you take the latitude of the locality from the table of the latitudes of the cities, it

[100] The times of a seasonal hour are its length in degrees of the equator; an equal hour is 15 times.

[101] The 12 seasonal hours of the day take $180^\circ + 2\Delta$ of the equator, where Δ is the difference of the day; one of them is therefore $15 + \Delta/6$ times.

is approximate, and not as accurate as one found by observation.

Chapter 15. On finding the altitude of the sun at noon on any day you want

Ar. p. 44

He said: If you wish to find the altitude of the sun at noon on any day you want, find the declination of the degree in which the sun is. If it is northern, subtract it from the latitude of the locality; if it is southern, add it to it. Subtract the latitude of the locality, after the addition or the subtraction, from 90; the remainder is the altitude of the sun at noon. If the declination is greater than the latitude of the locality, you know that the sun is to the north of the zenith: add 90 to the latitude of the locality and subtract the declination of the degree; the remainder is equal to the altitude above the northern horizon.

If you wish to find the altitude of the sun at noon in another way, subtract the latitude of the locality from 90; the remainder is the altitude of the beginning of Aries. If the declination is northern, add it to the altitude [of the beginning of Aries]; if it is southern, subtract it from it. The altitude of the beginning of Aries after the addition or the subtraction is the altitude of the sun at noon. If it exceeds 90, subtract it from 180; the remainder is the altitude above the northern horizon.[102]

Chapter 16. On finding how many hours of the day have passed, by measuring the sun, and on finding the ascendant

Ar. p. 44
Ar. p. 45

He said: If you wish to know how many hours of the day have passed, by measuring the sun, find the altitude of the sun at noon on that day, and then half the arc of the day on that day. Then measure the altitude of the sun with a quadrant or by the shadow. When you know the altitude, at whatever time of the day you like, find the versed sine of half the arc of the day, in the way set out at the beginning of the book, in the chapter on finding the versed sines from the arcs. Then take the sine of the altitude of the sun at the time of the measurement, multiply it by the versed sine of half the arc of the day, and divide the product by the sine of the altitude at noon. Subtract the quotient from the versed sine of half the arc of the day, and find the arc of the remainder as a versed sine, as I described for taking the arc of the versed sines; keep the arc you obtain. If the measurement was made before noon, subtract this arc from half the arc of the day; if it was made after noon, add it to half the arc of the day. Half the arc of the day, after the addition or the subtraction, is the amount of the sphere that has turned from sunrise to the time of the measurement. Divide it by the times of the hours of the day, I mean of that day, found from the degree of the sun; the result is the number of seasonal hours of the day that have passed. If you divide it by 15, they are equal hours.[103]

If you wish to find the ascendant from the amount of the sphere that has turned, add the amount of the sphere that has turned to the times of ascension opposite the degree of the sun in the climate; from the sum you find the ascendant, in the way that we have explained earlier in the book. If you wish, take the arc that you obtained as a versed sine: it is the distance of the sun from the line of the middle of the sky. Divide it by the times of the hours of the day, and subtract the result from 6 hours if the measurement was made before noon, or add it to 6 hours if the measurement was made after noon; the result is the number of seasonal hours of the day that have passed, which you can convert into equinoctial hours if you wish. If you wish to find the ascendant from this arc, subtract it from the ascension of the degree of the sun in the right sphere if the measurement was made before noon, and add it to the ascension if the measurement was made after noon. With the ascension you obtain you find the ascendant and the middle of the

[102] The words in brackets are supplied by Nallino; the codex lacks them.

[103] In modern terms vers t = vers H · (1 − sin h / sin h_0), where t is the distance of the sun from the meridian in times (the hour angle), H half the arc of the day, h the altitude and h_0 the altitude at noon. The rule is exact.

sky, by taking the arc of these ascensions in the climate and in the right sphere and finding the degrees of the signs that correspond to them, in the way described in the chapter on finding the degrees of the signs from the times of ascension.[104]

Chapter 17. On finding the altitude from the hours

Ar. p. 46

He said: If you wish to find the altitude from the hours of the day that have passed, take the hours from sunrise to the given hour. If they are equinoctial hours, multiply them by 15; if they are seasonal, multiply them by the times of the hours of that day. Look at what you obtain in either way: if it is less than half the arc of the day, subtract it from half the arc of the day; if it is more, take half the arc of the day from it. The result is the distance of the sun from the middle of the sky. Find its versed sine and subtract it from the versed sine of half the arc of the day. Multiply the remainder by the sine of the altitude of the sun at noon on that day, divide the product by the versed sine of half the arc of the day, and find the arc of the quotient, as I described to you for taking the arc of the sines. The arc you obtain is the amount of the altitude above the horizon at the given hour. [If the hour is] before noon, the altitude is on the east side; if it is after noon, it is on the west side.[105]

Chapter 18. On finding the distances of the stars from the equator, and the parts of the signs that culminate with them[106]

Ar. p. 46
Ar. p. 47

He said: Suppose you wish to find the distances of the stars from the equator, and the part of the signs that culminates [with them], from their positions in longitude and latitude. Every star that is on the circle of the signs, that is, that has no latitude, behaves like the sun in its declination from the equator, which is its distance from it. But for a star that has a latitude on one side or the other, its distance from the equator falls short of its latitude and the declination of the part in which it is, when the two are added together or one is subtracted from the other, as the case requires. For the latitude of a star is measured on an arc that passes through the poles of the ecliptic and the position of the star in longitude and latitude, while its distance from the equator is measured on an arc that passes through the poles of the equator and the position of the star. Therefore a star that has a latitude culminates with a part of the signs other than its own, except one that in its course coincides with the point of the beginning of Cancer or of the beginning of Capricorn. Only then do its distance from the equator and its latitude lie on one arc, so that its distance is equal to the declination with the latitude of the star added to it or subtracted from it; and so it culminates with that one of the two points in which it is. Of the stars that have latitude, one between the beginning of Cancer and the end of Sagittarius culminates after the culmination of the degree in which it is, if its latitude is to the north of the belt of the signs, and before the culmination of the degree in which it is, if its latitude is to the south. One between the beginning of Capricorn and the end of Gemini culminates, if its latitude is to the north, with parts that precede the part in which it is, I mean before the culmination of the part in which it is; and if its latitude is to the south, with parts that follow the part in which it is, I mean after the culmination of the part of the signs in which it is.[107]

[104] The ascension in the right sphere is counted from the beginning of Capricorn (chapter 5): taken in the table of the climate, it gives the ascendant; taken in the table of the right sphere, the degree in the middle of the sky.

[105] In modern terms $\sin h = \sin h_0 \cdot (\text{vers } H - \text{vers } t) / \text{vers } H$, with the letters of the note to chapter 16: the same relation, solved for the altitude.

[106] The translation reads «تجوز» for «نجوز» (p. 47, line 3), a misprint that the corrigenda do not list.

[107] Schiaparelli, in Nallino's commentary, found the second sentence obscure. With «the declination of the part» understood as its second declination (see the next note), it is a correct statement: the declination δ of a star is never greater than its latitude β and the second declination β' of its degree taken together, since $\sin \delta = \sin(\beta \pm \beta') \cdot \cos \varepsilon / \cos \beta'$ and $\cos \beta' \geq \cos \varepsilon$; it is equal to them only when $\beta' = \varepsilon$, at the beginnings of Cancer and Capricorn, as the following sentences say.

If you wish to find the distance from the equator of one of the stars that have latitude, and the parts of the signs with which it culminates, find the latitude of the star and its direction, and the declination of the part in which it falls. If the latitude and the declination are in the same direction, add them together; if they are in different directions, subtract the smaller from the greater. The result is the adjusted latitude; note the direction in which it falls. Then take the sine of this adjusted latitude, multiply it by the sine of what the whole inclination falls short of 90, divide the product by the sine of what the part falls short of 90, that is, of the complement of its declination, and find the arc of the quotient. The arc you obtain is the distance of the star from the equator, in the direction of the adjusted latitude. Keep it.[108]

Ar. p. 48

Then take the distance of the degree in which the star is from the beginning of Cancer or the beginning of Capricorn, whichever the degree is nearer to, before it or behind it. That is: take the times of ascension in the right sphere opposite the part in which the star is. If they are less than 90, take them as they are; if they are more than 270, subtract them from 360. What you obtain in either case is the distance of the star from the beginning of Capricorn. If they are more than 90, up to 180, subtract them from 180 and take the remainder; if they are more than 180, up to the full 270, take 180 from them. What you obtain in either case is the distance of the degree from the beginning of Cancer. Keep whichever it is, and find its sine as well. Then take the sine of the latitude of the star, multiply it by the sine of the whole inclination, and divide the product by the sine of the complement of the distance of the star from the equator. Multiply the quotient by the sine of the distance of the degree in which the star is from the beginning of Cancer or of Capricorn in the right sphere, which you have kept, divide the product by the radius, and find the arc of the quotient. The arc you obtain is the difference of the passage of the star through the middle of the sky.[109]

If the star is between the beginning of Cancer and the end of Sagittarius and its latitude is northern, add the difference of its passage through the middle of the sky to the times of ascension in the right sphere opposite the part in which the star is; if its latitude is southern, subtract the difference of its passage from these times, and proceed as before. If the star is between the beginning of Capricorn and the end of Gemini and its latitude is northern, subtract the difference of its passage from these times; if its latitude is southern, add the difference of its passage to the times. Then take the degree of the signs that stands opposite the times of ascension of the star's degree in the right sphere, after the addition or the subtraction, in the ascensions of the right sphere: that is the part of the signs that culminates with the star, I mean [a degree] of the sign in which you found the number of times, if God wills.

[108] The «declination» of the part here must be its second declination: the arc of the circle through the poles of the ecliptic between the degree of the star and the equator. With it the rule is exact, $\sin \delta = \sin(\beta \pm \beta') \cdot \cos \varepsilon / \cos \beta'$ (Nallino, after Schiaparelli, who derives it from Menelaus' theorem). Al-Battānī calls both declinations by the same name, and does not say here how the second is found.

[109] The translation keeps the reading of the codex and of Plato of Tivoli, «by the radius». Nallino, following Schiaparelli, prints in its place «by the sine of what the declination of the degree in which the star is falls short of 90», holding the rule to be false otherwise. The emendation is not needed. The distance of the star's degree from the solstitial point is taken here in the right sphere, as an arc of the equator, and the sine of that distance is $\cos \alpha$, where α is the right ascension of the degree; with it the rule of the codex is exact: $\sin \Delta = \sin \beta \cdot \sin \varepsilon \cdot \cos \alpha / \cos \delta$, where Δ is the difference of the passage. Schiaparelli's own derivation, $\sin \Delta = \sin \beta \cdot \sin \varepsilon \cdot \cos L / (\cos D \cdot \cos \delta)$, with L the longitude and D the declination of the degree, is the same rule, since $\cos L = \cos \alpha \cdot \cos D$; his objection takes the distance to be measured on the ecliptic. With the emendation the result is too large by the factor $1/\cos D$: for a star at longitude 60° and latitude 10° north, the codex's rule gives 2.45°, the true value, and the emended one 2.61°.

Chapter 19. On finding the arc of the day of a star, and the times of its hours, in every locality

Ar. p. 48
Ar. p. 49

He said: Suppose you wish to find the arc of the day of any star you like, that is, its stay above the earth from the time it rises until it sets, which is the part of the equator that rises from the rising of the star to its setting. Multiply the sine of the distance of the star from the equator by the sine of the latitude of the locality, and divide the product by the sine of what the latitude of the locality falls short of 90. Multiply the quotient by the radius, divide the product by the sine of what the distance of the star from the equator falls short of 90, and find the arc of the result. The arc you obtain is the difference of the quarter of the circle of the star. If the distance of the star from the equator is northern, add the difference of the quarter of its circle to 90; if it is southern, subtract it from 90. The result after the addition or the subtraction is half the arc of the day of the star; keep it. Then take a sixth of it: the result is the times of its hours above the earth. Then double half the arc of its day: the result is the whole arc of the day of the star above the earth. Subtract it from 360: the remainder is the whole arc of its night below the earth. Likewise subtract the times of its hours of the day from 30: the remainder is the times of its hours of the night, which are below the earth. As for a star that has no latitude, its arc of the day is the arc of the day of the degree in which it is, without any difference, because it runs exactly the same course as the sun.[110]

Chapter 20. On finding the degree of the ecliptic with which a star rises and the one with which it sets, in every locality[111]

Ar. p. 49

He said: If you wish to find the degree of the signs with which a star rises and the one with which it sets, take half the arc of the day of the star and half the arc of the day of the degree that culminates with it, and find the difference between them: this is half the difference of the two days; keep it. Then look: if half the arc of the day of the star is greater than half the arc of the day of the degree that culminates with it, subtract half the difference of the two days from the times of ascension opposite the part that culminates with the star in the given climate; if half the arc of the day of the star is the smaller, add half the difference of the two days to those times. The times of the part that culminates with the star in the climate, after the addition or the subtraction, are the times of ascension of the degree with which the star rises in that locality. Find the degree of the signs opposite them in the ascensions of the climate: it is the part with which the star rises in that locality, of the sign in which the number of times fell.

Ar. p. 50

If you wish to find the degree with which that star sets, take the times opposite the degree that is the counterpart of the degree culminating with the star, in that climate. Then look: if half the arc of the day of the star is greater than half the arc of the day of the degree that culminates with it, add half the difference of the two days to those times; if half the arc of the day of the star is the smaller, subtract half the difference of the two days from those times. The times after the addition or the subtraction are the times of ascension of the degree opposite to the degree with which the star sets. Take the degree of the signs that stands opposite them in the ascensions of the climate: it is the degree opposite to the degree with which the star sets. Find the counterpart of that degree: it is the degree of the ecliptic that sets with the setting of the star. It is known that if the star has no latitude, its passage through the middle of the sky does not differ [from that of its degree], and it rises and sets with the part of the ecliptic in which it is.

Or, if you wish, add the whole arc of the day of the star to the times of ascension, in the climate, of the degree with which it rises; the sum is the ascension of the degree that is the counterpart

[110] The difference of the quarter of the circle of the star is its ascensional difference, the share of chapter 13: $\sin \Delta = \tan \varphi \cdot \tan \delta$, where δ is here the distance of the star from the equator.

[111] The print has «الّني» for «الّتي» four times on p. 50 (lines 1, 6, 10 and 12), a dot missing.

of the degree with which it sets. Find the degree of the signs that corresponds to it: the star sets with the counterpart of that degree. You can also find the degree with which the star rises and the one with which it sets in another way. Take the times of ascension in the right sphere of the part that culminates with the star, and both add half the arc of the day of the star to them and subtract half its arc of the day from them. The one added to is the ascension of the counterpart of the degree with which it sets, and the one subtracted from is the ascension of the degree with which it rises, in the climate. Take the degrees of the signs opposite them according to the rule.[112]

Chapter 21. On finding how many hours of the night have passed, by measuring a star, and on finding the ascendant[113]

Ar. p. 50
Ar. p. 51

He said: If you wish to know how many hours of the night have passed, by measuring a star, find the degree that culminates with the star, half the arc of the day of the star, and the degree with which the star rises, as has been explained to you before this chapter. Find also the altitude of the star in the middle of the sky. That is: take the distance of the star from the equator; if it is northern, add it to the altitude of the beginning of Aries in that locality, and if it is southern, subtract it from it. The altitude of Aries after the addition or the subtraction is the altitude of that star in the middle of the sky. Whenever it comes out greater than 90, subtract it from 180: the remainder is its altitude above the northern horizon, and the star is then on the north side of the zenith. Then find the versed sine of half the arc of the day of the star, multiply it by the sine of the altitude of the star at the time of the measurement, and divide the product by the sine of the altitude of the star in the middle of the sky. Subtract the quotient from the versed sine of [half] the arc of the day of the star, and find the arc of the remainder as a versed sine; keep the arc you obtain. If at the time of the measurement the star is on the east side of the middle of the sky, subtract this arc from half the arc of the day of the star; if the star is on the west side, add it to it. Half the arc of the day of the star, after the addition or the subtraction, is the amount of the sphere that has turned from the rising of the star to the hour of the measurement; keep it.[114]

Then look: if the part with which the star rises is between the degree of the sun and the degree of its counterpart, the star rose by day; if it is between the counterpart of the degree of the sun and the degree of the sun, it rose by night. If it rose by day, take the times of ascension in the given climate of the degree with which the star rose, and subtract them from the times of ascension opposite the counterpart of the degree of the sun, which is opposite to it, in that climate. Subtract the remainder from the amount of the sphere that has turned since the rising of the star, which you have kept; the remainder is the amount of the sphere that has turned from sunset to the hour of the measurement. Divide it by the times of the hours of the night; the result is the number of seasonal hours of the night that have passed. If the star rose by night, subtract the times of ascension opposite the counterpart of the degree of the sun from the times opposite the part with which the star rose in that climate, and add the remainder to the amount of the sphere that has turned since the rising of the star; the sum is the amount of the sphere that has turned from sunset to the hour of the measurement. Divide it by the times of the hours of the night, in the same way; the result is the number of seasonal hours of the night that have passed. If you divide the amount of the sphere that has turned since sunset by 15, the result is the number of equal hours of the night that have passed.

Ar. p. 52

If you wish to find the ascendant from the amount of the sphere that has turned since the rising of the star, add the amount of the sphere that has turned since the rising of the star to the times of ascension opposite the part with which the star rises in the climate, and with the sum find the

[112] The ascensions in the right sphere are counted from the beginning of Capricorn (chapter 5), so that they can be entered directly in the table of the climate.

[113] The print has «الّني» for «الّتي» on p. 52, line 1, a dot missing.

[114] This is the rule of chapter 16, applied to a star. «Half» is lacking in the Arabic at its second occurrence; Nallino's translation supplies it.

ascendant and the middle of the sky by the rule given before. If you wish to find the ascendant in another way, take the arc that you obtained as a versed sine and subtract it from the times of ascension, in the right sphere, of [the part] that culminates with the star, if the star is on the east side, and add it to these ascensions if it is on the west side. With the times you obtain after the addition or the subtraction you find the ascendant in the same way, and the middle of the sky as well.

If you measure with the moon in particular, you need to know its parallax, so that you have its visible position correct, I mean the position in which it is seen, in longitude and latitude. From it you then find its visible distance from the equator, the visible part that culminates with it, half the arc of the day of its visible position on the ecliptic, half the arc of the day of the visible part that culminates with it, and the visible degree with which it rises. When you know all this, take the altitude of the moon and work with it, after finding the altitude of the moon in the middle of the sky from its visible distance from the equator.

Chapter 22. On finding the altitude of the stars from the hours of the night, in every locality[115]

Ar. p. 52

He said: If you wish to find the altitude of any star you like from the hours, take the degree that is in the middle of the sky at that time, and the rising and the setting degree. Then find the distance of the star you want from the line of the middle of the sky at that time. That is: take the times between the part of the middle of the sky and the part that culminates with the star, in the right sphere. If the degree that culminates with the star is to the east of the degree of the middle of the sky, subtract the times of ascension of the degree of the middle of the sky from the times of ascension of the degree that culminates with the star; if it is to the west, subtract the times of ascension of that degree from the times of ascension of the degree of the middle of the sky. The result after the addition or the subtraction is the distance of the degree that culminates with the star from the line of the middle of the sky.

Ar. p. 53

If you wish to find this in another way, take the times opposite the part with which the star rises in the climate, and take the times opposite the counterpart of the part with which the star sets. If the part that culminates with the star is to the east of the middle of the sky, subtract the times opposite the degree with which the star rises from the times opposite the rising part; if the degree is to the west of the middle of the sky, subtract the times opposite the rising part from the times opposite the counterpart of the part with which the star sets. What you obtain in either case is its distance from the horizon. Subtract it from half the arc of the day of the star; the remainder is the distance of the star from the line of the middle of the sky.

When you know the distance of the star from the line of the middle of the sky, on whichever side it is, find the versed sine of this distance and subtract it from the versed sine of half the arc of the day. Multiply the remainder by the sine of the altitude of the star in the middle of the sky, divide the product by the versed sine of half the arc of the day of the star, and find the arc of the quotient as a direct sine, as you take the arc of the sines; the arc you obtain is the altitude of the star at that hour. It is known that if the part with which the star rises follows the part that is rising at that time, the star has not yet risen; if it precedes the rising part, the star has risen and is above the earth. If the part with which the star sets follows the setting part, the star has not yet set and is above the earth; if it precedes it, the star has set. For a star does not rise until the part with which it rises has risen, and does not set until the part with which it sets has set. As for the moon, you find its visible altitude from its distance from the equator, if God wills.[116]

[115] The print has «الّني» for «الّتي» on p. 52, line 15, a dot missing.

[116] This is the rule of chapter 17, applied to a star: $\sin h = \sin h_0 \cdot (\text{vers } H - \text{vers } t) / \text{vers } H$, where h_0 is the altitude of the star in the middle of the sky.

Chapter 23. On finding the azimuth of any star from its altitude and its place in the sphere

Ar. p. 53
Ar. p. 54

He said: If you wish to find the azimuth of any star you like on the horizon circle, find the altitude of the star at that time, the distance of the star from the equator and the latitude of the locality. Then proceed with them in the way that I described to you in the chapter on finding the azimuth of the altitude and of the shadow, leaving out nothing, except that you use the distance of the star from the equator in place of the declination of the degree of the sun. If you wish to find the azimuth of the moon, do this with its visible distance from the equator, so that you have the azimuth of the moon on the circle of altitude correct; and likewise for every star whose azimuth on the horizon circle you wish to find, if God wills.

Chapter 24. On finding the distance of a star from the equator, and the part that culminates with it, from the parts of the signs that rise or set [with it] and from the amplitude of its rising and setting on the horizon circle

Ar. p. 54

He said: Suppose you wish to find which degree of the signs culminates with a star, and the distance of the star from the equator, from the azimuth of its rising or setting and the part of the ecliptic that rises or sets with it, when these are known from some measurements. Multiply the sine of the altitude of the beginning of Aries in that locality by the sine of the azimuth of the rising or setting of the star, divide the product by the radius, and find the arc of the quotient. The arc you obtain is the distance of the star from the equator, on the side of the azimuth.[117]

With it find half the arc of its day, in the way described for finding half the arc of the day of a star from its distance from the equator. Then look: if the star is on the eastern horizon, add half the arc of its day to the times of ascension, in the climate, of the degree that rises with it; if it is on the western horizon, subtract half the arc of its day from the ascensions of the part with which it itself sets. Enter what you obtain, in either case, in the ascensions of the right sphere and take the degree of the signs opposite it: that is the part that culminates with the star.[118]

Chapter 25. On finding the part in which a star is, and the latitude of the star, from its distance from the equator and the degree of the signs that culminates with it

Ar. p. 54
Ar. p. 55

He said: Suppose you wish to find the part in which a star is, and the latitude of the star, from its distance from the equator and what culminates with it, when these are known from measuring the altitude of the star in the middle of the sky and the degree that culminates with it, or from measuring it at the horizon. When the altitude of a star in the middle of the sky has been taken and the difference between it and the altitude of the beginning of Aries in that locality is known, this is its distance from the equator, on the side on which it falls: if the altitude is greater than the altitude of the beginning of Aries, the distance is to the north, and if it is less than the altitude of the beginning of Aries, the distance is to the south. The part that culminates with it at that time is found from the part of the signs that culminates at that hour, by measuring another star whose position is known, or by other measurements with which the part of the middle of the sky can be found. If it can be known only from the rising or setting of the star on the horizon circle and the part that rises or sets with it, [it is found] as we have explained in the preceding chapter,

[117] In modern terms $\sin \delta = \cos \varphi \cdot \sin A$, where A is the amplitude; this is the rule of chapter 7 solved for the declination.

[118] The rule for the eastern horizon is exact, with the ascensions in the right sphere counted from the beginning of Capricorn (Nallino). The rule for the western horizon is wrong as it stands: the ascension to be taken is that of the counterpart of the part that sets with the star, as in chapter 20, since that counterpart rises when the star sets. With the setting part itself the result is wrong by 180° plus twice the ascensional difference of that part. Nallino's commentary states the rule as printed, without remark.

from whatever allows the rising or setting part of the ecliptic and its azimuth on the horizon to be known.

Ar. p. 56

When you know this, in whichever way you found it, take the declination of the part that culminates with the star and the distance of the star from the equator. If both are on the same side, subtract the smaller from the greater; the remainder is the adjusted distance. Keep it, and find its sine and the sine of what this distance falls short of 90. Then find the sine of the whole inclination and the sine of what the whole inclination falls short of 90. Then subtract the sine of the complement of the whole inclination from 120; the remainder is the longest chord. Then subtract the declination of the part that culminates with the star from 90, find the sine of the remainder, and subtract that too from 120; the remainder is the exceeding chord. Keep all these by their names. Then multiply the sine of the whole inclination by the radius, and divide the product by the sine of what the declination of the part that culminates with the star falls short of 90: the result is the sine of the adjusted inclination; keep it with its name and its direction. Then find its arc, subtract the arc from 90 and find the sine of the remainder: this is the sine of the complement of the adjusted inclination. Then multiply the sine of the adjusted inclination that you kept by the sine of the adjusted distance mentioned earlier in this chapter, and divide the product by the sine of the complement of the adjusted distance. Multiply the quotient by the exceeding chord and divide the result by the longest chord. Multiply the quotient by the sine of the complement of the whole inclination and divide it by the sine of the complement of the adjusted inclination. Multiply the quotient by the sine of the distance of the degree that culminates with the star from the beginning of Cancer or the beginning of Capricorn, whichever it is nearer to on either side, I mean behind it or in front of it, in ascensions of the right sphere. Divide the result by the radius and find the arc of the quotient: the arc you obtain is the variation of the star. Keep it.

Then look: if the part that culminates with the star is between the beginning of Cancer and the end of Sagittarius, and the distance of the star from the equator is northern, subtract the variation of the star from the times of ascension in the right sphere of the part that culminates with the star; if it is southern, add it to them. If the star is between the beginning of Capricorn and the end of Gemini, and its distance from the equator is northern, reverse this and add the variation of the star to those times; if it is southern, subtract it from them. Find the degree of the signs opposite the times obtained after the addition or the subtraction, in the ascensions of the right sphere: the result is the degree and minute of the signs in which the star is, if God wills.

If the distance of the star from the equator and the declination of the part that culminates with the star are on different sides, the way to work is to add them together: the sum is the adjusted distance. Multiply the sine of the whole inclination by the sine of the distance of the star from the equator, divide the product by the sine of what the adjusted distance falls short of 90, multiply the quotient by the sine of what the declination of the part that culminates with the star falls short of 90, and divide the product by the radius: the result is the sine of the adjusted inclination. Multiply it by the sine of the distance of the star from the equator as well, divide the product by the sine of the complement of the distance of the star from the equator, multiply the quotient by the exceeding chord, and divide the result by the longest chord: the result is the dividend; keep it with its name. Then find the arc of the sine of the adjusted inclination that you obtained, subtract the arc from 90, take the sine of the remainder and multiply it by the dividend that you kept. Divide the product by the sine of the complement of the whole inclination, and multiply the quotient by the sine of the distance of the part that culminates with the star from the beginning of Cancer or the beginning of Capricorn, whichever it is nearer to in front of it or behind it, in the right sphere. Divide the product by the radius and find the arc of the quotient: the arc you obtain is the variation of the star. Follow the first rule with it, until you know the part of the ecliptic in which the star is.[119]

Ar. p. 57

[119]Schiaparelli, in Nallino's commentary, judges these rules for the longitude false. He suggests that they are not al-Battānī's, but were written to fill a leaf lost from the exemplar from which the Escorial codex and Plato of Tivoli's

If you wish to find the latitude of the star and the direction of its latitude, multiply the sine of the distance of the star from the equator by the sine of what the declination of the degree in which you have found the star falls short of 90, divide the product by the sine of what the whole inclination falls short of 90, and find the arc of the quotient. Then look: if the arc is greater than the declination of the degree in which you found the star, subtract the declination of that degree from it; if the arc is the smaller, subtract it from the declination of that degree. The result after the addition or the subtraction is the latitude of the star. If you wish to know the direction of the latitude, look: if the arc is greater than the declination of the degree in which the star is, the latitude is on the side of that declination; if it is less, the latitude is on the side opposite to the declination of the degree of the star.[120]

Chapter 26. On finding the distances between the stars according to their configuration in the sphere and the positions of one relative to another; by it the true distance of the moon from the sun is found, according to the latitude of the moon[121]

Ar. p. 57

He said: We wish to show how the amounts of the distances between the stars are found on the greatest circle, the one that lies between the two poles of the ecliptic, which is the circle of the signs. It has been shown by proof that in every quadrilateral inscribed in a circle, the products of each two opposite sides, one by the other, added together, are equal to the product of one of its diagonals by the other. In every quadrilateral inscribed in a circle in which two of the sides are parallel, the two remaining sides are opposite and, for the reason given above, equal; when they are produced they meet in a point. Its diagonals too are equal, and the product of one of them by the other is equal to the product of one of the parallel sides by the other together with the product of one of the opposite sides by the other.[122]

Ar. p. 58

Since this is so, we draw a line for a piece of the ecliptic and mark its ends A and B, and from the points A and B we draw two lines that meet at Z; let the point Z be one of the poles of the ecliptic, whichever it is. Each of the lines AZ and BZ is then the line of a quarter of the circle that passes through the poles of the ecliptic and the positions of the two stars. We put one of the two stars at the point A of the ecliptic, and the other inclined from the ecliptic in latitude, at the point Ṭ; its position on the ecliptic is known to be the point B. The arc BṬ is the latitude of the star, and we draw the line AṬ, which is the amount of the distance between the two stars. The line AṬ and the arc AṬ are found by drawing from the point Ṭ a line parallel to the line BA, the line ṬH, and from the point Z, the pole, a line to the middle of the line BA, the line ZG, and by marking the sign M at the place where it cuts the line ṬH. The quadrilateral ṬBAH is then a figure of four sides, of which the sides BA and ṬH are parallel, and ṬB and HA equal and opposite; produced, they meet at the point Z. It is clear on the sphere that each of the arcs ZA, ZB and ZG is a quarter of a circle; therefore the arcs ZṬ, ZM and ZH are equal, and so each of the arcs ṬB, MG and HA is equal. By what has been said before, the line ṬM has also been shown to be half the line ṬH. Since the right-angled triangle BGZ is similar to the small right-angled triangle ṬMZ, the line ṬM is known in amount from the line BG, which is already known; and since they are in one triangle, the ratio of ṬM to BG is as the ratio of ZṬ to ZB, and also as the ratio of ZM to ZG. When the line ṬM is known, the whole line ṬH is known, since it is twice ṬM.[123]

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version both derive; al-Battānī could have found the longitude by inverting the rules of chapter 18. A numerical check bears out the judgement. When the star and its culminating degree are on the same side of the equator, the first rule comes close to the true variation without being exact: for a star at longitude 60° and latitude 10° north it gives 2.48°, against the true 2.45°, and for one at 150° and 25° north 10.63°, against 10.47°. When they are on opposite sides, the second rule is far off: for a star at longitude 200° and latitude 12° north it gives 0.07°, against the true 4.53°.

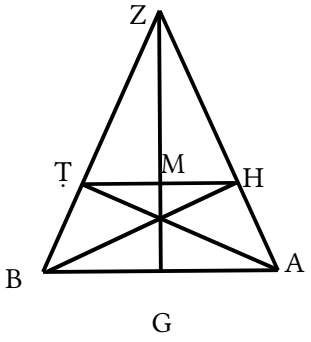
[120] The «declination» of the degree is here again its second declination β′ (chapter 18). With it the rule is exact (Nallino, after Schiaparelli): $\sin x = \sin \delta \cdot \cos \beta' / \cos \varepsilon$, and the latitude is the difference between x and β′.

[121] The translation follows Nallino's corrigenda in reading «ضرب», «product», for «صرب» (p. 57, line 16).

[122] The first theorem is Ptolemy's theorem on the quadrilateral inscribed in a circle (*Almagest* I 10). For «inscribed in a circle», in the second sentence, the codex has «on a sphere or on a plane»; the reading is Nallino's.

[123] The sentence from «therefore the arcs ZṬ, ZM and ZH» to «is equal» is printed by Nallino in brackets as the

Let the arc AB between the two stars in longitude be 60 parts, so that the arc BG is half of that, 30 parts. We take the latitude of the star whose position in longitude is the point B to be 30 parts: this is the arc BṬ. Then by what has been said about the sphere the arc MG is also 30 parts, and so the arc MZ that remains is 60 parts; its sine, which is the line MZ, is [51 parts] 57 minutes 42 seconds, approximately. The arc BG has been shown to be 30, and its sine, the line BG, is also 30 parts. The whole arc ZG is a quarter of a circle, and the line ZG is its sine, which is 60 parts, the radius. So if we take of the line BG the part given by the ratio of the line ZM to ZG, there remains the ratio of the line ṬM to the line BG; this is what we wished to show.[124]



Ar. p. 59

He said: The computation of this is to multiply ZM by BG, which gives 1558 parts 51 minutes, approximately. When this is divided by the line ZG, the result is 25 parts 58 minutes 51 seconds, approximately, which is the amount of the line ṬM. So the arc ṬM is 25° 39½′, and the whole arc ṬH is twice that, 51° 19′. The quadrilateral BṬHA is then known in its sides, and the diagonal ṬA is known too, by what has been said before about finding the full chords that fall in these sides. Since the sine ṬM has been shown to be 25;58,51, the line ṬH, which is the full chord of ṬH, is twice that, 51 parts 57 minutes 42 seconds. The full chord of the arc BA is twice the sine BG, which is 60 parts. The full chord of the arc ṬB is 31 parts 3 minutes 30 seconds: this is the chord of the 30 parts that are the latitude of the star. The line ṬB is equal to it: it is the sine of 15 parts, doubled, and these 15 are half the arc ṬB. When the side BA is multiplied by the side ṬH parallel to it, the product is 3117 parts 42 minutes, and the product of ṬB by HA, which is equal to it, is 964 parts 37 minutes, approximately. Added together, they equal the product of ṬA by itself, since ṬA is equal to HB. So ṬA by itself is 4082 parts 19 minutes, and its root is 63 parts 54 minutes, approximately, which is the amount of the line ṬA. So the arc ṬA, the arc of the full chord, is 64° 19′, and this is the true distance between the two stars, whose difference in longitude was only 60 degrees.[125]

Further, if the two stars are both at the point B in longitude, and one of them at the point Ṭ in latitude, the distance between them is the amount of the latitude alone, the arc ṬB. If one of them is at the point Ṭ and the other at the point H, and they have equal latitudes in this figure, the distance between the two stars is the arc ṬH. Likewise, if one of them were at the point H and the other at D, the distance between them would be known, by drawing the line DK parallel to the line BA and to the line ṬH and finding the amount of DK in the way we have described: the quadrilateral DṬHK then becomes known in its sides, and so the line HD between the two stars,

addition of an unskilled reader. Its equalities hold if M lies on the circle of latitude through Ṭ and H; but the arc ṬH of the computation below is an arc of the great circle through Ṭ and H, whose middle is nearer to the pole. The computation does not depend on them: the ratio of ṬM to BG is that of ZṬ to ZB, and the sine of ZṬ is the 51;57,42 used below.

[124] «51 parts» is supplied by Nallino; the codex lacks it. Nallino takes the passage on the arcs MG and MZ to be corrupt, and reads «the arc ZṬ that remains is 60 parts, and its sine, the line ZṬ, 51;57,42»; and in the last sentence he understands: if BG is reduced in the ratio of ZM to ZG, the result is ṬM.

[125] Nallino reads ZṬ and ZB for ZM and ZG in the first sentence, with the same numbers. For 51° 19′ the codex has 51° 59′. The true distance is 64° 20′, since $\cos d = \cos 30° \cdot \cos 60°$.

the diagonal of the quadrilateral, becomes known too. The distance of the star at the point D from the one at the point A is known from the quadrilateral DBAK.[126]

Ar. p. 60

Likewise, if you wish to know the distance between two stars, look. If one of them has no latitude, like the sun or another of the stars that are on the belt of the signs, and the other has a latitude on either side, take the amount of degrees of longitude between them: this is the first side. Then take half of it and find its sine, multiply it by the sine of what the latitude of the star falls short of 90, divide the product by the radius, and keep the result as it is. Then take its arc and double it: this is the second side. Then find the full chord of the latitude of the star by the rule given to you at the beginning of the book, which is to take the sine of half the latitude and double it: the result is the full chord of the latitude. Then find the full chord of the first side and the full chord of the second side as well; the fourth side is equal to the third, whose full chord is the chord of the latitude. When you have done this, multiply the full chord of the first side by the full chord of the second side, and add to it the full chord of the latitude multiplied by itself, which is the product of the third side by the fourth. Take the root of the sum and find its arc as you find the arcs of full chords, that is, take half of it, find its arc and double the arc: the result is the distance between the two stars.[127]

If both stars have latitude on the same side, and the two latitudes are equal, find the arc of the second side: it is the amount between them. If the latitudes on the same side are unequal, subtract the smaller from the greater: the remainder is the third side, and the fourth side is equal to it; keep it. Then subtract the latitude of each of them from 90, find the sine of the remainder, and multiply it by the sine of half the degrees of longitude between them. Divide what results for each of them by the radius, find the arc of the quotient and double it: the result is the amount of each of the two sides of longitude; the longer of them is the first side and the shorter the second. Find their full chords, which are twice what results from each of them by the division, multiply one of the two chords by the other, and add to the product the third side multiplied by itself. Take the root of the sum, take half the root, find its arc and double the arc: the result is the distance between the two stars.[128]

If the latitudes of the two stars are on different sides, add the two latitudes together: the sum is the third side, and the fourth side is equal to it. Then subtract each of the two latitudes from 90 and find the sine of each, which is the sine of what each of them falls short of 90. Multiply it by the sine of half the degrees of longitude between them and divide what results for each of them by the radius; double the quotient: what results for each of them is the full chord of the first side and of the second side. Multiply one of them by the other, and add to the product the full chord of the third side multiplied by itself. Take the root of the sum and take half of it; find the arc of the result and double the arc: the result is the distance between the two stars.[129]

Ar. p. 61

It is known that when the two stars are together in one degree, and only one of them has a latitude, or both have latitudes, on the same side or on different sides, the distance between them is only the amount of latitude between them; and when neither has a latitude, the distance between them is the amount of the degrees of longitude, whatever it is. This chapter is needed above all for working out the directions in nativities.[130]

[126] The codex has no figure with the points D and K; Nallino adds one to his translation.

[127] In modern terms, with chords in a circle of radius 1: $\mathrm{crd}^2\, d = \mathrm{crd}\, \Delta\lambda \cdot \mathrm{crd}\, s + \mathrm{crd}^2\, \beta$, where $\Delta\lambda$ is the difference in longitude, β the latitude and s the second side, $\sin \tfrac{1}{2}s = \cos\beta \cdot \sin \tfrac{1}{2}\Delta\lambda$. The rule is exact: it amounts to $\cos d = \cos\beta \cdot \cos\Delta\lambda$.

[128] «The third side multiplied by itself» is its full chord multiplied by itself, as in the next paragraph.

[129] The rules of these two paragraphs are exact as well: they amount to $\cos d = \sin\beta_1 \cdot \sin\beta_2 + \cos\beta_1 \cdot \cos\beta_2 \cdot \cos\Delta\lambda$.

[130] The directions (*tasyīrāt*) are the astrological progressions of a nativity (Nallino's *profectiones*).

Chapter 27. On finding the length of the year and the motion of the sun in it

Ar. p. 61

He said: The ancients disagreed about the length of the year. Some of the ancient people of Egypt and Babylon said that it is 365 days, a quarter of a day and 1/120 of a day. Ptolemy says that they reckoned it from the sun's leaving one of the fixed stars until it returns to it, and he finds fault with them for this and says that it is most absurd. For if this view were allowed, nothing at all would prevent us from saying that the year is the time from the sun's leaving Saturn, or another of the wandering stars, until it returns to it, and this view is very plainly false. The year is only the time from the sun's leaving an unmoving point of the sphere until it returns to it, either from one of the two equinoctial points to the same point, or from one of the two solstitial points to the same point; for no starting points on the ecliptic are more fitting than these points.[131]

Hipparchus worked with a year of 365 days and a quarter of a day only, although it had become clear to him that it is less than that, as Ptolemy reports of him when he collected his views. For he said that the year is in truth 365 days and less than a quarter of a day, because he found that the summer solstice came before the time of the full quarter day in excess of 365 days. This made him uncertain about the motion of the sun, so that he imagined that it had another sphere, eccentric to the centres of the two spheres.[132]

Ar. p. 62

The ancients took this mostly from summer observations, made when the sun passes the summer solstitial point. These are not found to be as accurate as observations made when the sun passes one of the two equinoctial points, especially the autumnal equinoctial point, because the air is clearer and purer at that time than at the time of the spring equinox. For when the sun passes the solstitial point it moves slowly in declination, and when it passes the two equinoctial points its motion in declination is very fast. That is why Ptolemy relied only on the autumnal observations and based his reckoning on them. One of the observations of Hipparchus on which he relied, and of whose truth he had no doubt, is the observation that he reported: he found that the sun passed the autumnal equinoctial point in the year 178 after the death of Alexander, on the third of the five epagomenal days, at midnight at Alexandria, of the night whose morning was the fourth of the epagomenal days; and he was sure of it.[133]

Ptolemy observed 285 Egyptian years later; this is the observation that he says in his book he verified and made exact with the greatest care. He found that the sun passed the autumnal equinoctial point in the third year of the reign of Antoninus, which is the year 463 after the death of Alexander, on the ninth day of Athyr, one of the months of the Copts, about one hour after sunrise at Alexandria. When he took the time between the two observations, he found it to be in truth 285 Egyptian years, 70 days and a quarter of a day and 1/20 of a day, in place of the 71 days and a quarter that should accumulate from the full quarters in these 285 years. The ratio of this one day less 1/20 of a day, by which the time of the observation came before the time of the quarter day in excess of 365 days, to the 285 years between the two observations is as the ratio of one day to 300 years. So the length of the year found from these two observations is 365 days and a quarter of a day less 1/300 of a day, which is one part and a fifth of a part out of 360 parts.

Ar. p. 63

He says that he also took the ancient summer observations made before Hipparchus, namely the observation made in the time of Apseudes, king of Athens, in which the sun passed the summer solstitial point 108 Egyptian years before the death of Alexander, in the morning of the twenty-first of Pharmouthi, one of the months of the Copts, in that year; and that he observed the sun himself and found that it passed the summer solstitial point in the year 463 after the

[131] 365 1/4 + 1/120 days is 365 days 6 hours 12 minutes, a year measured by the fixed stars (Nallino). Ptolemy discusses the length of the year in *Almagest* III 1.

[132] Hipparchus found 365 1/4 − 1/300 days, that is 365 days 5 hours 55 minutes 12 seconds (Nallino).

[133] Ptolemy does not say that Hipparchus made this observation at Alexandria; Hipparchus observed at Rhodes, which he took to lie on the meridian of Alexandria (Nallino). The epagomenal days are the five days added after the twelve months of 30 days of the Egyptian year.

death of Alexander, on the eleventh of Mesori, one of the months of the Copts, about two hours after midnight of the night whose morning was the twelfth of that month. Between these two observations there were about 571 Egyptian years, 140 days and a half and a third of a day, in place of the 142 days and a half and a quarter of a day that would accumulate from the quarters of the said years if the quarters were full in the years. So he found that the summer solstice came before the time of the full quarter by one day, two thirds and a quarter of a day, and the ratio of this one day, two thirds and a quarter of a day to the said 571 years is as the ratio of two full days to 600 years. This agreed with what he had worked with, that the observation comes before the time of the full quarter day by one day in every 300 years, although these summer observations are not as reliable as the autumnal ones, for the reason we have given. It is clear that the observation made before Hipparchus was earlier than the observation of Hipparchus by about the time between the observation of Hipparchus and that of Ptolemy, for it was 286 years before Hipparchus.[134]

Ar. p. 64

Then we observed at the city of al-Raqqa. One of our autumnal observations, on which we rely and whose accuracy we trust as far as the instrument showed us, is the observation made 743 years after the autumnal observation of Ptolemy mentioned above. When we measured, we found that the sun passed the autumnal equinoctial point in the year 1194 of the years of Dhū l-Qarnayn, which is the year 1206 after the death of Alexander, about four hours and a half and a quarter of an hour before sunrise on the nineteenth day of Aylūl, one of the months of the Rūm, which is the eighth day of Pachon, one of the months of the Copts. Since the meridian circle of Alexandria comes before the meridian circle of al-Raqqa by about two thirds of an equal hour, there are between the two observations, I mean our observation and the autumnal observation of Ptolemy, 743 Egyptian years, 178 days and a half and a quarter, less about two fifths of an hour, in place of the 185 days and a half and a quarter of a day that should accumulate from the quarters in these years if the quarters were full between the two observations. When we divide these 7 days and two fifths of an hour, by which the time of the observation came before the time of the quarter day in excess of 365 days, by the 743 years between the two observations, the share of one year comes to 3 parts 24 minutes of the 360 parts that make up the revolution of a day and its night. When we subtract this from the time of the quarter day, which is 90 parts, there remains as the excess over the full 365 days 86 parts 36 minutes; so the true length of the year is 365 days, 14 minutes and 26 seconds [of a day], approximately.[135,136]

When we divide the 360 parts of the circle of the sphere by the length of the year that has been found, the mean motion of the sun in a day and its night comes to 0;59,8,20,46,56,14; in the 30 days that make the Egyptian month, 29;34,10,23,28,6,47; and in the 365 days that make the Egyptian year, 359 parts 45 minutes 46 seconds 25 thirds 32 fourths 2 fifths 31 sixths, approximately. We have compounded these motions and entered them in the tables for the collected years and the single years, the months, the days and the hours, in the era of the Arabs and the era of the Rūm, so that the position of the sun by its mean motion, which is called the mean of the sun, can easily be found at any time we want, in whichever of the two eras we like. It is clear that the length of the year that we obtained by observation is less than the length that Ptolemy gives by 2 parts and a fifth of a part, and so the motion of the sun that we found exceeds the motion that Ptolemy gives by 0;0,0,3,33,43,43 in a day, and in the Egyptian year by 0;0,21,40,10,53,56, approximately,

[134] The observation in the time of Apseudes, archon of Athens, is the summer solstice observed by Meton and Euctemon, 27 June 432 BC (Nallino). For 108 years the codex has 168.

[135] The years of Dhū l-Qarnayn are those of the Seleucid era, which al-Battānī begins on 1 September; the observation fell before sunrise on 19 September 882 (Nallino). Two thirds of an hour is 10° of longitude; al-Battānī's own table of cities gives al-Raqqa 73° 15′ and Alexandria 60° 30′, a difference of 12° 45′ (Nallino).

[136] The minutes and seconds of the year are sixtieths of a day: 365;14,26 days is 365 days 5 hours 46 minutes 24 seconds. Hipparchus' and Ptolemy's year is 365;14,48 days, 2;12 parts of a day of 360 parts longer.

if God wills.[137,138]

Chapter 28. On finding the anomaly of the motion of the sun, and what appears with it of the position of its farthest distance from the centre of the earth[139]

Ar. p. 64
Ar. p. 65

He said: Having finished with the length of the year and the mean motion of the sun, we try to explain the anomaly that appears in the course of the sun, and its greatest amount, and what appears with it of the position of the point of its farthest distance from the centre of the earth on the ecliptic. In this we follow the method that Ptolemy used in his book, from the sun's passage through the quarters of the ecliptic, by measurements that we made in successive years, renewing the observations as carefully as we could. We established that the sun passes from the autumnal equinoctial point to the spring equinoctial point in 178 days and about 14½ hours, and that it passes from the spring equinoctial point to the autumnal equinoctial point in a longer time: what we found by observation, as far as our diligence reached, is that its true course, which is seen by measurement, takes 186 days and about 14¾ equal hours. It became clear from what we have said that the point of its farthest distance is in this half.[140]

Then we observed and found that the sun passes from the beginning of Aries to the beginning of Cancer, that is, from the spring equinoctial point to the summer solstitial point, in 93 days and about 14 equal hours, a little less than that. It is also clear that its passage from the spring equinoctial point to the summer solstitial point takes a longer time than its passage from the summer solstitial point to the autumnal equinoctial point. So we knew that the point of the farthest distance, and the centre of the eccentric sphere on which the point of the farthest distance lies, fall, on the ecliptic, in this quarter, which takes longer than the remaining quarter. We found the mean motion of the sun in the 186 days 14¾ hours to be 183 parts 56 minutes 12 seconds, and in the 93 days 14 hours 92 parts 14 minutes 10 seconds, approximately.[141]

Since this is so, we draw a circle for the ecliptic, ABGD, about the centre H, with the diameters AG and BD at right angles. We take the point A as the spring equinoctial point; then the sign B is the summer solstitial point, the sign G the autumnal equinoctial point, and the point D the winter solstitial point. In the quarter of the circle on which AB lies, for the reason given above, we mark the sign Z and take it as the centre about which we draw the circle of the sun's eccentric sphere, inside the first circle. We write on it KLMN, on the diameters KM and LN, which intersect at the centre Z at right angles. We write the sign F where the lines BD and KM meet; the sign LA where the diameter AG cuts the circle KLMN on the side of [the point A]; and the sign Ḍ where the diameter BD cuts the circle KLMN on the side of [the point B]. We draw the perpendicular of the arc LA K from the point LA to the sign Y on the diameter KM, and also the perpendicular ḌḤ. We draw the line HZ, which passes through the two centres, and extend it to the ecliptic, which is the circle ABGD, and we write the sign Ṭ where it meets it, and the sign ʿ where it cuts the circle KLMN.[142]

Ar. p. 66

[137] Sexagesimal numbers with many places are written here in the usual notation: 0;59,8,20,46,56,14 is 0 degrees, 59 minutes, 8 seconds, 20 thirds, 46 fourths, 56 fifths and 14 sixths. Ptolemy's daily motion is 0;59,8,17,13,12,31 and his yearly motion 359;45,24,45,21,8,35; the two differences given here are exact. In the yearly difference the text has 50 fifths; the difference is 53, as Nallino's translation has it.

[138] The collected years are the periods of years used as arguments of the tables, the single years the years within a period. The era of the Arabs is that of the Hijra; the era of the Rūm is the Seleucid era of the Syrians.

[139] The heading prints «الثامن العشرون» without «و». The translation reads «مركزه» for «مركوه» (p. 68, line 2) and «التي» for «الني» (p. 70, line 20), misprints that the corrigenda do not list.

[140] Al-Battānī has not shown before why the farthest distance lies in the half that takes longer; Nallino, after Schiaparelli, suspects that something has fallen out. Ptolemy's method is that of *Almagest* III 4.

[141] Here and on p. 66 the text writes 183;56,2, but it gives 3;56,12 for the excess over 180 and 1;58,6 for its half; Nallino's translation has 183;56,12. With al-Battānī's daily motion the mean motion in 186 days 14¾ hours is 183;56,13. In 93 days 14 hours it is 92;14,26, as the tables give; Nallino keeps 92;14,10, which the computation below also uses, and takes the 16 seconds for the «little less» of the observed time.

[142] The words in brackets are supplied by Nallino. LA is the letter lām-alif, used here as a sign of the figure.

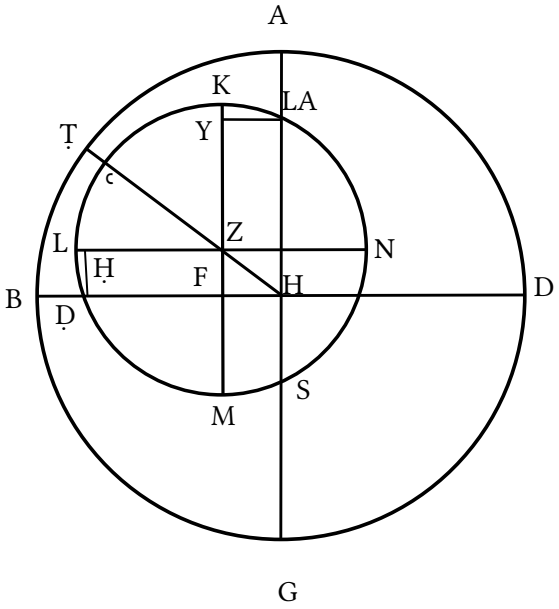
It is established that the arc AB is 90 parts, and that the arc KL of the eccentric circle is also 90 parts. The point LA of the eccentric circle is its point of the beginning of Aries, and the point Ḍ the place of the beginning of Cancer. So the arc LA K L Ḍ M S of the eccentric is the amount of the eccentric circle that the sun passes over by its mean motion between the beginning of Aries and the beginning of Libra, which, as was said before, is 183 parts 56 minutes 12 seconds. The arc K L Ḍ M is half the eccentric circle, 180 parts; so each of the two arcs KL and SM is half of the 3 parts 56 minutes 12 seconds by which the sun's mean motion exceeded 180, that is, 1 part 58 minutes 6 seconds: the arc LA K is this amount, and the arc MS is equal to it. It is known too that the arc LA K L Ḍ is what the sun passes over on the eccentric by its mean motion from the beginning of Aries to the beginning of Cancer, so that the arc LA L Ḍ of the eccentric circle is of the 92 parts 14 minutes 10 seconds; and since the arc LA K L of it is known from what was said before, 91 parts 58 minutes 6 seconds, the arc LḌ is 16 minutes 4 seconds. Clearly the perpendicular LA Y is half the chord of twice the arc LA K, and the perpendicular ḌḤ half the chord of twice the arc LḌ; so the perpendicular LA Y is about 2 parts 3 minutes 39 seconds and the perpendicular ḌḤ about 16 minutes 45 seconds, the sines of the arcs LA K and LḌ. Since the line KM is parallel to the line AG, the line HF is equal to the line LA Y; and since the line LN is parallel to the line BD, the line ZF is equal to the line ḌḤ. So the side HZ of the right-angled triangle HFZ is known. The line HF multiplied by itself is about 4 parts 14 minutes 48 seconds, and the line ZF multiplied by itself 4 minutes 41 seconds; and the line HZ, which subtends the right angle, multiplied by itself is equal to HF and ZF, each multiplied by itself, added together. So HZ multiplied by itself is 4 parts 19 minutes 29 seconds, and its root is 2 parts 4¾ minutes: this is the line HZ between the two centres. In the measure in which a quarter of the circle about the right-angled triangle HFZ is 90 parts and its radius 60 parts, the arc HZ is about 1 part 59 minutes; this is the whole anomaly of the motion of the sun that appeared from these observations.[143]

Ar. p. 67

Next we seek the amount of the arc BṬ of the ecliptic; when it is known, the remaining arc ṬA is known in amount. For the point ʿ is the point of the farthest distance of the sun's eccentric sphere: when we drew the line HZ through the two centres and extended it to the ecliptic, it cut the circle KLMN at the point ʿ and the ecliptic at the point Ṭ. So we wish to know the ratio of the line HZ to the line HṬ, which is the radius, and the amount of the arc BṬ of the ecliptic. It has been shown that the line HZ is 2 parts 4¾ minutes in the measure in which the radius is 60 parts; and in the measure in which the radius is 60 parts the line HṬ is that too, since it is equal to the line HB. So the line HṬ measures HZ about 28 5/6 times. Further, as was shown first, if the line HZ is made 60 parts, the line FZ is in that measure about 8 parts 4 minutes; for this is what results when it is multiplied 28 5/6 times. Or, if you wish, multiply the line FZ by the line HṬ, the radius of the sphere: by the order of the multiplication the product is 16 degrees 45 minutes. Divide this by the line HZ, which has been shown to be 2 parts 4¾ minutes; the result is 8 parts 4 minutes, which is the sine of the amount of the angle BHṬ. So the arc BṬ is about 7 parts 43 minutes. Clearly the point of the farthest distance of the eccentric sphere, which is the point ʿ, falls 7 parts 43 minutes from the summer solstitial point towards the preceding side of the ecliptic, that is, 82 parts 17 minutes from the beginning of Aries. This is what we wished to show.[144]

[143] «The two arcs KL and SM»: the arc meant is LA K, as the next words say and Nallino's translation has it. The greatest equation of the sun is 1;59,10 in al-Battānī's tables, and Ibn Yūnus gives the same value for him (Nallino).

[144] The farthest distance (apogee) of the sun is thus at Gemini 22° 17′. Ptolemy (*Almagest* III 4) put it at Gemini 5° 30′.



The observation on which we based this measurement was in the year 1194 of the years of Dhū l-Qarnayn, when we observed the course of the sun from the beginning of Aries to the beginning of Cancer and to the beginning of Libra.[145]

Ar. p. 68

What remains for us is to distribute this anomaly over the degrees of the signs, to find its share degree by degree and to enter it in the tables, so that the equation of the motion of the sun is easily found when it is needed. Ptolemy has shown and made clear that the irregular motions are seen and supposed in two ways. One is to imagine that the star has a sphere whose centre is the centre of the ecliptic, and on this sphere another sphere attached to it, whose centre runs on the circle of the first sphere and turns on it. This second sphere is a small sphere that does not surround the earth, and the large sphere is the one that carries the centre of the small sphere in the order of the signs, by the amount of the motion in longitude that is known for the star in the order of the signs, in either of the two ways. The star itself moves on the epicycle, which is the small sphere, either towards the preceding side or towards the following side; or this small sphere carries the star to either side, whichever it is. This motion is the motion of the proper anomaly of the star. The second of the two ways is to imagine that the star has a sphere whose centre is the centre of the ecliptic, and another sphere of the same size, whose centre is not the centre of the first sphere but outside it, and whose circle cuts the circle of the first sphere in two places; the star is on this eccentric sphere, the sphere carrying the star or the star moving on it. Whichever of the two ways you imagine, it comes to the same for this anomaly and what appears from it. We begin with the first way and give an example of it.[146]

Ar. p. 69

We draw a figure for the circle of the signs, ABGD, about the centre H, and we take the centre of the epicycle to be at first at the point A. We draw about it a circle for the epicycle, ṬZ, and draw the diameter AG and extend it to the point Ṭ, which is the point of the farthest distance of the epicycle. We take the position of the sun on the epicycle to be the point Z, draw from it a perpendicular to the line AṬ, write M at its foot, and draw also the line AZ, which is equal to the line AṬ, since each of them is the radius of the circle of the epicycle. It has been made clear earlier in this chapter that the radius of the epicycle is the line HZ found in the first figure, which is 2 parts 4¾ minutes. Since what we have described is clear, consider the motion of the sun on the epicycle towards the opposite of the order of the signs, that is, towards the preceding side, or the epicycle carrying the sun to this side, every day by as much as the mean motion of the sun in a day, in the measure in which the circle of the epicycle is 360 parts. The mean motion of the sun

[145] That year began on 1 September 882 (Nallino).

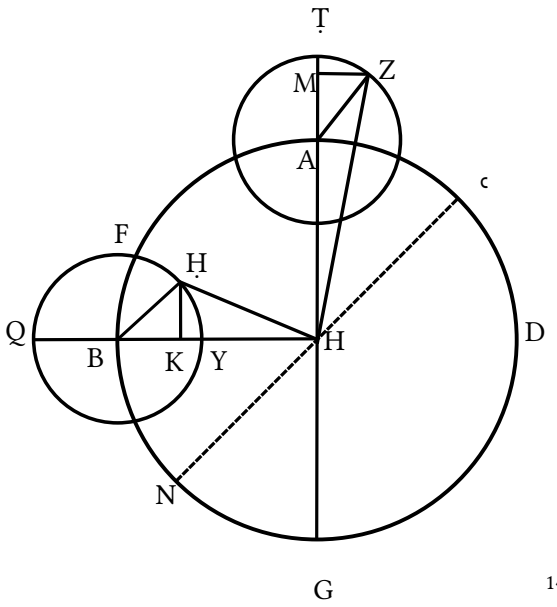
[146] *Almagest* III 3. The «sphere» on which the epicycle is carried is the circle of the deferent.

that appears by measurement is the motion of the centre of the epicycle towards the following side, which is the motion we have posited, also in the measure in which the circle ABGD is 360 parts. Now we take the arc ṬZ, between the sun and the point of the farthest distance of the epicycle, to be 30 parts in the measure in which the epicycle is 360, draw the line HZ in this figure, and seek the arc of the line ZM, which is the anomaly of the motion of the sun there. It has been shown that the line HA is the radius of the parecliptic sphere, 60 parts by the measure on which we have based the computation, in which the diameter AG is 120; so the line HṬ, from the centre of the parecliptic sphere to the point of the farthest distance of the epicycle, from which the motion on the epicycle begins, is 62 parts 4 minutes 45 seconds. Since the triangle ZMA is right-angled, AZ multiplied by itself is equal to AM and ZM, each multiplied by itself, added together, and the angle MAZ is known; so the line ZM is known, and when ZM is known, the remaining side AM of the triangle is known, which is the sine of what the angle ZAṬ and the arc ZṬ fall short of a quarter of their circle. So the line HM is known in amount; and the triangle ZMH is right-angled, and its line HZ, which subtends the right angle, is known for this reason, so that the line ZM is known in amount in its measure, and the arc on it is the arc of the anomaly. If the arc ZṬ is 30 parts, as we took it, its sine is also 30 parts in the measure in which the line AZ, the radius, is 60 parts; but in the measure in which the line AZ is 2 parts 4¾ minutes, the line ZM is 1 part 2 minutes 22½ seconds, the remaining line AM 1 part 48 minutes 2 seconds, and the line HM 61 parts 48 minutes 2 seconds. Clearly then [the line HZ] is about 61 degrees 48 minutes 35 seconds. In the measure in which the line HZ is only 60 parts, the line ZM is 1 part 33 seconds, and the arc on it about 57 minutes 49 seconds; this is the amount of the arc ṬZ, which is the anomaly of the motion of the sun. So the arc ʿA of the ecliptic is 29 degrees 2 minutes 11 seconds, whereas it was 30 parts; for the centre of the epicycle has moved from the point ʿ to the point A as the sun has moved on the epicycle from the point Ṭ to the point Z.[147]

Ar. p. 70

Further, we take the centre of the epicycle at the point B and draw about it a circle for the epicycle, ḤYQ. We take the position of the sun on it to be the point Ḥ, and the arc QḤ that the sun has passed over from the point Q, the farthest distance, to be 150 parts; so the arc YḤ that remains, from the position of the sun to the point of the nearest distance, is also 30 parts. We draw the line HḤ and the perpendicular ḤK. Clearly again each of the triangles BKḤ and ḤKH is right-angled, and each of the two sides BḤ and BH is known: BḤ is the radius of the epicycle and BH the radius of the ecliptic. The angle ḤY and the arc ḤY are given, so the perpendicular ḤK is known as well. If the arc ḤY is 30 parts, as we took it, its sine is also 30 parts, and the arc on KB, which completes the quarter, is 60, whose sine is 51 parts 57 minutes 41 seconds. In the measure in which the line BḤ is 2 parts 4¾ minutes, the perpendicular KḤ is 1 part 2 minutes 22½ seconds, and the line BK that remains is, in that measure, 1 part 48 minutes 2 seconds; so the line HK is about 58 degrees 11 minutes 58 seconds.

[147] The parecliptic sphere (*al-falak al-mumaththal bi-falak al-burūj*, «the sphere made like the ecliptic») is the sphere concentric with the ecliptic, in the plane of which the deferent lies (Nallino). «The arc of the line ZM» is the arc of the circle about the right-angled triangle of which ZM is the sine. The words in brackets are supplied by Nallino.



[148]

The line HḤ therefore comes to about 58 parts 12 minutes 34 seconds. But in the measure in which the line HḤ is 60 parts, the perpendicular KḤ is 1 part 4 minutes 17 seconds, and the arc on it 1 part 1 minute 24 seconds, in the measure in which the circle about the right-angled triangle BKḤ is 360; this is the arc of the anomaly, which is the arc ḤY. So the arc NB of the ecliptic is 31 parts 1 minute 24 seconds. This is what we wished to show.[149]

Ar. p. 71

He said: We show it also in the second way, which is seen on the eccentric sphere. We draw a circle for the ecliptic, ABG, on the diameter AG about the centre H, and a circle for the eccentric sphere, ZMḤ, about its centre, the point Ṭ, so that the diameter AG passes through the two centres. The point Z is then the farthest distance from the parecliptic sphere, and the point Ḥ the nearest distance from it. We take the position of the sun on the eccentric sphere to be at first the point M, the arc ZM that the sun has moved on the eccentric being 30 parts; so the angle ZṬM is also 30 parts. It has been shown that the line HṬ between the two centres is 2 parts 4¾ minutes. Since this is as we have drawn, we draw the line ṬM, which is the radius of the eccentric sphere, and the line HM; we extend the line ṬM in a straight line to the point L, and draw from the point L a perpendicular to ML, the perpendicular LH. The triangle ṬLH is right-angled, and its angle LṬH is equal to the given angle ZṬM. The arc on HL of the circle about the triangle ṬLH, the circle being 360, is 30 parts, and its sine also 30 parts in the measure in which the line ṬH between the two centres is 60 parts; and the line LṬ that completes the quarter remains 51 degrees 57 minutes 41 seconds, since the arc LṬ completes the quarter, that is, is 60 parts. But in the measure in which the line ṬH between the two centres is 2 parts 4¾ minutes, the line HL is 1 part 2 minutes 22½ seconds, and the line LṬ that completes the quarter 1 part 48 minutes 2 seconds; so the whole line LM is 61 parts 48 minutes 2 seconds. The triangle MLH is right-angled, so its line HM, which subtends the right angle, is known: it is 61;48,35. But in the measure in which the line HM is 60 parts, HL is 1 part 33 seconds, and the arc on it 0;57,49, the circle about the triangle ṬLH being 360. So the arc AB of the ecliptic remains about 29;2,11.[150]

Ar. p. 72

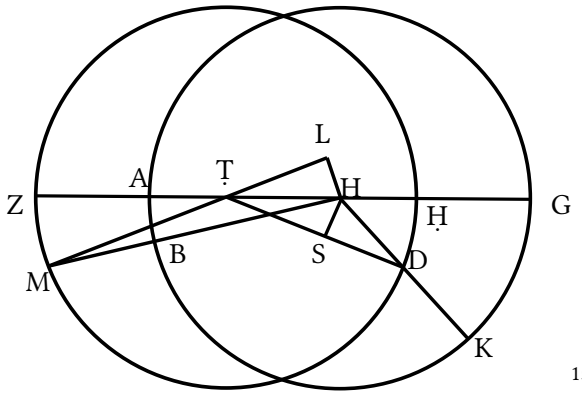
Further, we put the sun at the point D of the sun's eccentric sphere and take the arc ZD to be 150; so the arc DḤ that remains, from the position of the sun to the nearest distance, is also 30 parts. We draw the two lines HK and ṬD, each of which is the radius of its sphere, and draw also

[148] In the figure the sun on the first epicycle is printed «ن», N; the text calls it Z (ز). The dashed line N^{c} is the diameter of the ecliptic parallel to the line of the farthest and the nearest distance; F is not named in the text.

[149] For the arc NB the codex has FB; the reading NB is Nallino's.

[150] The translation reads «33 seconds» with the codex. Nallino prints «33 minutes». The codex is right: HL is 1;2,22½ × 60 / 61;48,35 = 1;0,33 parts, the value found for the same line ZM in the first way, and only this gives the arc 0;57,49 that follows.

the perpendicular HS. Since the triangle ṬSH is right-angled, and its side HṬ between the two centres is known, and the side HS and the angle DṬḤ are known, the side ṬS and the remaining angle ṬHS are known; so the line DS that remains is known, and HD, which subtends the right angle of the right-angled triangle HSD, is known. Since the arc DḤ and the angle ḤṬD are known and given, 30 parts as has been shown, their sine is also 30 parts; likewise the arc HS of the circle about the right-angled triangle HSṬ is 30, if this circle is 360, and its sine, which is the perpendicular HS, is also 30 parts in the measure in which the line HṬ, the radius of this circle, is 60 parts. But in the measure in which the line ṬH is 2 parts 4¾ minutes, the perpendicular HS is 1 part 2 minutes 22½ seconds, and so the line SṬ that remains of the sides of the triangle is 1 part 48 minutes 2 seconds. The line ṬD, the radius of the eccentric sphere, is 60; when SṬ is taken from it, the line SD that remains is 58 parts 11 minutes 58 seconds, and the line HD, which subtends the right angle of the triangle HSD, is about 58 parts 12 minutes 34 seconds. In the measure in which the line HD is 60 parts, the perpendicular HS is 1 part 4 minutes 17 seconds, and the arc on it 1 part 1 minute 24 seconds; this is the amount of the anomaly. So the arc KG of the ecliptic is about 31 parts 1 minute 24 seconds. This is enough about this anomaly; it is what we wished to show.[151]



152

Ar. p. 73

He said: In this way we have computed it degree by degree and entered it in the tables, counted from the point of the farthest distance. In this way also the single equation of the moon is computed, and the mean equation of the planets, which is the radius of the epicycle of each of them, when its sine is taken and it is distributed in this way. Whenever you wish to compute this, look at the degrees that the star, or the sun, or the moon, has passed over on the epicycle from the point of the farthest distance, which is called the anomaly of the sun, the moon and the other stars. If this anomaly is less than 180, work with it; if it is more than 180, subtract it from 360 and work with the remainder. The procedure is to take the degrees that you obtain in either way. If they are less than 90, take their sine and the sine of what remains of them to 90, and multiply each of the two sines by the radius of the epicycle of the star, which is the sine of the whole equation. Divide each product by the radius. Add what results for the sine of the complement of the degrees to the 60 that are the radius, multiply the sum by itself, add to it what results for the sine of the degrees multiplied by itself, take the root of the sum and keep it. Then return to what resulted for those degrees, multiply it by the radius, and divide it by the root that you kept. If the degrees with which you work are more than 90, take 90 from them, take the sine of the remainder and the sine of what remains of it to 90, multiply each of the two sines by [the radius of] the epicycle, and divide each product by the radius. Subtract what results for the degrees from 60, multiply the remainder by itself, add to it what results for the complement

[151] The text has «1 part 4 minutes 24 seconds» for the arc of the anomaly; the arc whose sine is 1;4,17 is 1;1,24, as in the first way, and the 31;1,24 that follows requires it (so Nallino's translation).

[152] In the figure the point at the right end of the line of the centres is printed «ب», B, a second time; the text calls it G (ج).

of the degrees multiplied by itself, take the root of the sum and keep it. Then return again to what resulted for the complement of the degrees, multiply it by the radius and divide by the root that you kept. Find the arc of the quotient. The arc you obtain, in whichever of the two ways, I mean from the first way, which results from the division of the first quantity, or from the division of this second quantity, is the share of those degrees of the anomaly with which you worked, for whichever star you intended, in the anomaly of its motion: it is the equation of the star. The radius of the epicycle of the sun is 2;4,45, the radius of the epicycle of the moon 5;15, the radius of the epicycle of Saturn 6;29,50, the radius of the epicycle of Jupiter 11;30,0, the radius of the epicycle of Mars 39;27,22, the radius of the epicycle of Venus 43;9,0, and the radius of the epicycle of Mercury 22;30,30. This is what has been shown by the observations, and the computation agrees with it; it is the sine of the mean equation of all the stars, if God wills.[153,154]

Chapter 29. On finding the inequality of the days with their nights, and on converting the one kind into the other

Ar. p. 73
Ar. p. 74

He said: Many people, and the common folk, think that the days with their nights are equal in length, I mean that each day with its night is 24 hours for them; but in truth it is not so. One mean day with its night is the rising of all the 360 times of the equator from the horizon circle or the meridian circle, together with the further times of the equator that rise with the 59 minutes that the sun moves by its mean motion in a day and night. One irregular day with its night is the rising of all the 360 times of the equator together with what further rises with the irregular course of the sun in a day and night, which is necessarily either more or less than 59 minutes. A beginning taken from the horizon circle changes and differs in every place with the difference of the ascensions of the signs there, while the beginning from noon is fixed in one state, without change or difference, because the signs pass through the meridian circle alike in every locality. Therefore the beginning of the days in the reckoning of the stars and the computation of their positions is not taken from sunrise or sunset, but from noon or midnight. Further, all the motions of the stars set down in the tables are set down for mean days of equal length. If what accumulates of the difference between the irregular days with their nights and the mean days with their nights is neglected, in the course of the sun and of the other stars it does not amount to anything from which a perceptible error appears; but in the moon it is very evident, because of its speed.

For the greatest amount that accumulates between the irregular days and the mean days is about half an hour, and in this time the motion of the moon is at times 18 minutes; between the days that exceed the mean days and the days that fall short of them it is twice that. This anomaly is composed of two parts: one is the anomaly of the motion of the sun, which is the equation; the other is the anomaly of the passage of the signs through the middle of the sky, since they do not all rise there with the same amount. The greatest amount that accumulates from the anomaly of the motion of the sun is about 3 parts and a quarter and a tenth, and the greatest amount that then accumulates from the passage of the signs through the middle of the sky about 4 parts and a quarter and a fifth; so what accumulates from the two together is 7 parts 48 minutes, which is about half an hour and a fifth of a tenth of an equal hour. The place of the shortfall is from about two thirds of Aquarius to about the beginning of Scorpio, and the place of the excess from about

[153] «Anomaly» renders al-Battānī's *ḥāṣṣa*, the portion of motion that falls to the star on its epicycle; most other astronomers call it *al-khāṣṣa*, «the proper [motion]» (Nallino's glossary). In modern terms, with r the radius of the epicycle and a the anomaly, the rule is $\sin q = r \sin a / \sqrt{((60 + r \cos a)^2 + (r \sin a)^2)}$, the radius being 60; it is exact, and its second form, for $a > 90°$, is the same rule written with $a - 90°$. The single equation of the moon is its first anomaly; the mean equation of a planet is the equation produced by its epicycle at its mean distance.

[154] Nallino corrects the codex in four of the radii: the sun (codex 2;5,45), the moon (6;15), Mars (19;25,22) and Mercury (29;30,30). For Mars the printed text has 39;25,22; the translation follows his emendanda, printed in Part II, and his translation, which read 39;27,22. Ptolemy's values, in the same parts, are: sun 2;30 (the eccentricity), moon 5;15, Saturn 6;30, Jupiter 11;30, Mars 39;30, Venus 43;10, Mercury 22;30.

the beginning of Scorpio to about two thirds of Aquarius.[155]

Ar. p. 75

In this book we have set out the mean motions in the tables on the basis that the assumed position of the sun is at 18 degrees 19 minutes [of Aquarius] by its mean motion, and at 20 parts of it by its true motion, in which it is seen; and with this day and its night you compare the other days of the year in this book.

He said: If you wish to convert irregular days into the mean days with which the mean motions are taken from the tables, take [the arc] between the first, assumed mean position of the sun and the second position to which it moves, also by its mean motion, and keep the number of degrees. Then take also [the arc] between its first true position, in which it was, and the second position to which it moves by its true motion, in times of the ascensions of the signs in the right sphere. If the number of these times is greater than the degrees of the mean motion that you kept, find what part of an equal hour the difference between them is, and add it to the given irregular days; if the number of times is smaller, subtract it from them. The days that you obtain after the addition or the subtraction are those converted from the irregular days into the mean days, in whichever of the two reckonings is used for the beginning of the days, I mean from noon or from midnight. If you wish to convert from the mean days obtained from the tables into the irregular days that are found, do the reverse: add the difference to the mean days when the number of times is smaller, and subtract it from them when the number of times is greater. The mean days after the addition or the subtraction are those converted from the mean days into the irregular days that are found.

On the basis that we have laid down in this book for the assumed position of the sun, the number of times is always the smaller, until a long period of time has passed in which the position of the farthest distance of the sun, as we found it on the ecliptic, has changed much, and so what arises from the anomaly of the motion of the sun has changed. Since this is as we have described, we have added 18 minutes to the mean position of the moon at the root of the computation. We have taken the share of each degree of the signs in the amount of the inequality of the days with their nights and entered it in the tables of the ascensions of the right sphere, in the column that follows the ascensions in each sign. When we take what stands opposite the true degree of the sun in that column, find its amount in equal hours and subtract it from the irregular days, what remains is the mean days with which the motions are found from the tables; and when it is added to the mean days, the result is the irregular days that are found by measurement.[156]

Chapter 30. On the description of the spheres of the moon, the anomaly of its motions, the increase and decrease of its light, the causes of the two eclipses, and the distances of the two luminaries from the centre of the earth, their diameters and the sizes of their bodies when they are compared with the earth[157]

Ar. p. 76

He said: When the motions of the moon were investigated, two anomalies were found in it. One of them is single by itself: it appears alone, only at the times of the conjunctions with the sun and the oppositions to it that take place by the mean course of the sun and the moon, and it depends on the position of the moon on its epicycle. The second anomaly appears through the moon's distances from the sun; it is compounded with the first anomaly, so that the two act together. This is grasped by demonstration with geometrical lines, by imagining that the moon has four spheres. The first is made like the ecliptic: it lies level beneath it, moves with its motion without deviating from it, and has its centre at the centre of the ecliptic. The second sphere is inclined to it towards the north and the south; it is as large as this parecliptic sphere, and the two have one

[155] This is the inequality that is now called the equation of time. Al-Battānī counts it from its least value: 7;48 times of the equator are 31 minutes of an hour, close to the range of about 31 minutes over which the equation of time varies.

[156] The root of the computation is the epoch of the tables.

[157] The last words, «with the earth», are Nallino's conjecture: only the tops of the letters can be read in the codex.

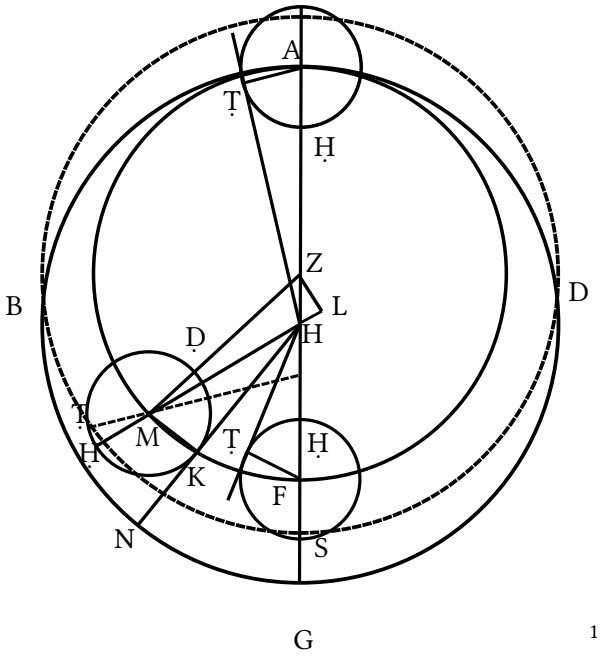
centre. Its greatest inclination to either side is about 5 parts, which is the amount of the moon's distance from the belt of the signs in latitude. The motion of this inclined sphere, against the order of the signs, is about 3 minutes a day. This is the motion of the two nodes. One of them is called the head: from it the moon passes in latitude towards the north. The other is called the tail: from it the moon passes towards the south. These two nodes are the places where the inclined sphere and the parecliptic sphere intersect.

Ar. p. 77

Inside this inclined sphere is a third sphere, whose centre lies outside the centre of the other two. It hangs from the inclined sphere and touches it at a point that is its highest place; that point is called the farthest distance from the earth. It moves within the inclined sphere against the order of the signs, about 11 parts 12 minutes a day. There is a fourth sphere, called the moon's own epicycle. Its centre is on the circle of the eccentric sphere, and this centre moves on the circle of the eccentric sphere in the order of the signs about 24 parts 23 minutes [a day]. It begins its motion from the point of the farthest distance of the eccentric sphere, which is taken together with the mean position of the sun. Therefore the centre of the epicycle falls on the point of the farthest distance twice in the lunar month, once at the mean conjunction and again at the opposition. The moon moves on the circle of the epicycle about 13 degrees 4 minutes a day, against the order of the signs on the epicycle, beginning from the point of the farthest distance that is seen from the centre of the eccentric sphere. Since the centre of the epicycle falls on the circle of the inclined sphere at these two times, as we have said, nothing prevents us from taking it, in this way, to move on the circle of the inclined sphere about 13 degrees 14 minutes a day; this is its motion in latitude. The node, at the intersection of the two spheres, carries it back against the order of the signs by the 3 minutes that we said are the motion of the inclined sphere, so that its motion in longitude, in the order of the signs, remains about 13 degrees 11 minutes a day. The motion of the moon on the epicycle is the first motion mentioned.[158]

It is clear from what we have described that nothing enters the motion of the moon from the eccentric sphere at these two times, since then the moon is not distant from the mean position of the sun or from the point opposite it. So nothing of the second anomaly is mixed with the single anomaly until the moon moves away from the sun; at those distances the second anomaly, which comes from the eccentric sphere, is compounded with it according to the moon's distances from the sun. This is the figure of the four spheres of the moon; the demonstration will be clear to you from the example of their drawing.

[158]The daily motions are Ptolemy's, rounded. The 24;23 is twice the daily excess of the moon's motion in longitude over the sun's, and the eccentric moves back by 24;23 − 13;14 + 0;3 = 11;12 (Nallino). The 13;4 is the daily motion in anomaly, 13;3,53,56,... in *Almagest* IV 3.



159

Ar. p. 78

He said: I drew a circle for the parecliptic sphere, marked ABGD, with H at its centre, and a circle for the inclined sphere, also marked ABGD; let its centre also be the point H, as is the case on the globe, which turns on its two poles. We draw the diameter AS and mark on it the centre of the eccentric sphere at the point Z, between the centre of the two spheres and the sign A. About the centre Z we draw a circle for the eccentric sphere, marked AMF, with the radius AZ. We take the arc AM, as much as we wish, as the motion of the centre of the epicycle from the point A, which is the point of the farthest distance and the position of the sun, to the point M. We take the point M as the centre of the epicycle and draw about it a circle for the epicycle, marked ḤṬḌK. We draw the two lines HMṬ and ZMḤ. Then the point Ṭ of the epicycle is the position of the farthest distance that is seen from the centre, the point H, which is the centre of the earth and the centre of the ecliptic, and Ḥ is the position of the true farthest distance, which is seen from the centre Z, I mean from the centre of the eccentric sphere. So the arc ṬḤ is clearly the anomaly of the motion of the moon in its proper course on the epicycle; it is the anomaly set down in the third column of the tables of the equation of the moon. We take the motion of the moon on the epicycle to be from the point Ḥ to the point Ṭ, then to Ḍ, and make the position that it has reached on the epicycle the sign K. We draw the line HKN touching the epicycle, and draw the line MK, the radius of the epicycle as displaced by the distance of its centre from the point A of the eccentric sphere. Since the moon is on the line that touches the epicycle, the radius of the epicycle is the whole single anomaly together with what is compounded with it of the second anomaly, according to the distance of the moon from the sun, which is at the point A. It is clear in this figure that when the moon is in the first half of the epicycle, on which are ḤṬḌ, its true position on the ecliptic, which is seen from the centre H, is less than its mean position in longitude, which is the position of the centre of the epicycle; so the equation is subtracted from the mean position of the moon when the anomaly is less than 180 degrees. When it is in the second half, on which are ḌKḤ, its true position on the ecliptic is greater than its mean position; so when its anomaly is more than half the circle, the equation is added to the mean position of

[159] The dashed circle about Z, as large as the circle of the ecliptic, is not mentioned in the text; Schiaparelli found no use for it (Nallino's commentary). The point L and the lines ZL and LH belong to the computation on p. 80.

the moon, if God wills.[160,161]

Ar. p. 79

As for the single equation, which appears at the times of the conjunctions and the oppositions and is set down in this book of ours in the second column of the tables of the equation, we have shown how it is computed, in the way in which we worked in computing the equation of the sun and distributing it in the tables. The greatest amount that this single anomaly of the moon reaches is 5 parts 1 minute, and its sine, which is then the radius of the epicycle, is about 5¼ parts: this is the ratio of the 60 of the radius to the 5¼ parts. This is what Ptolemy stated and showed from the lunar eclipses, in which the true position of the moon must be opposite the true position of the sun, in which the sun is seen on the ecliptic. At those times [the arc] between the position of the moon by its mean course and the degree opposite the true degree of the sun is the single anomaly of the motion of the moon according to the position of the moon on the epicycle, and so the whole of this single anomaly becomes known. We too have observed many lunar eclipses, established their exact times and middles, and found the amount of this single anomaly to be as we have described.[162]

Ar. p. 80

As for the second anomaly, which is found from the distance of the moon from the sun, the greatest amount that they found reaches about 2⅔ parts. Compounded with the 5 parts 1 minute of the single anomaly, the two anomalies together come to about 7 parts 40 minutes. This happens when the centre of the epicycle is at the point F, at a distance of half the circle from the point A; the radius of the displaced epicycle is then about 8 parts, the sine of the 7⅔ parts. From what we have described it becomes clear that the line HZ between the two centres is 10 parts 19 minutes. The demonstration of this is as I shall describe. We again draw a circle for the epicycle, marked ṬḤ, about the point A, which is the farthest distance on the eccentric sphere, and draw the line HṬ touching the epicycle, and the line AṬ. Since the moon on the tangent line completes the whole single anomaly, which has been shown to be 5;1 in the measure in which the four right angles are 360, its sine is 5;15 in the measure in which the radius is 60 parts, the radius of the parecliptic, inclined, sphere in this figure. Again we take the centre of the epicycle at the sign F, which is the point of the nearest distance of the eccentric sphere, draw about it a circle for the epicycle, also marked ṬḤ, and draw the line HṬ touching the epicycle, and the line FṬ. Since the moon on the tangent line, I mean at the position Ṭ, completes the whole of the two anomalies, which is 7;40, its sine is about 8 in the measure in which the four right angles are 360 and the radius, the line HA, is 60. The line FṬ is equal to the line AṬ, and it was shown that the line AṬ is 5¼ parts in the measure in which the line HA is 60 parts. But since the centre of the epicycle has come to a place where the measure changes, because of its nearness to the point H, which is the centre of the earth and the true place of sight, [FṬ] appears in the measurement as about 8 parts, in the measure in which the line HA is 60. So in the measure in which the 8 parts are 60 parts, the 5¼ parts are 39 degrees 22 minutes, and that is the amount of the line HF, from the centre of the earth to the point of the nearest distance of the eccentric sphere. Likewise, by the inverse of this ratio, in the measure in which the 8 parts are 5¼ parts, the 60 parts are 39 parts 22 [minutes]. Since the amount of the line HF has become clear, 39 parts 22 minutes, when it is added to the line HA, which is 60, it comes to 99;22, which is the whole diameter of the eccentric

[160] In the printed figure the line from H through M ends at Ḥ, and Ṭ lies on the dashed line drawn through M from the point of the diameter that is as far from H as Z is, on the other side. The true farthest distance of the epicycle is seen from that point in Ptolemy's theory (*Almagest* V 5), and al-Battānī himself uses it on p. 82. The text, with the codex and Plato of Tivoli, draws HMṬ and ZMḤ instead and has the true farthest distance seen from the centre Z of the eccentric; Nallino's translation, following Schiaparelli, corrects the text to agree with the figure. The arc ṬḤ is the same in either reading.

[161] The «displaced» radius of the epicycle is its radius measured in parts of which the distance of its centre from the earth, HM, is 60; it grows as the centre moves from A towards F: 5;15 at A, 7;12 at the doubled distance of 120, about 8 at F, as is shown below.

[162] *Almagest* IV 6. The single equation is computed as the equation of the sun is in chapter 28.

sphere; and half of it, the radius of the eccentric sphere, is 49;41.[163,164]

Ar. p. 81

Since the radius of the epicycle according to its displacement from the sun is known, and the distance between the centres and the radius of the eccentric sphere are known, what remains to complete our knowledge of this is to explain how to compute the arc ṬḤ, which is set down in the third column, and to explain what is compounded of the single equation with the second between the two distances, in the way in which I have set it out in the tables and entered it in the fourth and fifth columns. In the fourth: if these 2 degrees 40 minutes, which are entered in the fifth, are taken as 60, how much is compounded of them, and what is its ratio to the 60. This is known as I shall describe. We extend the line MH to the sign L and join the point L to the point Z; then the ratios of the sides of the triangle MLZ are known, and its angles. We take the arc AM, as Ptolemy took it in this matter, as 120 parts; this is the doubled distance of the moon from the sun. Since we take the ratio of the sines to the radius, the angle LZH is 30 parts, and the angle ZHL, which completes the quarter, 60 parts, in the measure in which the circle that goes round the triangle ZHL is 360 parts. Also the sine of the angle LZH is 30 parts, and the sine of the angle ZHL about 51;58, in the measure in which the line HZ is 60; but in the measure in which the line HZ is 10;19, the line HL is about 5;10 and the line ZL 9;16. Again, when the line HKN in the figure touches the epicycle and the position of the moon on the epicycle is the point K, the first anomaly together with what is compounded with it of the second anomaly is greatest. Since the line MK is the radius of the epicycle and the line ZM the radius of the eccentric sphere, by which you know these amounts, the ratio of LM is known from the ratio of ZM and ZL, and so the whole line ML is 48;53. When the line HL, which has been shown to be 5;10, is taken from it, the line HM from the centre remains 43;43. The line MK, the radius of the epicycle, has been shown to be 5;15; but in the measure in which the line HM from the centre is 60, the line KM, the radius of the displaced epicycle, is about 7 parts 12 minutes, and the arc on it about 6 parts 54 minutes; that is the amount of the arc MK. When the 5 parts 1 minute, the amount of the whole single anomaly, are taken from it, what is compounded with it of the second anomaly remains, 1 part 53 minutes. If the 2⅔ degrees, approximately, are 60, this 1 degree 53 minutes is 42;38 of the 60; that is what is set down under 120 in the fourth column, and it results from the ratio of the minutes to the one part, which is the ratio of 0;42,38 to 60. When these 42 and 38 are extended and made 60, the 1 part 53 minutes become 2 parts 39 minutes, which is what is set down in the fifth under 120.[165,166,167]

We also find [the arc] between the true farthest distance and the mean distance, which is the arc ṬḤ, in the figure set out, by taking the doubled distance of the moon from the sun by their mean course as 90;30, as Ptolemy took it in the figure to which his demonstration led, as we have said, so that the motion of the moon on its epicycle from the point Ṭ is 333;12. We draw this circle as an example of it.[168]

[163] The greatest second anomaly is Ptolemy's (*Almagest* V 3), and so is the distance between the centres (V 4). His extreme equations of the moon, 5;1 at the syzygies and 7;40 at the quadratures, compare with the modern 5;2 and 7;34 (Schiaparelli, in Nallino's commentary); the second anomaly is the inequality now called the evection.

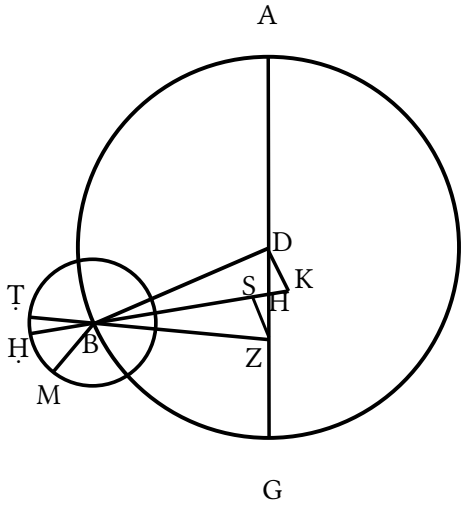
[164] «Half» is supplied by Nallino. After «the point F» Plato of Tivoli has «which is the nearest distance of the eccentric sphere», words that Nallino takes to have fallen out of the codex. At the end of the paragraph Plato also has «so the distance between the two centres is 10;19 in the measure in which AH is 60 parts», that is, 60 − 49;41.

[165] The text prints LHZ for the angle of 30 parts, in this sentence and the next (the codex has AHM); the angle of 30 parts is the angle at Z, LZH, as in Nallino's translation. Ptolemy (*Almagest* V 7) makes the angle AHM, not the arc AM, 120 degrees, which is what gives the angle ZHL of 60 (Schiaparelli).

[166] On the base HZ = 10;19 the sine of 60 parts is 8;56 (10;19 × 51;58 / 60), and the line ML = 48;53 that follows agrees with it, not with the text's 9;16, which would give 48;49. Plato of Tivoli has 18;56.

[167] The 42;38 is reckoned with the greatest second anomaly 2;39, that is 7;40 − 5;1, rather than with the rounded 2⅔: 60 × 1;53 / 2;39 = 42;38, while 60 × 1;53 / 2;40 = 42;22,30. In al-Battānī's tables the 2;39 of the fifth column stands under 110, not 120 (Nallino). Schiaparelli, in Nallino's commentary, takes the last sentence for an interpolation by a reader who did not understand the computation.

[168] These are the values of Ptolemy's example in *Almagest* V 6.



Ar. p. 82

He said: This is the circle of the eccentric sphere, marked ABG, about the centre D, with the diameter AG, on which the centre of the ecliptic falls, at the sign H. The circle about the point B is the one called the epicycle, marked MḤṬ. We draw the line BM and the line HBḤ, extend the line BH to the point K, and join the point K to the point D. Then the angle KDH of the triangle is the half part by which [the doubled distance] exceeds 90, and the arc HK is half a part in the measure in which the circle that goes round the triangle DKH is 360 parts; its sine is 0;31,25 in the measure in which the line HD, the radius, is 60. The angle KHD remains 89;30, and so the arc KD is 89;30 and its sine about 60. But in the measure in which the line DH between the centres is 10;19, the line HK is about 5 minutes and the line KD about 10;19. Further, because of the difference of the centres that we have described, we make the line HZ equal to DH, the line HS equal to HK, and the line SZ equal to the line DK. The line DB, from the centre of the eccentric sphere to its circle, has been shown to be 49;41 in the measure in which the line MB, the radius of the epicycle, is 5;15. From what we have described, the whole line BK is 48;36. When HK, which has been shown to be 5 minutes, is taken from it, the line HB remains, in that measure, 48;31; and since the line HS is also 5 minutes, the line SB remains, in that measure, 48;26. From the ratio of SZ and SB you know the ratio of the line BZ, which is therefore about 49;31 in the measure in which the line SZ was about 10;19. When the line BZ is extended and made 60, the line SZ is in that measure about 12;32, and the arc on it about 12;1, in the measure in which one right angle is 90; that is the amount of the arc ṬḤ. So the true motion of the moon on the epicycle, which is seen from the centre of the sphere, counted from the point Ḥ, is 345 parts 13 minutes. Therefore, when the doubled distance is less than 180, the arc ṬḤ is added to the anomaly of the moon obtained from the table, and when it is more than 180, the arc ṬḤ is subtracted from that anomaly. For the centre of the epicycle is at first between the point A of the eccentric sphere and the point F, on the side of M; then it passes to its other half, between F and A, on the side of D. These 12 parts 1 minute are what is set down in the third column under 90½ parts.[169,170,171]

Ar. p. 83

He said: As for the motion of the moon in longitude, we find it as it stands in Ptolemy's book, after we add to it what we found in correcting the motion of the sun, and so we have entered it in the tables. Likewise its motion in anomaly is the motion set down in Ptolemy's book for the anomaly of the moon, with nothing added to it or taken from it. As for its motion in latitude, we found that it exceeds in Ptolemy's book by 27 minutes; so we distributed these minutes over the time between him and us, subtracted them from the motion in latitude, and entered in the

[169] Here Z is the point of the diameter that is as far from H as D is, on the other side; the true farthest distance Ṭ of the epicycle lies on the line from Z through B (Ptolemy, *Almagest* V 5).

[170] The line SZ is 10;19 × 60 / 49;31 = 12;30, as Plato of Tivoli has it, and its arc 12;1; the 12;32 of the text would give 12;3.

[171] The letters A, F, M and D of the last sentences are those of the first figure (Nallino).

tables what remained. We have had no need to set down tables of the doubled distance between the sun and the moon, since we know the distance between the sun and the moon by their mean course at any time we wish; when we double it, that is what we use in place of what is obtained from the tables, without addition or subtraction. Likewise we found that the latitude of the moon reaches about 5 parts, which is set down in the seventh column of the tables of the equation, and we entered it as it was; except that its share exceeds, in our time and his, by a half and a quarter of a part, so we subtracted that from its proper course, and what remains is what you find there, if God Most High wills.[172,173]

As for the cause of the lunar eclipse, it is that the sun is greater than the earth and the earth greater than the moon. The rays of the sun go out along the sides of the earth until they end in the air on the other side in the shape of the head of a pine, and so the shadow of the earth is called the pine-shaped shadow; its end lies beyond the sphere of Mercury. When the passage of the moon is at one of the two nodes of its sphere at the times of its opposition to the sun, its centre comes to the centre [line] of the ecliptic: it falls on the belt of the signs, faces the sun across a diameter of the sphere, with no latitude by which it is inclined away from the sun, and is directly opposite it. Then the earth comes between the moon and the sun and screens it from it, so that the moon falls into the pine-shaped shadow of the earth that we have mentioned; its light is darkened, and it is eclipsed according to how far or near its path is to the belt of the signs, which is the middle of the shadow. If it has no latitude at all and is in the node itself, its passage at the middle of the eclipse is on the middle line of the shadow, and its eclipse is then the most complete of eclipses and the longest in duration. For this reason the moon never has its light complete: it receives all the light only when the sun falls on the half of its circle that faces it, and that happens only when it is on the diameter of the sun, with half the sphere between them, which it can reach only under the darkness, at the middle of the eclipse. If it is inclined from the path of the sun in latitude to the north or to the south, less than half the circle of the sphere lies between it and the sun, and its opposition to the sun is not along a straight diameter. So when it is opposite the sun while inclined from its path in latitude, its eclipse is according to the amount of its latitude, up to the greatest latitude at which its circle can touch the circle of the shadow.[174,175,176]

Ar. p. 84

It is clear from what we have said that none of the stars can be eclipsed at their opposition to the sun, since the shadow does not reach their centres; and Mercury does not move as far from the sun as opposition, to fall into the shadow then. But the moon may eclipse the other stars and hide them from sight when its visible path agrees [with theirs] in longitude and latitude, for then it falls on the line that goes out from the eye to the star in question. The stars also eclipse one another, when the lower of them passes under one of those above it and comes onto its course in longitude and latitude. If the sun were equal to the earth, the breadth of the shadow would be the same from its bottom to its top; it would have no end in the air, but would rise without end. The eclipse of the moon would then be of one and the same amount at the top of its epicycle and at the bottom, it would linger in the eclipse longer than it is found to linger, and all the stars would

[172] The correction of the motion of the sun is that of chapter 27; adding it to the moon as well keeps Ptolemy's mean distance of the moon from the sun. In 100 Roman years al-Battānī's mean motion of the moon in longitude exceeds whole revolutions by 307;57,46, Ptolemy's by 307;21,37; Hansen's tables of the nineteenth century give 307;53,40 (Schiaparelli, in Nallino's commentary).

[173] «Its share» (*ḥiṣṣa*) is the argument of latitude; Nallino reads it for the codex's *ḥāṣṣa*, «its anomaly». Schiaparelli explains the ¾ part as the correction of the motion in latitude accumulated over the roughly 1200 years from the epoch of the era of Dhū l-Qarnayn, at the rate of 27 minutes in the roughly 750 years between Ptolemy and al-Battānī (27′ × 1200 / 750 = 43′).

[174] The reading «head» (*jumjuma*) is uncertain (Nallino). *Ṣanawbarī*, «pine-shaped», is the Arabic word for conical.

[175] In this chapter (p. 92) the shadow reaches 254⅔ radii of the earth from the centre of the earth; in chapter 50 the distances of Mercury run from 64⅙ to 166 radii of the earth and those of Venus from 166 to 1070, so that the shadow passes the sphere of Mercury and enters that of Venus (Nallino).

[176] The translation follows Nallino's emendanda (Part II, p. XXX), «its centre becomes the centre of the ecliptic», for the printed «its centre comes to be in the ecliptic»; the codex has «it becomes the centre of the ecliptic».

be eclipsed at their opposition to the sun. If the sun were smaller than the earth, the top of the shadow would be broader than its bottom, and it would rise in the sky without end; the higher the breadth of the shadow rose, the more the moon and the stars would be eclipsed, lingering in the eclipse for days that differ according to their course compared with the course of the sun.[177]

As for the sun, the cause of its eclipse is the moon. When at the times of the conjunctions its centre happens to be seen on the belt of the signs, it comes between the eyes of those who look at the sun and the sun, since it falls on the line that goes out from the eye to the sun, being nearer to the earth than the sun is; just as a small thing can always be seen to hide a large thing when it is nearer to the eyes. The amounts of the eclipse are according to the latitude of the moon as it is seen, up to the limit at which its circle cannot hide anything of the circle of the sun. For this reason the eclipse of the sun has different amounts in places of different latitude, while the eclipse of the moon is seen of one amount in all places.

As for the distances of the sun and the moon, their diameters and the sizes of their bodies compared with the earth, to find them Ptolemy put forward two lunar eclipses, on which he based the measurement. He carried it out on the assumption that the moon hides the whole sun from sight when it is at its farthest distance from the earth at the times of the conjunctions and is seen on the belt of the signs, and he gave the diameter of the sun no perceptible variation, relative to the moon, between its far and near distances from the earth, but made it the same as the moon's. He mentioned in his book no observations of solar eclipses and did not use them, and we do not know what prevented him. But in the amounts of the solar eclipses that we observed we did not see anything that requires the circle of the moon to cover the circle of the sun and hide it from sight in the ratio mentioned. On the contrary, we also found a clear and evident change in the diameter of the sun, relative to the moon, between its farthest and nearest distances, as geometrical measurement requires, although in itself it is small and imperceptible.[178,179]

Ar. p. 85

As our demonstration of what we have described we take two well-known solar eclipses that we observed in our time. In one of them the sun and the moon were in the region of their farthest distance; in the second the sun was in the region of its nearest distance and the moon near its mean distance. The middle of the first eclipse, as we found it by observation, was in the year 1202 of Dhū l-Qarnayn, which is the year 1214 from the death of Alexander, one seasonal hour after noon on the eighth day of Āb, in the city of al-Raqqa, and more than two thirds of the sun was eclipsed to the sight. By our computation the sun at the time of the conjunction was in Leo 20;54 by its mean course and in 19;14 of it by its true course. The moon by its mean course was in Leo 17;7, and in truth, necessarily, with the degree of the sun; so its proper course on the epicycle from the position of the true farthest distance was 333;13. Its mean motion in latitude was 174;43, and in truth 176;51. The visible conjunction, which is the middle of the eclipse, was about an eighth of an hour after the time of the true conjunction; so its motion in latitude at the middle of the eclipse was 176;55, its visible latitude 6 minutes to the south, and its true latitude about 16 minutes to the north. By Ptolemy's computation and on that ratio, the part of the sun eclipsed should have been more than a half and a quarter, and the middle of the eclipse should have been about an hour before the time at which it occurred by measurement.[180,181]

Ar. p. 86

[177] Mercury and Venus are never in opposition; Nallino suggests that «upper» has fallen out after «all the stars».

[178] *Almagest* V 14.

[179] The translation reads «منعه», «what prevented him», with Plato of Tivoli and Nallino's translation, for the printed «صنعه», «what he did» (p. 84, line 18). Bailly and Delambre, cited by Nallino, explain that Ptolemy did not rely on solar eclipses because his lunar theory put the distance of the moon, and with it its parallax and apparent diameter, far from the truth.

[180] 8 August 891 (Nallino). Nallino's translation has «apogee» for «nearest distance» in the description of the second eclipse; the Arabic is right: the sun, at Aquarius 8;35, was 46 degrees from its nearest distance at Sagittarius 22;17.

[181] The translation reads 333;13 for the printed 333;17 (p. 85, line 7), a misprint that the corrigenda do not list: Nallino's apparatus and translation have 13 minutes (the codex has 332;57), and Schiaparelli's recomputation of the eclipse from al-Battānī's tables, in Nallino's commentary, gives 333;12,57. It also confirms Nallino's 17;7 for the mean position of the moon (the codex has 14;7), and puts the true conjunction 1 hour after noon in true time, 47 minutes

The middle of the second eclipse, as we found it by observation in the city of Antioch in the year 1212 of the years of Dhū l-Qarnayn, which is 1224 from the death of Alexander, was about 3⅔ equal hours before noon on the 23rd day of Kānūn II, and the part of the sun eclipsed was a little more than half to the sight. The middle of the eclipse at al-Raqqa, as its time was taken for us, was 3 hours and less than half an equal hour before noon, and less than two thirds of the sun was eclipsed to the sight. By our computation the mean position of the sun at the time of the true conjunction was Aquarius 7;9, and in truth 8;35 of it. The moon by its mean course was in Aquarius 12;49, and in truth, necessarily, with the degree of the sun; so its course in anomaly from the point of the true distance on the epicycle was 126;37, its mean motion in latitude 173;25, and in truth 169;11. The middle of the eclipse to the sight was about a half and a third of an equal hour before the time of the conjunction; its visible latitude was about 10 minutes and its true latitude about a degree less a minute, but its motion in latitude at the middle of the eclipse was 168;45. By Ptolemy's computation and on those ratios the whole sun should have been eclipsed, and the middle of the eclipse should have been about two hours later than the time found. This is a defect of a kind that may not be tolerated in computation at all.[182,183,184]

We also mention two lunar eclipses of our time on which measurement can be based for what we intend in this kind [of inquiry]. The first of them was in the year 1194 of the years of Dhū l-Qarnayn, which is the year 1206 from the death of Alexander, on the 23rd day of Tammūz. We observed it, and the middle of the eclipse in the city of al-Raqqa was 8 equal hours and a little after noon of that day; a little more than a half and a third of the diameter of the moon was eclipsed. By our computation the sun was in Leo 5;21 by its mean course, and in truth in 4;2 of it; the mean position of the moon was Aquarius 8;45, and in truth, necessarily, opposite the degree of the sun. So its motion in anomaly from the point of the mean farthest distance on the epicycle was 113;0, and in truth 114;9; its mean motion in latitude was 190;49, and in truth 186;5, so that its latitude at the time of the opposition was about 32 minutes to the south. By Ptolemy's computation the part of the moon eclipsed should have been a half and a third and an eighth of its diameter, and the time of the middle of the eclipse should have preceded the time at which we found it by about a half and a quarter of an equal hour.[185,186,187]

Ar. p. 87

The second eclipse was in the year 1212 of the years of Dhū l-Qarnayn, which is the year 1224 from the death of Alexander. We observed it ourselves: the middle of the eclipse in the city of Antioch was about 15⅓ equal hours after noon of the second day of Āb, and at al-Raqqa about 15 hours and a third and a quarter of an hour after noon, which is the time of the opposition; a little less than the diameter of the moon was eclipsed. By our computation the sun was in Leo 14;10 by its mean course, and in truth in 14;36 of it; the mean position of the moon was in Aquarius 19;24, and in truth, necessarily, opposite the true degree of the sun. So its course in anomaly from the point of the mean farthest distance on the epicycle was 110;7, and in truth 111;5; its mean motion in latitude was 190;10, and in truth 185;21, so that its true latitude at the middle of the eclipse, which is the time of the opposition, was about 28 minutes. By Ptolemy's computation

after noon in mean time.

[182] 23 January 901, about 8:20 in the morning, true time (Nallino).

[183] Al-Battānī thus puts somewhat more than 10 minutes of time, 2½ degrees of longitude, between Antioch and al-Raqqa, and for the second lunar eclipse below about a quarter of an hour (Nallino); the difference of their longitudes is 2° 55′, 11⅔ minutes of time.

[184] Schiaparelli's recomputation from al-Battānī's tables (Nallino's commentary) confirms the numbers as Nallino prints them; the codex has 126;32, 199;11 and 198;45 for 126;37, 169;11 and 168;45.

[185] 23 July 883 (Nallino).

[186] 113;0: for the minutes the text prints a sign that Nallino's translation and Schiaparelli read as the zero of the codex; Nallino's apparatus suggests 113;8, Halley's reading. Schiaparelli, recomputing the eclipse from al-Battānī's tables, finds 113;13 for the mean anomaly and 114;13 or 114;14 for the true anomaly, for the 114;9 of the text; Nallino's translation has 113;13 and 114;14.

[187] Oppolzer's canon of eclipses gives this eclipse a magnitude of 11.2 digits, against the 10 digits, ⅝ of the diameter, observed by al-Battānī; for the next, 12.6 digits against nearly 12 (Schiaparelli, in Nallino's commentary).

and on those ratios only a half and a third of the diameter of the moon should have been eclipsed, and the time of the middle of the eclipse should have preceded the time at which it occurred by observation by about a half and a third of an equal hour.[188,189]

So the times and amounts of the eclipses and the positions of the two luminaries differed in everything that we have mentioned, and we found the like of that, and less and more, in many solar and lunar eclipses whose times we observed and whose amounts we established. We have contented ourselves with mentioning these two lunar eclipses. In both of them the sun was in the region of its farthest distance, and the moon in both was at one and the same place in its distance, about 1½ parts less than its mean distance; the latitude of the moon in both was on one side, and between its first latitude and its second there was 0;3,50. The difference between the two eclipses came to one part of 8 and a half and a quarter parts of the diameter of the moon. When that is multiplied by these minutes between the two latitudes, the whole diameter of the moon in these two eclipses comes out about 0;33,30. If the ratio of the diameter of the shadow to the diameter of the moon is the ratio that Ptolemy stated, which is about 2⅗ times the whole diameter of the moon, the radius of the shadow at the place of the moon's passage is about 0;43,30. In the measure in which the 36 minutes 10 seconds that the moon moves in an hour at the times of the conjunctions and oppositions, its greatest course at those times, are 35⅓ minutes, the amount of the diameter of the moon then, the 30 minutes 12 seconds that the moon moves in an hour at those times, its least course then, are about 29½ minutes; this happens when it is at its farthest distance, and that is then the diameter of the moon. Ptolemy, however, worked on the basis that it is 0;31,20, and carried out on that basis the computation and the derivation of the distances and the diameters.[190,191,192]

Ar. p. 88

Since what we wanted has become clear, and the diameter of the moon according to its passage at each of the two distances has become evident, and we have taken the ratio of the diameter of the shadow to the diameter of the moon to be the ratio mentioned, the radius of the shadow at the farthest passage of the moon is by it about 0;38,30 when the sun is at its farthest distance; when the sun is at its farthest distance and the moon at its nearest distance, the radius of the shadow is about 46 minutes. So it is clear that the radius of the shadow at the farthest passage of the moon is less than what Ptolemy worked with by about 2⅙ minutes, because of the excess that fell in the diameter of the moon in his computation. As for the radius of the shadow at the nearest passage, it is equal in amount in the two computations. It must also be that the radius of the shadow has, between the farthest and the nearest distance of the sun, a variation that measurement requires, of about 50 seconds: the radius of the shadow when the sun is at its nearest must be less by this amount than when it is at its farthest distance.[193]

[188] 2 August 901, 15⅓ hours after noon in true time at Antioch, that is, 3 August at about 3:20 in the morning (Nallino).

[189] The mean position of the sun must be Leo 16;10: with al-Battānī's equation of the sun this gives the true position 14;36 of the text, and his tables give 16;10 (Schiaparelli, in Nallino's commentary; Nallino's translation has 16;10). Schiaparelli also reads 110;11 for the mean anomaly (the text has 110;7, the codex 110;4) and 111;9 for the true anomaly (the text 111;5): in the Maghribi script of the codex the numerals 9 and 5 are easily confused.

[190] The anomalies of the sun in the two eclipses were 42 and 52 degrees. The distances of the moon, in parts of which its mean distance is 60, were 58.04 and 58.32, that is, 1.96 and 1.68 parts less than the mean. The difference of the latitudes, 4 minutes by the rounded values given above, is 3;49 by an exact computation: al-Battānī computed the eclipses more precisely than he reports them (Schiaparelli, in Nallino's commentary).

[191] The magnitudes reported above, a little more than ⅞ of the diameter and a little less than the whole, differ by about ⅛, which would give 6 rather than 8¾, and a diameter of the moon of only 23 minutes. Schiaparelli, in Nallino's commentary, takes the passage as a check that the observations do not contradict a diameter of 33½ minutes, not as a measurement of it.

[192] Al-Battānī takes the apparent diameters of the moon to be in the ratio of its hourly motions. The hourly motions given here, 36;10 and 30;12, are not quite those of his tables, 36;4 and 30;18 (Schiaparelli). Ptolemy's 0;31,20 is in *Almagest* V 14.

[193] Ibn Yūnus objected that al-Battānī and al-Nayrīzī make the radius of the shadow vary by only about 7 minutes between the farthest and the nearest passage of the moon, much less than the geometry of the shadow requires; he agreed that the nearness of the sun changes it by no more than a minute (Nallino's commentary).

As for the solar eclipses, Ptolemy worked in them as we have mentioned. He made the diameter of the moon, when it is at its farthest distance, subtend an arc of the ecliptic of 0;31,20, and held that it then hides the whole sun at the times of the visible conjunctions, when it is on the belt of the signs to the sight. For this reason he made the diameter of the sun equal to the diameter of the moon, so that, though the sun is many times greater, the moon hides it; and he gave the diameter of the sun, relative to the moon, no change between its two distances, as he did for the moon. But it had become clear to us by measurement that the diameter of the moon at its farthest passage subtends an arc of the sphere of 29½ minutes, and that it cannot hide the whole sun from sight when it is at its farthest distance, since the sun's diameter is greater than its own, subtending 0;31,20. That is when the sun is at its farthest distance, when its course in an hour is 0;2,23; when it is at its nearest distance, its course in an hour is 0;2,33. In the measure in which the 2 minutes and 23 seconds are 31⅓ minutes, the 2 minutes and 33 seconds are about 33⅔ minutes; so we find that the diameter of the sun, relative to the moon, changes between its two distances by 2⅓ minutes. By this the amounts of the solar eclipses came out right for us, and it became clear that the radius of the shadow at the farthest passage of the moon subtends an arc of the circle of the sphere of 0;38,30, and its radius at the nearest passage of the moon an arc of about 46 minutes.[194,195]

Let us try to show the distance of the sun and what appears with it. This cannot be done in the way close to the view of Ptolemy on which he worked, except by repeating the figure as it is and with the ratios given in his book, and then apportioning it according to what we found by our observations; for whenever we work with what we found by our observations, the ratios do not come out right and the distance is multiplied many monstrous times. So we draw this figure and then follow it with the explanation. Picture what is drawn for you in it, and picture how it is worked, and you will get it right, if God wills.[196]

Ar. p. 89

I began, with God's help, and drew three circles, their centres on a straight line, facing one another, some greater than others. On the greatest of them, which is the circle of the sun, I wrote ABG, and D at its centre; on the one next to it in size, which is the circle of the earth, KLM, and N on its centre; and on the smallest circle, in the middle, which is the third, the circle of the moon, HZḤ, and Ṭ on its centre. I placed it between the circle of the sun and that of the earth. I drew the two lines of the rays at the ends of the diameter of the sun, the signs A and G, touching the circle of the earth at the points K and M and meeting on the other side at the point S, so that the triangle ASG is the cone. The line DS divides it into two halves, so that it becomes two triangles, each of them right-angled. From the centre of the earth, the sign N, I also drew two lines that touch the circle of the moon at the signs Ḥ and H and pass through the signs A and G, touching the circle of the sun at them, because of the solar eclipses in which the moon hides the whole sun from the sign N. We draw the diameter AG, and the diameter HḤ, which we extend to the point Q; we draw also the diameter KM. We take the position of the moon when it is at its farthest distance from the earth at the times of the lunar eclipses to be the point F, and make the line ṬN equal to the line FN. Since the line DS passes through all the centres, it is clear in this figure that the line DA is the radius of the sun, the line ṬḤ the radius of the moon and the line MN the

[194]The translation follows the corrigenda in reading «تَغْيِيرًا», «change», for «تَعْيِيرًا» (p. 88, line 6).

[195]From these values al-Battānī concludes that the moon at its farthest distance cannot hide the whole sun, so that an eclipse of the sun can leave a ring of the sun visible, which Ptolemy's equal diameters exclude (Delambre, cited by Nallino). The eclipse of 8 August 891 described above was annular along a band across Asia Minor and Mesopotamia, not far from al-Raqqa (Oppolzer's canon, cited by Schiaparelli). Al-Battānī's diameters of the sun, 31;20 at its farthest and 33;40 at its nearest distance, compare with the modern 31′ 32″ and 32′ 36″, and his diameter of the moon at its farthest distance, 29;30, with the modern 29′ 26″ (Nallino's commentary).

[196]Ptolemy's figure and numbers, which follow, are those of *Almagest* V 15. The figure rests on the relation that the parallaxes of the sun and the moon together equal the apparent radii of the shadow and the sun together. With Ptolemy's values (radius of the shadow 0;40,40, of the sun 0;15,40, parallax of the moon at 64;10 radii of the earth about 53½ minutes) the parallax of the sun comes to about 2¾ minutes; with al-Battānī's radius of the shadow, 0;38,30, to about ⅗ of a minute, and the distance of the sun grows almost fivefold.

radius of the earth. We also draw the line ʿFḌ, which is the diameter of the shadow; so the line FḌ is the radius of the shadow.[197,198]

Ar. p. 90

What Ptolemy worked with, and on which the course of the computation rests, is that the line AS is 60 parts in the measure in which the diameter is 120 parts; the triangle ADS is right-angled and the distance is great, so the line DS is also about 60 parts in that measure. The angle ḤNṬ is 0;15,40 in the measure in which the four right angles of the circle that goes round the right-angled triangle ADS are 360, and the angle ḌNF is in this measure 0;40,40. The sine ṬḤ, which subtends the angle ḤNṬ, is 0;16,24, and that is the line ṬḤ; the sine of the angle ḌNF is 0;42,35, and that is the line FḌ, in the measure in which the line FN is 60 parts. But in the measure in which the line MN, the radius of the earth, was one part, and in which the line FN, the distance of the moon from the centre of the earth, was 64;10, the line ṬḤ is 0;17,33 and the line FḌ 0;45,38; so the ratio of FḌ to ṬḤ is the ratio of 2⅔ to one, approximately. The line ṬN is equal to the line FN, and therefore the line FḌ and the line ṬQ together are twice the line MN. When the line FḌ, which has been shown to be 0;45,38, and the line ṬḤ, which has been shown to be 0;17,33, are added together, the sum is 1 part 3 minutes 11 seconds of the double; when the line MN, which is the single one, is taken from it, the line ṬN remains, 3 minutes 11 seconds, and the line DL, which completes the double, is 56 minutes 49 seconds. Likewise the line ṬN is 0;3,11 and the line DṬ 0;56,49 when the whole line DN is made one part, and the line DN measures the line ṬN about 18⅘ times. Likewise the line DA measures the line ṬḤ also about 18⅘ times, which is the ratio of the line DN to the line ṬN. It has been shown that the line ṬN is 64;10 in the measure in which MN is a line of one part; so the line DN, the distance of the sun from the centre of the earth, is about 1210 times the line MN, the radius of the earth. The diameter of the sun is about 18⅘ times the diameter of the moon, but the diameter of the earth is about 3⅔ times the diameter of the moon; so the diameter of the sun is 5½ times the diameter of the earth.[199,200]

Ar. p. 91

The cube, which has length, breadth and depth, that results from multiplying one by itself and then by one is one: I mean the cube of the earth. The cube that results from multiplying 5½ by itself and then by 5½ is 166 and a quarter and an eighth. The cube that results from multiplying 18⅘ by itself and then by 18⅘ is 6644½. The cube that results from multiplying the diameter of the moon, which is the line HḤ, by itself and then by the diameter of the moon is one part of 39¼ of the cube of one. So the size of the sun is 166 and a quarter and an eighth times the size of the earth, and 6644½ times the size of the moon; and the size of the earth is 39¼ times the size of the moon.[201,202]

Further, if the line MN is one part, the line FḌ is 0;45,38, and the line FN in the same measure is 64;10. If the whole line SN is made one part, the line FS of it is 0;45,38 and the line FN 0;14,22, which remains to complete the one part. Since the line SF is [to the line FN] in the ratio of 45 minutes 38 seconds to 14 minutes 22 seconds, the line SF is 203 and a half and a third in the

[197] The text calls the greatest circle «the circle of the earth» as well; it is the circle of the sun.

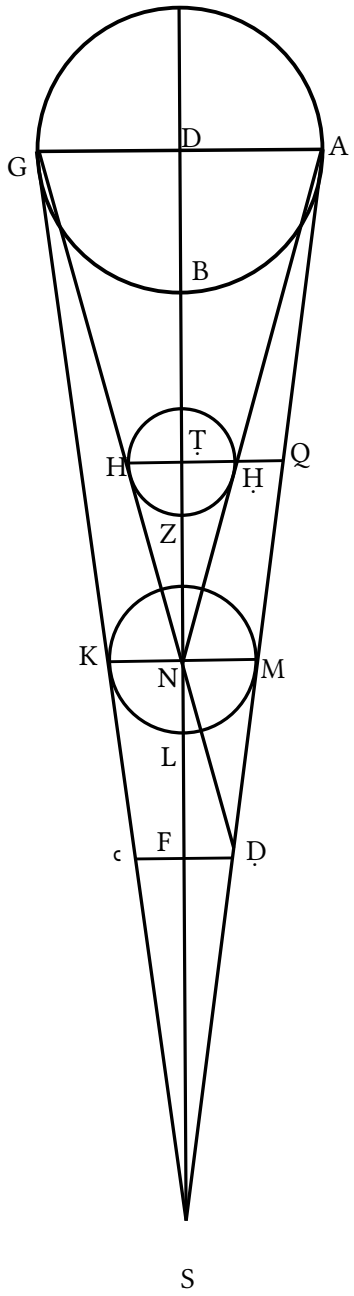
[198] The lines from A and G touch the circles near the ends of the diameters drawn, not at them; the difference is imperceptible, and Ptolemy says that the lines through the points of contact are equal to the diameters to the senses (Nallino).

[199] Nallino takes «that the line AS is 60 parts … about 60 parts in that measure» and «of the circle that goes round the right-angled triangle ADS» for interpolations, found also in Plato of Tivoli: neither AS nor DS is 60 parts, as what follows shows. The angles are the apparent radii of the moon and of the shadow at the moon's farthest distance (*Almagest* V 14).

[200] «Twice» is supplied by Nallino. The line that completes the double, 2 − 1;3,11 = 0;56,49, is ḤQ; the text calls it DL. The words «the line ṬN remains, 3 minutes 11 seconds, and» anticipate the next sentence, where DN is taken as one part; Nallino's translation leaves them out and reads ḤQ (they are in the codex and in Plato of Tivoli). DN is to DṬ as MN is to ḤQ (Nallino). Ptolemy finds the same 1210 radii of the earth in *Almagest* V 15.

[201] The translation reads «والعُمْق», «and depth», for «والغَمْق» (p. 90, line 15), a misprint that the corrigenda do not list.

[202] Ptolemy (*Almagest* V 16) makes the sun about 170 times the size of the earth, the quotient of 6644½ by 39¼; al-Battānī cubes the rounded ratio 5½.



The line DN is not in the figure of the codex or in that of Plato of Tivoli; Nallino added it.

measure in which the line FN is 64⅙. When the line FN is added to the line SF, the line SN, from the end of the shadow to the centre of the earth, is about 268 times the radius of the earth; but [the line] from the centre of the sun to the end of the shadow, the line DS, is 1478 times the radius of the earth, the line MN. These are the ratios that Ptolemy derived, and the distances, on the basis of the size of the diameter of the shadow and the sizes of the diameters of the moon and the sun with which he worked.[203]

Ar. p. 92

Since we have made clear the difference that appeared in this, that the diameter of the moon at its farthest passage is only 0;29,30, the radius of the shadow at the farthest passage of the moon 0;38,30, and the diameter of the sun, as has been said, 0;31,20, on this basis we find that the diameter of the moon falls short of the diameter of the sun by a minute and a half and a third. If we take its amount out of the 5 minutes and a half and a third by which the moon varies in its distance from the earth and its nearness to it, we find it to be about a third less a fifth of a tenth of that. When we take [that part] of the 10⅓ parts that are the whole diameter of the epicycle, by which the distance of the moon from the earth varies at the times of the conjunctions and oppositions, it comes to about 3⅕ parts. When we subtract that from the 64;10 that is the farthest distance of the moon from the earth, we find that the distance of the moon from the centre of the earth at the place where its diameter is 0;31,20, where it can hide the whole sun from sight, is about 60 and a half and a third and a tenth and a third of a tenth times [the radius of the earth]. When the diameter of the moon is 0;31,20, the radius of the shadow at the place of the moon's passage is about 0;40,40, which is close to the amount with which Ptolemy worked. When we multiply the 18⅘ by the 60 and a half and a third and a tenth and a third of a tenth, the distance of the sun from the centre of the earth when it is at its farthest distance is 1146 times the radius of the earth; and when we divide that by the 4½ parts that are, approximately, between the diameter of the earth and the diameter of the sun, [the line] from the centre of the earth to the end of the shadow is 254⅔ times the radius of the earth. Since the whole diameter of the epicycle of the sun, which is twice the distance between the centres, as appeared earlier in this book, is about 4⅙ parts, the variation of the distance of the sun from the centre of the earth is about 76 times the radius of the earth, and half of that, 38, is the share of the mean distance. So the nearest distance of the sun from the centre of the earth is 1070 times the radius of the earth, its mean distance 1108, and its farthest distance 1146, as has been shown; and the moon hides the sun from sight when there are about 1085 times the radius of the earth between it and the sun. This is the ratio that appeared to us from the solar eclipses.[204,205]

As for the moon's receiving light, it is from the sun, and the increase and decrease of its light are according to its distance from it and its nearness to it. For of every round body the eye falls only on the half that faces it. So the half of the globe of the moon that faces the earth is the half that faces the sun when we see the moon full of light, and that is at the times of the middle of the lunar month. When the half that faces the earth is other than the half that faces the sun, and the half that faces the sun is the remaining one, we see no light in it at all; that is the time of the dark of the moon. In between, we see the light in it according to how much of the light of the half that faces the sun falls in the half that faces the eyes. The farther it goes from the sun from the time of the dark of the moon, the more light there is in it, and it increases until the moon is opposite the sun, about a diameter of the sphere away, and is full of light. Then it begins to decrease, as it did in increasing, until it vanishes at the end of the month.

Ar. p. 93

We draw an example of this: a circle for the sphere of the moon about the centre Y, with the diameter ZN. We extend the diameter ZN to the point B, take the point B as centre and draw about it a circle for the sun, on the diameter AG. Let the point Y be the centre of the earth, and

[203] The words in brackets are supplied by Nallino. SN is to FS as MN is to FD. Ptolemy's values are in *Almagest* V 15.

[204] The distance is 60 29/30 radii of the earth; 18⅘ × 60 29/30 = 1146, and 1146 − 60 29/30 = 1085 (Nallino).

[205] Al-Battānī's distances of the sun, 1070 to 1146 radii of the earth, are like Ptolemy's 1210 about twenty times too small: the mean distance of the sun is about 23,500 radii of the earth.

the line BY represent the distance of the sun from the earth. We take the point N of the inclined lunar sphere as centre and draw about it a circle for the moon at the time of the conjunction, when the centre of its circle is under the centre of the circle of the sun in direction, I mean on the line that passes through the centre of the sun and of the earth. Then the centre of the circle of the moon moves away from the point N on its sphere according to its course in a day, or in more than a day or less, until it reaches the opposition, where the centre of its circle is the point Z. We mark the centres of the circles of the moon between the points Z and N, one after another, with the sign D. From the two ends of the diameter of the sun, the signs A and G, we draw lines to all the circles of the moon, so that two lines touch each of the circles at the two ends of a diameter of the circle of the moon; we mark on them in all the circles the signs K and Ṭ, and join the points K and Ṭ of each circle by a line passing through the centre D. Then we colour red the bright half of the circle of the moon, which faces the sun, and blot out the other half, which the sun does not see. Then we draw from the point Y, the centre of the earth, to each of the circles of the moon two lines touching the circle of the moon, mark the places of contact on each circle with the signs M and L, and join them by a line passing through the point D of each circle. So we find the half of that circle on which the eye falls, which is the half that faces the earth; and according to how much of the bright half that faces the sun falls in this half that faces the earth and on which the eye falls, is the amount of light seen in the moon. It is clear in this figure that the points M and L of the circle of the moon fall, at the dark of the moon, at the place of the points K and Ṭ. When the moon moves away from the sun, a small segment of the half of its circle that faces the sun falls in the half that faces the earth; the more its distance from the sun increases, the more the segment grows, until the quarter of the month is reached, when the [lit] half amounts to half of the half that faces the earth. Then the segment continues to grow beyond that amount until the moon is opposite the sun, when the whole half that faces the sun is the half that faces the earth, and the points M and L come to the places of K and Ṭ.[206]

We shall draw this figure and place ten circles of the moon in it; the distance of the moon from the sun in each of the nine circles after the circle of the conjunction is 20 parts of the parts of the half circle, and the shape of the light in the circle of the moon is according to the amounts of these distances from the direction of the position of the sun, which is the point B. Since this is so, the increase and decrease of the light of the moon has become clear in this figure, as you see it on the plane of the circle. In the round body, however, it is twice each segment, because its angle, in the spherical way, is unlike this figure; we shall represent it in what follows, if God Most High wills, and success is from God.[207]

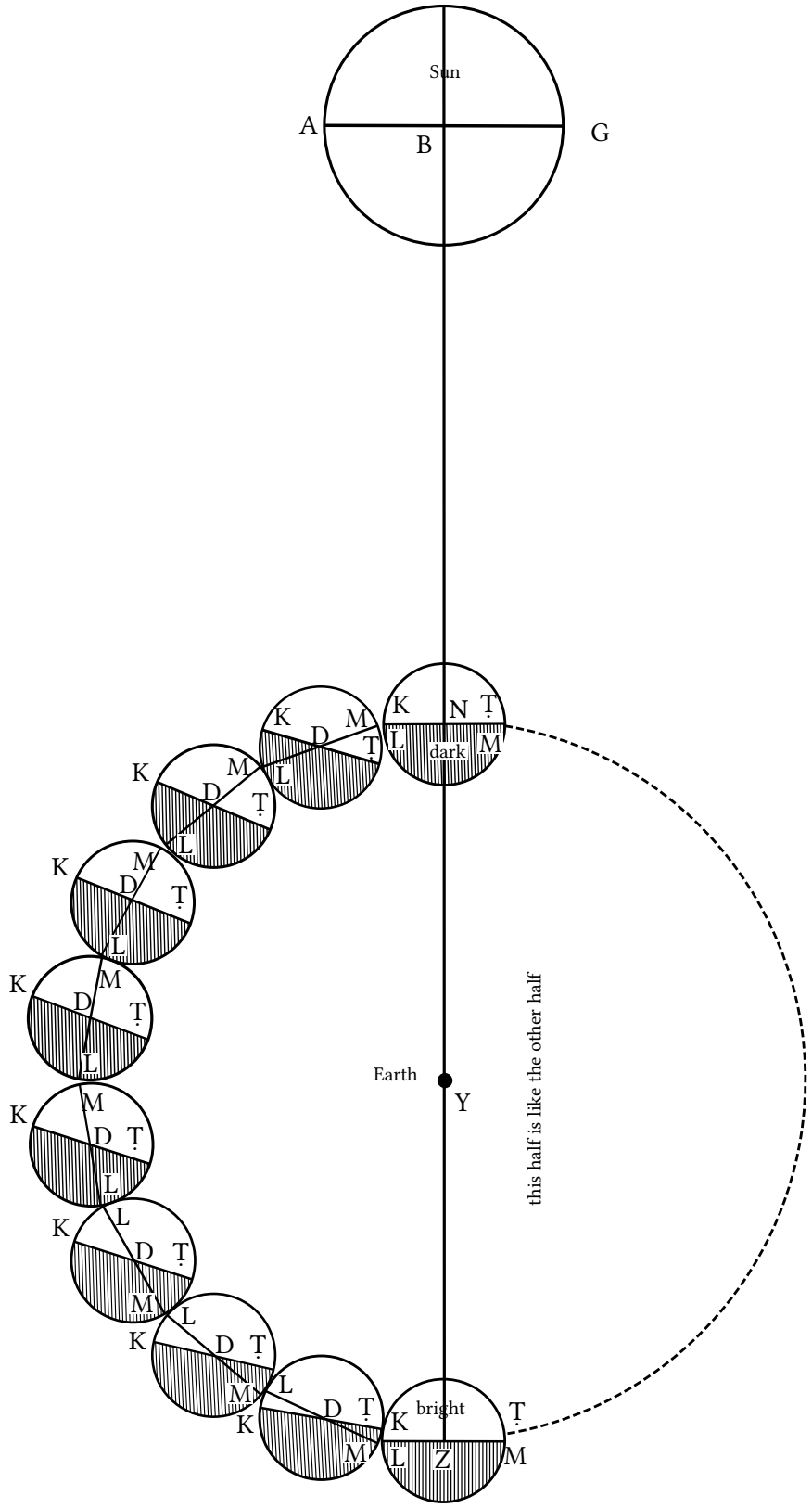
Ar. p. 94

It has become clear from what has appeared that, in the measure in which the circle of the moon in which the light is seen complete is 15, each 12 parts of its distance from the sun are one part, up to the full 180, which make up the whole 15; and if the circle of the moon is 12 parts, each 15 parts of the distance are one of the parts of the light. On these ratios we have drawn the two circles and explained them, if God Most High wills.[208]

[206] In the printed figure the dark halves are hatched and the bright halves left white. Its letters are Nallino's: the figure of the codex and that of Plato of Tivoli have none. He leaves out the lines from A and G that touch the circles of the moon at K and Ṭ, to keep the figure clear, and the lines from Y that touch them at M and L.

[207] That is, the circles stand 20 degrees apart: nine steps of 20 degrees lead from the conjunction to the opposition. For «what follows», Nallino suggests chapter 43.

[208] That is (Nallino): if the disc of the moon in the figure is made 15 degrees of its orbit, twelve discs fill the half orbit; if it is made 12 degrees, fifteen.



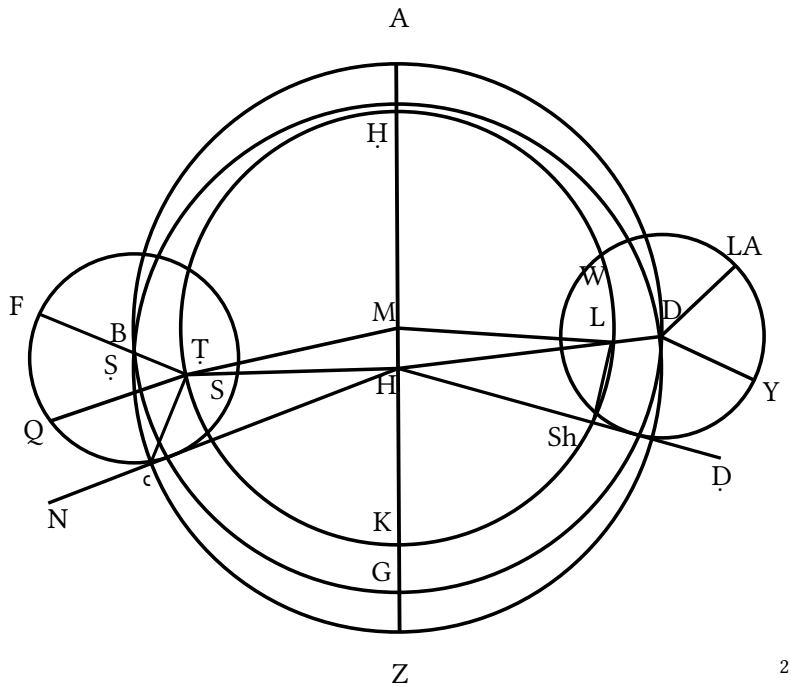
The words of the figure are «the sun», «dark» in the first circle, at the conjunction, «bright» in the last, at the opposition, «the earth», and, along the dashed arc on the right, «this half is like the other half»: the waning moon would repeat on the right the circles of the waxing moon.

Chapter 31. On the description of the spheres of the five stars and their states

Ar. p. 95

He said: As for the description of the spheres of the five stars and of their states, which have become known by way of demonstration from the anomaly of their course: each of them has four spheres, in the manner of the spheres of the moon. The first is made like the ecliptic: its centre is the centre of the ecliptic, it lies level beneath it, and its motion is like its motion. The second is the inclined sphere: its centre is the centre of the parecliptic sphere and its size the size of that sphere, and its greatest inclination from the circle of the parecliptic sphere, towards the north and the south, is the whole latitude of the star. Inside this sphere is another sphere, whose centre lies outside the centre of the two spheres; it hangs from it and touches it at a point, which is the point of the farthest distance. From the amount between the centres of the two spheres the equation of the anomaly and the centre of each of the stars is known, as was shown for the moon. The fourth sphere is the epicycle of the star. Its centre runs on this eccentric sphere from the point of the farthest distance in the order of the signs, by the amount of the mean motion of the star in longitude in a day; and the star moves on the epicycle, from the point of the distance that is seen from the centre of the ecliptic, also in the order of the signs, by the amount of the star's own motion in anomaly each day. The radius of each of the epicycles of the stars is the amount of [the sine of] its mean equation. It has a displacement in the lower and the upper part of the circle: it grows in the lower part, exceeding the mean, and shrinks in the upper part, falling short of the mean. This mean equation is what is set down in the sixth column of the tables of the equation of the star; the amounts of the shortfall are set down in the fifth column, and the amounts of the excess in the seventh. What is set down in the fourth column is the minutes in proportion to which [a part] of the excess and the shortfall is taken, in the way that I gave for the moon for the excess. What is set down in the third column is the equation of the anomaly and the centre, which results from the distance between the centres. These are the figures of the spheres that we said we would draw.[209]

Ar. p. 96



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[209] The words in brackets, «the sine of», are added by Nallino's emendanda (Part II, p. XXXI). The «centre» is the distance of the centre of the epicycle from the farthest distance of the eccentric, counted at the centre of the eccentric; the «equation of the anomaly and the centre», which the eccentricity produces, is applied to both (Nallino).

[210] Nallino prints the figure as the codex gives it, «most ineptly» in his words. Schiaparelli, in his commentary, points out that the circle of the ecliptic and the inclined circle cannot be drawn in one plane as two circles about different centres, that the nodes B and D are put 90 degrees from the line of the apsides, which is true of no planet, and that

He said: The first thing I drew was the circle of the parecliptic sphere, marked ABGD, about the centre H; another circle, for the inclined sphere, marked ḤBZD, whose centre is also the point H, as it falls on the globe; and a third circle, for the eccentric sphere, marked ḤṬKL, with M at its centre. Clearly the point Ḥ is the farthest distance of the eccentric sphere and the point K its nearest distance. We take the point Ṭ of the eccentric sphere as the centre of the circle of the epicycle and draw about it the circle of the epicycle, marked FQS. We draw from it the line MṬQ and the line HṬF, take the position of the star on the epicycle to be the point ʿ, and draw the line HʿN, along which the motion of the star on the ecliptic is seen. Clearly the diameter AZ passes through the centres. Again we take the point L of the eccentric sphere as centre and draw about it a circle for the epicycle, marked Y LA Sh; we draw the two lines M L Y and H L LA, put the star at the position Sh of the epicycle, and draw the line H Sh Ḍ, along which it is seen on the ecliptic.[211]

Ar. p. 97

It is clear in these circles that, when the point A is the direction of the farthest distance on the ecliptic, the star is at the point ʿ of the epicycle whose centre is Ṭ, and the centre of the epicycle is at the point Ḥ, the line that goes out from the point H passes through the centre M and the points A and Ḥ, and we find the point of the farthest distance on the epicycle at the sign Q, on the line MṬQ. For the line MṬQ is then in the place of the line HMA, and the place of Q is then the place of A; so the mean and the true farthest distance are the same, without any difference, since the line HṬF, along which the true point of the farthest distance is seen, then also comes into the place of the line HA. But when the centre of the epicycle comes to the place Ṭ, between the points Ḥ and K of the eccentric sphere, which is less than half its circle, the place of the true distance on the epicycle is the point F and the place of the mean farthest distance the point Q; from Q the course of the star on its own epicycle is counted, which is the arc Qʿ, and so its course in anomaly is greater than the arc Qʿ by the arc QF, which is the difference. Likewise the position of the centre of the epicycle that is seen from the point H on the ecliptic is less than the one seen from the centre M, again by the arc QF, since the point F is nearer to the point M than the point Q is. Likewise, when we put the centre of the epicycle in the second half of the circle, at the point L, the position of the centre of the epicycle in which it is seen on the ecliptic from the centre H becomes greater than the position in which it is seen from the centre M, by the arc LA Y, and the true farthest distance, which is seen from the centre H, becomes less than the mean farthest distance, which is seen from the centre M, again by the arc LA Y. For the star is at the mark Sh of the epicycle; its mean motion on its own epicycle is from the point Y to the point LA and then to the point Sh, and its true motion is from the point LA, so that the arc LA Sh of the epicycle is smaller than the arc Y LA Sh by the arc Y LA. Therefore the equation of the anomaly and the centre is added to the centre when the motion of the centre of the epicycle is between the point of the farthest distance of the eccentric sphere and its point of the nearest distance on the side of the sign L, and it is subtracted from the anomaly; and when the centre of the epicycle is in the second half of the eccentric sphere, on the side of the sign Ṭ, the equation of the anomaly and the centre, which is the arc Y LA, is subtracted from the centre and added to the anomaly. When the true position of the centre on the ecliptic is known, it gives the shares of the difference, of the excess and the shortfall, that belong to the epicycle at its displacement, since it is on the eccentric sphere.[212]

Ar. p. 98

Further, when the star is at the position ʿ of the epicycle, its distance from the point Q is less than

the lines of sight need not touch the epicycles. He suspects that the circle drawn about M as large as the ecliptic was at first the equant of Ptolemy's theory, the circle about which the centre of the epicycle moves uniformly.

[211] The text has S for Sh, in the letters of the second epicycle (the codex has ʿ) and for the position of the star on it; the figure and what follows have Sh, as Nallino's translation reads.

[212] The text has the centre of the epicycle move uniformly about the centre M of the eccentric, and so counts the «centre» at M; in Ptolemy's theory the uniform motion is about the equant point, which lies as far from M as H does, on the other side (Schiaparelli, in Nallino's commentary). On the side of Ṭ the equation is the arc QF; the text names the arc Y LA of the other half (Nallino).

half the circle of the epicycle, so the position in which it is seen on the ecliptic is greater than the position of the centre Ṭ on the ecliptic by the arc on the perpendicular ʿṬ. When the star is at the position Sh, the arc LA Sh of the epicycle is more than half its circle, so the position in which it is seen on the ecliptic is less than the position in which the centre L is seen by the arc that falls on the perpendicular L Sh. Therefore the mean equation of the star, corrected for the displacement of the epicycle, is added to the corrected centre when the corrected anomaly of the star is less than 180, and is subtracted from it when the corrected anomaly is more than 180; what results after that addition or subtraction is the distance of the star on the ecliptic from the point of the farthest distance of the eccentric sphere, whose position on the ecliptic is determined.[213]

As for the cause of the retrogradation of the wandering stars, it is that the change of the equation of the star, in its increase and decrease in one day, is at some of its places on the epicycle greater than its mean course, which is the motion of the centre of the epicycle on the eccentric sphere in a day. When the true position of the star is computed for a day, and its course for another day is then added to its mean position, and from that is subtracted an equation that exceeds the first equation by more than its mean course in that day, or to it is added an equation that is less than the first equation by more than the mean course of the star in the day, the position of the star on the ecliptic is less than the position in which it was at first. This can happen only when the star is in the nearer half of its epicycle, which is necessarily smaller than the upper half, since the arc of the eccentric sphere bounds these two halves: the lower half is the arc Sh W, and the upper half the arc W LA Y Sh. So when the course of the star from the point of the farthest distance on the epicycle reaches the point W, it is there like one standing set up in the circle, for it then falls on the line that touches the epicycle and goes out from the point H. The decrease and increase of the change of its equation in a day are then equal to its mean course in a day, and no motion appears in it until it passes the point W and falls in the lower half. Then its motion on the ecliptic continues to be seen towards the preceding side of the signs until it reaches the sign Sh and falls on the tangent line a second time, when it is like one standing raised in the circle on that side. When it rises from the point Sh, its motion begins to be seen in the order of the signs, as long as it is in the upper half of the epicycle. Yet the star in itself has no retrogradation in its course; this befalls it only for us, because of the difference of the centres of its spheres and its state on the epicycle.[214,215]

As for the sun and the moon, this does not befall them for us, because the course of each of them in a day is many times greater than the change of its equation in a day. So the state of retrogradation is not evident in them, although they too must in themselves have a course in the nearer half of their epicycles; but it is not perceived in them.

Ar. p. 99

We have tested the motion of each of these five wandering stars many times at places on their epicycles: at one, when the star was at their point of the farthest distance; at a second, when it was at the point of the mean distance; at a third, when it was at the point of the nearest distance; and at other places of the centre of the epicycle on the eccentric sphere, farther from and nearer to its point of the farthest distance. So we established what appeared of their mean motions in longitude in excess of the motions set down in Ptolemy's book, and also what appeared of the anomaly of their motions and of the positions of their farthest distances on their spheres

[213] «The arc on the perpendicular ʿṬ» is the arc whose sine is the perpendicular from Ṭ to the line of sight Hʿ, that is, the angle at H between the centre of the epicycle and the star; in the figure the line of sight touches the epicycle at ʿ (Schiaparelli, in Nallino's commentary).

[214] In modern terms, the motion in longitude in a day is the mean motion together with the change of the equation; when the change is negative and greater than the mean motion, the star goes backwards. The sentence from «When the true position» restates this obscurely (Schiaparelli, in Nallino's commentary).

[215] Delambre, cited by Nallino, points out that at the point of contact the star moves along the line of sight, so that its motion on the epicycle adds nothing to the motion of the centre: the star is not stationary there, and the stations fall between the points of contact and the nearest point of the epicycle. The points of contact are not where the epicycle cuts the eccentric either, as the figure itself shows (Nallino).

eccentric to the ecliptic. We corrected it and entered it in the tables, after adding to their course in longitude the correction that we found in each of them. Their equations we found close to those in Ptolemy's book, and the positions of their distances likewise, and we entered them as they were, except the farthest distance of Jupiter: we measured it by the moon many times, according to the visible position of the moon at the times of the measurements, and found that it was less than the amount set down by about 8 parts.[216]

Since the motions of the upper stars on their epicycles are what remains of the mean course of the sun when the mean motion of the star in longitude is taken from it, and the mean course of Venus and Mercury is equal to the mean course of the sun, their anomaly is obtained from the tables. We found the anomaly of Venus greater than its anomaly as set down by about 4½ parts, and that of Mercury by about 2½ parts. We divided that over the time between us and Ptolemy, and added what fell to one day to the course in anomaly of each of them for a day. As far as we were able, we have left nothing from which we saw that a defect might arise without putting it right, although their motions cannot be grasped exactly, as the motions of the two luminaries can, since their observations were made only when they came close to some of the fixed stars. Since their farthest distances also move with the motion of the sphere of the fixed stars, we have had no need to set down tables of the motions for the anomaly of each of the upper stars and for the course of the two lower stars, or of the measurement of their distances from the Heart of the Lion or from other fixed stars. We have also left out the mention of the observations of them that we made at the places mentioned, for the sake of brevity and so as not to lengthen the discourse in the explanation that we are attempting for each of them.[217]

Ar. p. 100

As for the latitudes of the three upper stars, I mean Saturn, Jupiter and Mars, they come close to the amounts that we found in Ptolemy's book, and we kept them as they were for finding their latitudes. But in the latitudes of Venus and Mercury we found a great discrepancy, which arose, as we think, from the procedure used to find the latitude; so we changed the method of working that we found for them in Ptolemy's book to one that we saw comes close to and agrees with the latitudes that we find for them by observation. It may be that what happened in the procedure in Ptolemy's book came from the translator of his Greek words, or from a fault in the copy from which the book was translated; and God knows best.[218]

[216] In the 740 years between the epochs of Ptolemy's tables (AD 137) and al-Battānī's (AD 880), al-Battānī's mean motions in longitude exceed Ptolemy's by 2° 54′ for Saturn, 3° 5′ for Jupiter and 4° 12′ for Mars, and are better than his (Nallino). Al-Battānī's farthest distances of Saturn and Mars are 20° 18′ greater than Ptolemy's (for Mars with Nallino's correction of the minutes), that of Jupiter only 12° 19′, the 8 degrees of the text. He had raised Ptolemy's longitudes of the fixed stars by 11° 10′; Nallino does not know why he used 20° 18′ for the farthest distances, and supposes that he relied on his own observations.

[217] Over the 740 years between the epochs of the two sets of tables, al-Battānī's anomalies exceed Ptolemy's by 4° 28′ for Venus and by 0° 57′ for Mercury; Nallino's translation therefore has 1 degree for Mercury, where the codex has 2½ and Plato of Tivoli 11½. Such tables, with the positions measured from the Heart of the Lion (Regulus), are in Theon's tables (Nallino).

[218] Al-Battānī's procedure for the latitudes of the five stars is in chapter 47.